THE EVOLUTION OF ENGINEERING TEAMS

Why Engineering Never Seems To Build "The Right Thing"

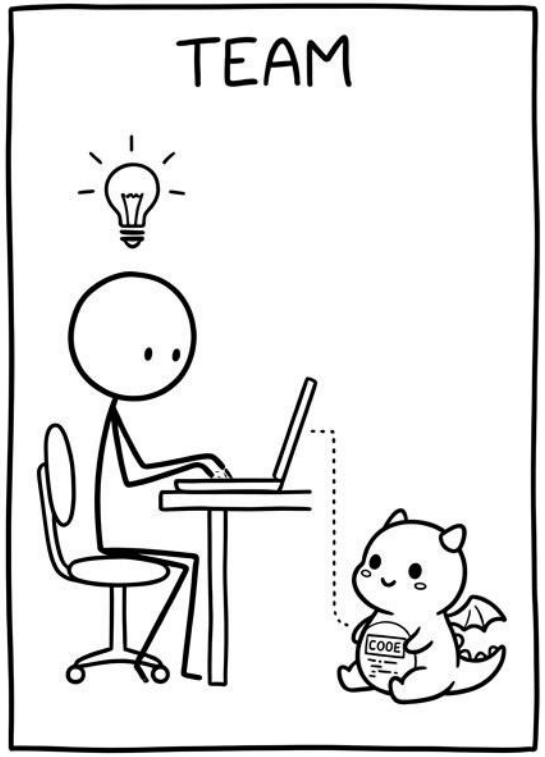
A Systems Explanation of Why It's So Damn Hard

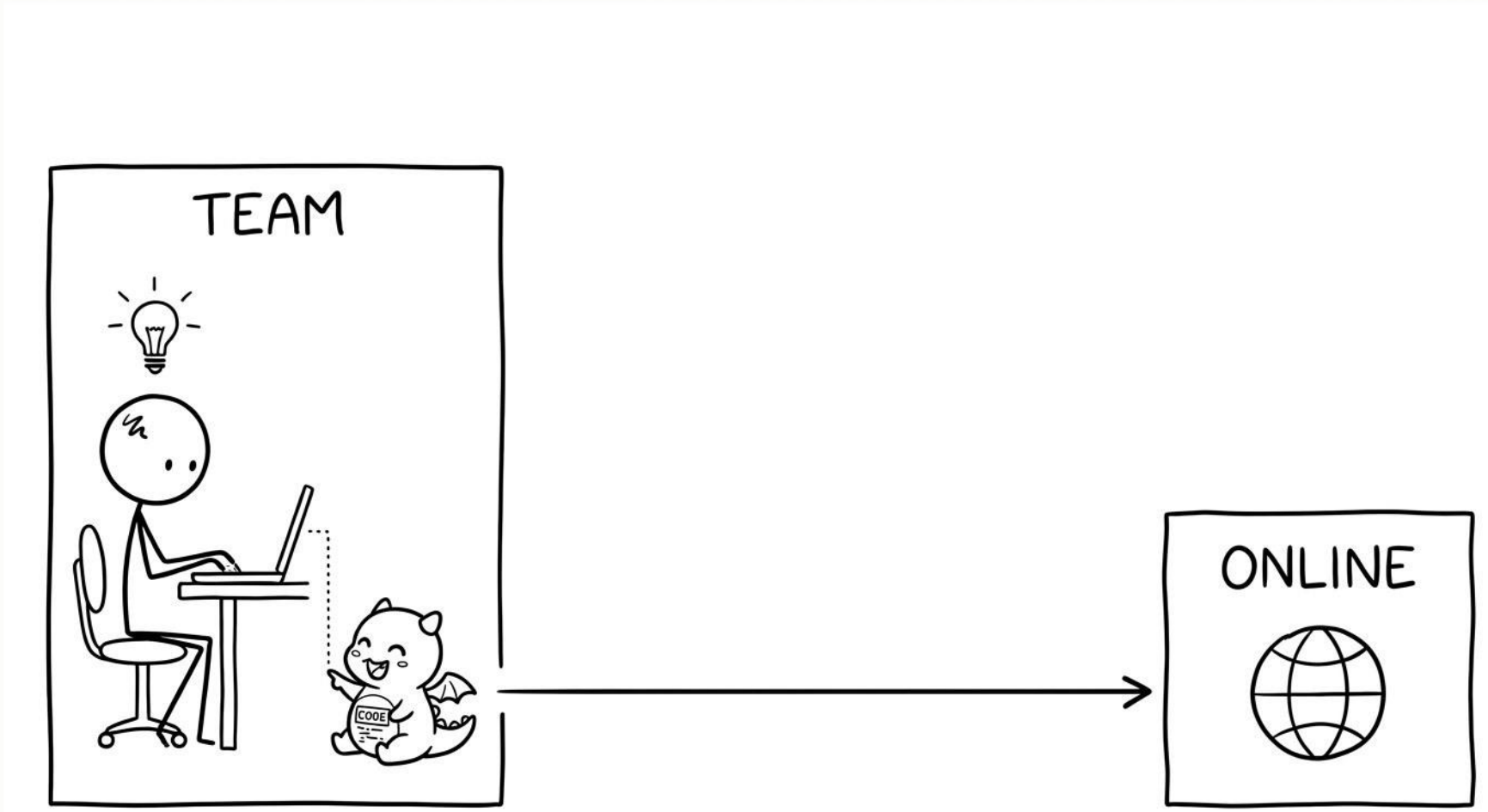
A field guide for non-engineers and new managers, in stick figures.

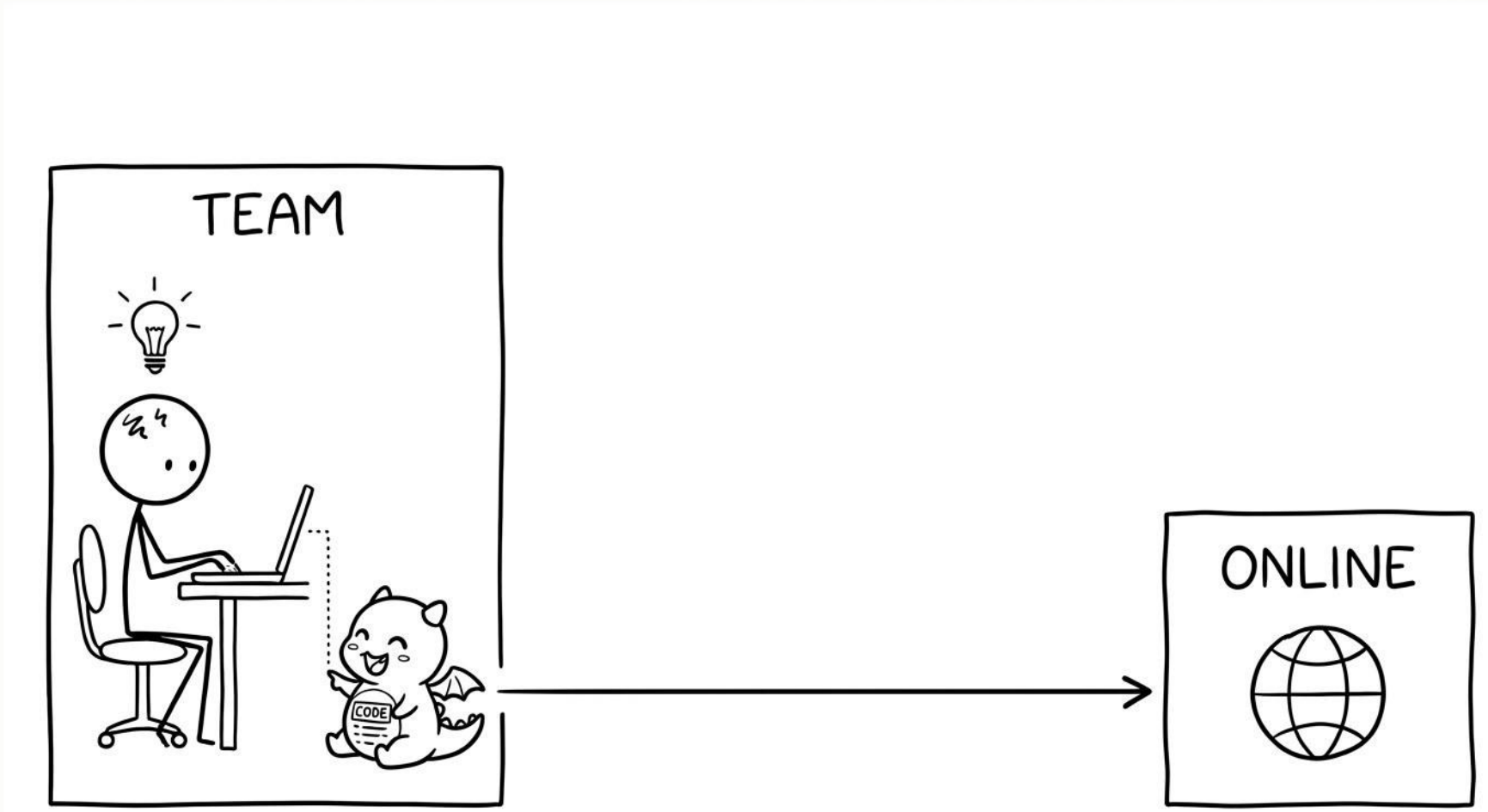
PART 1

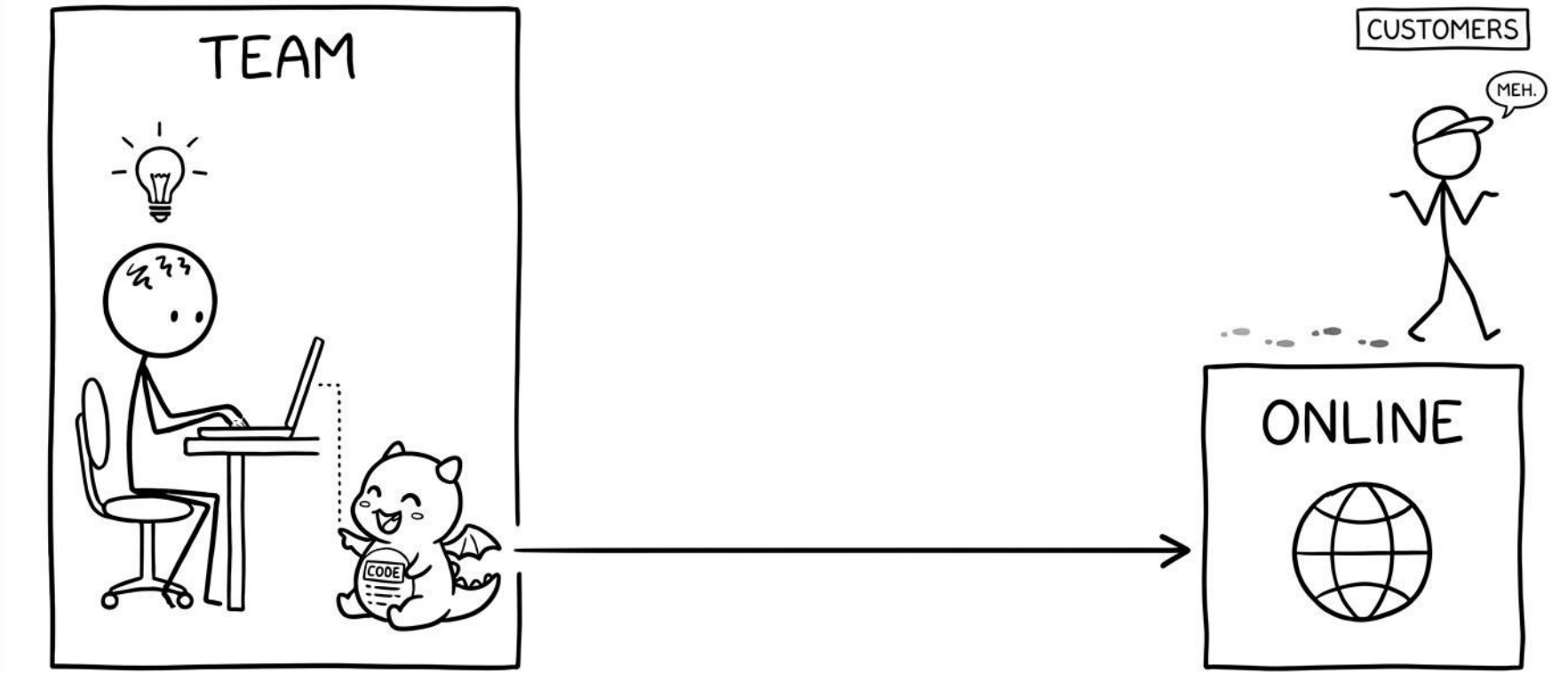
How an Engineering Org Grows

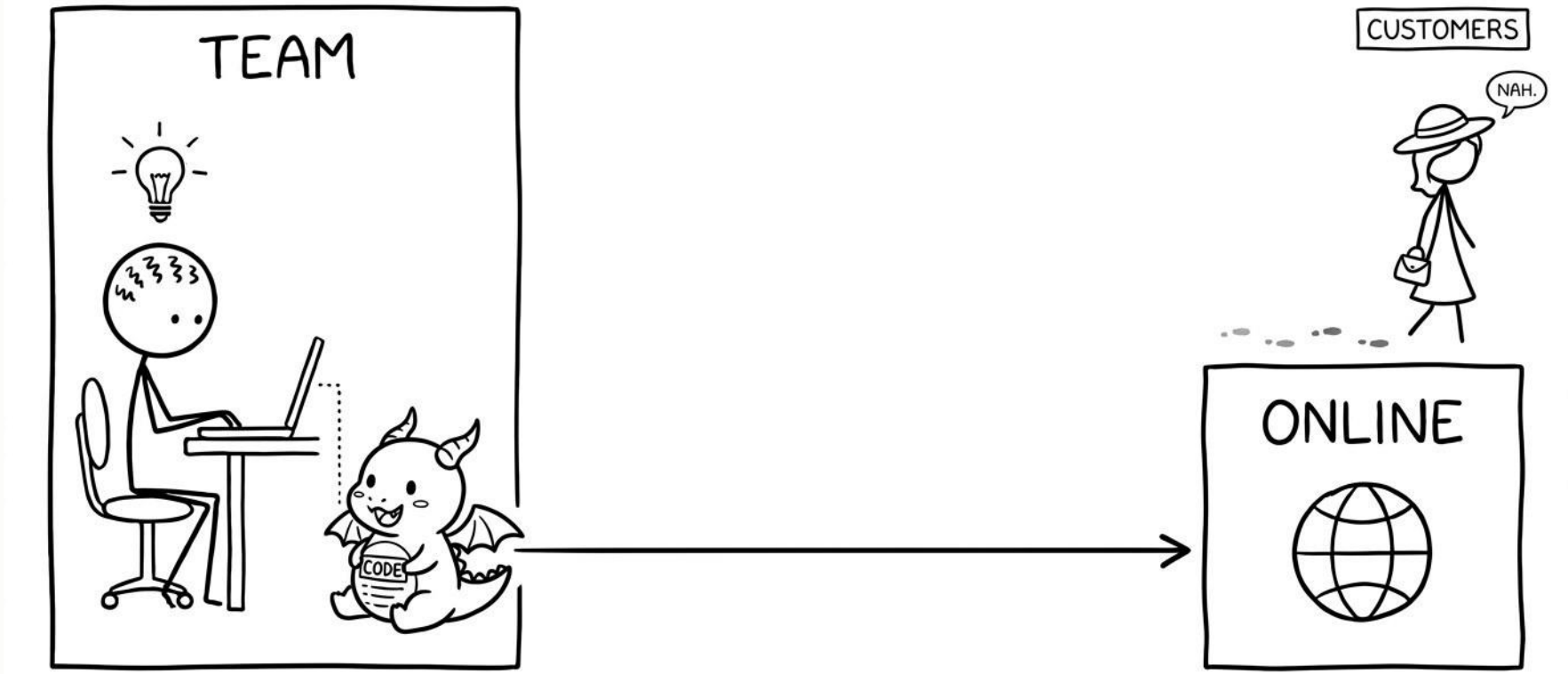
From one engineer to a whole org, one sensible step at a time

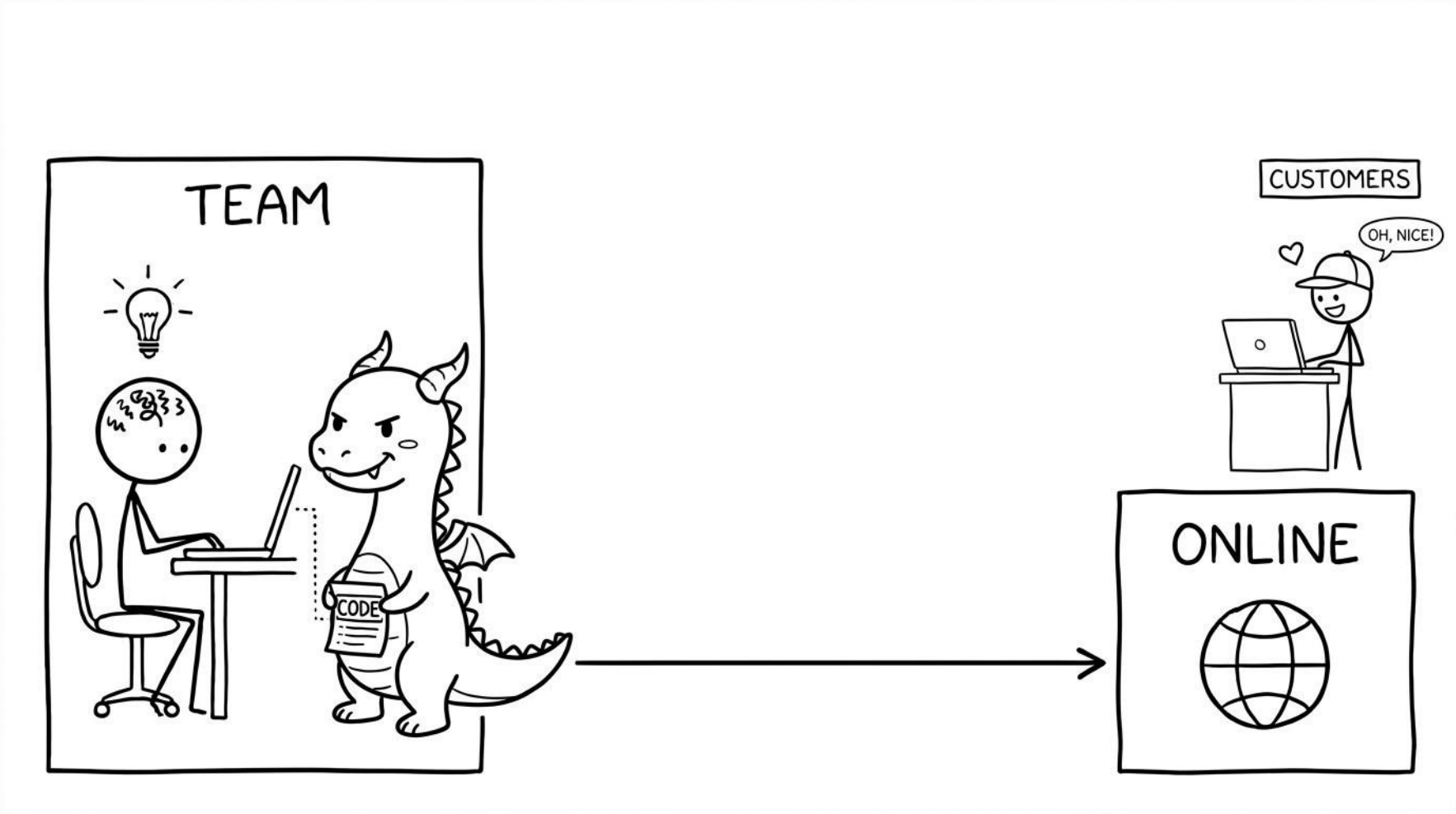

TEAM
COOE

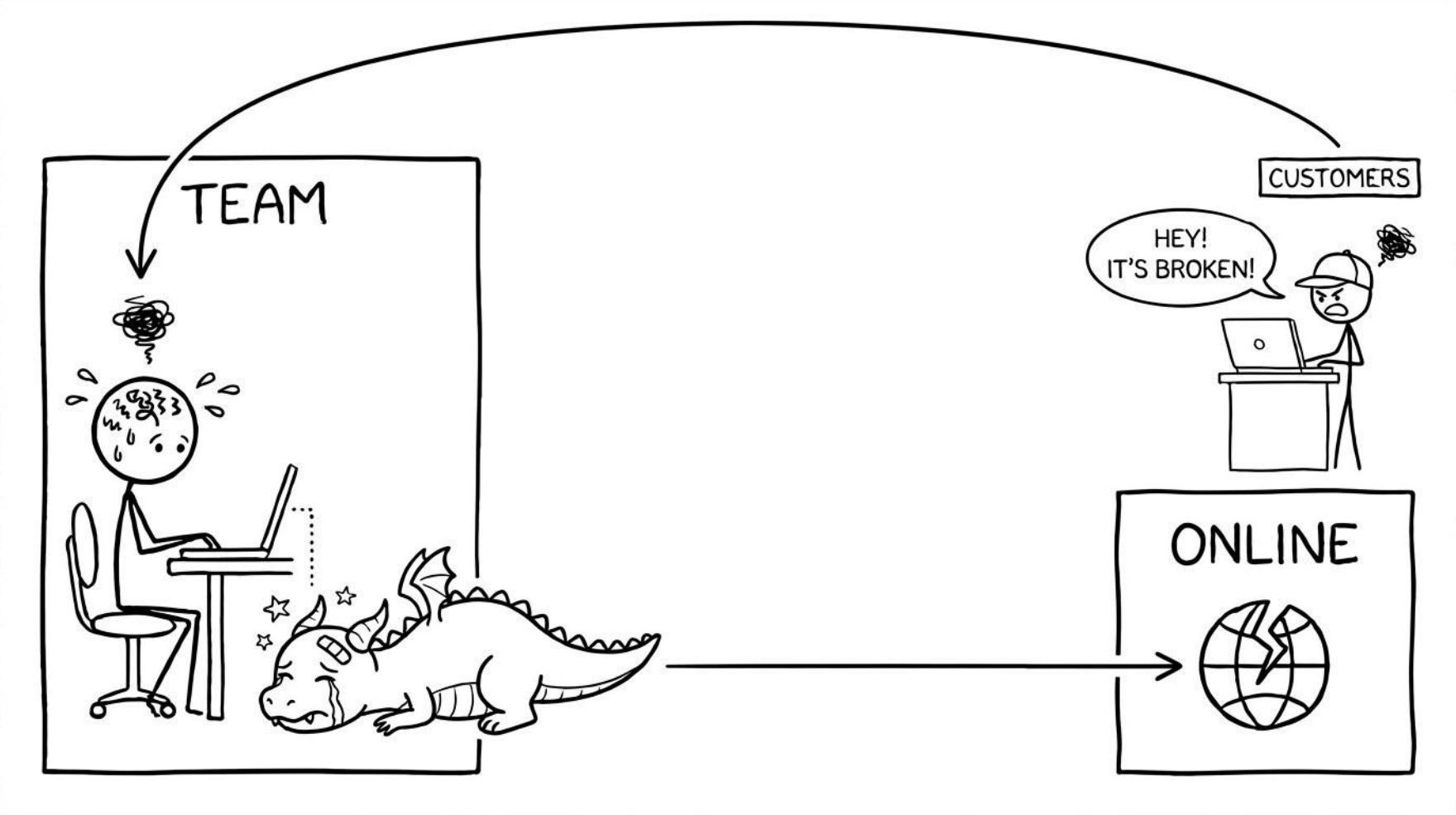

TEAM
CODE
ONLINE

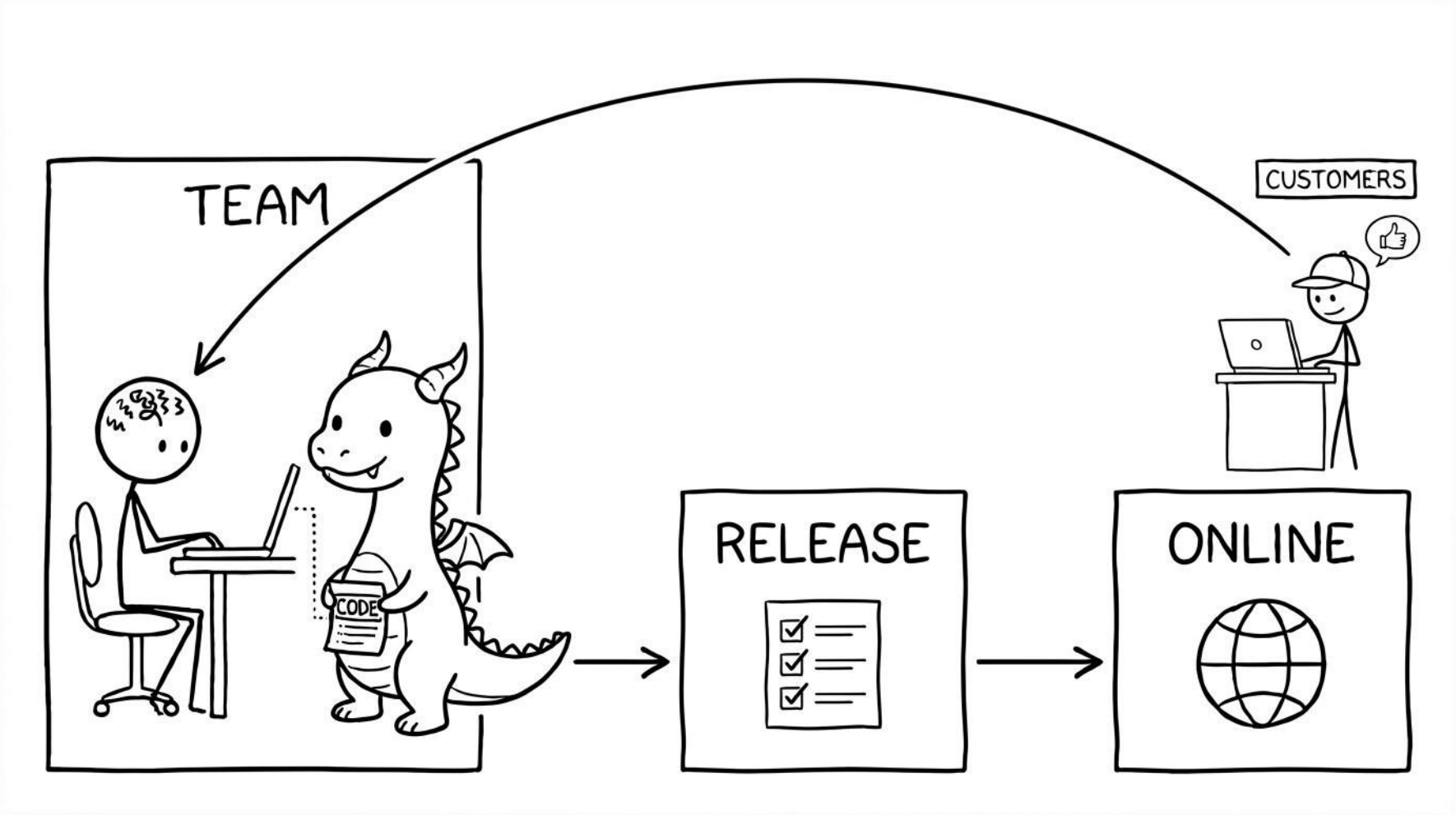

TEAM
CODE
ONLINE

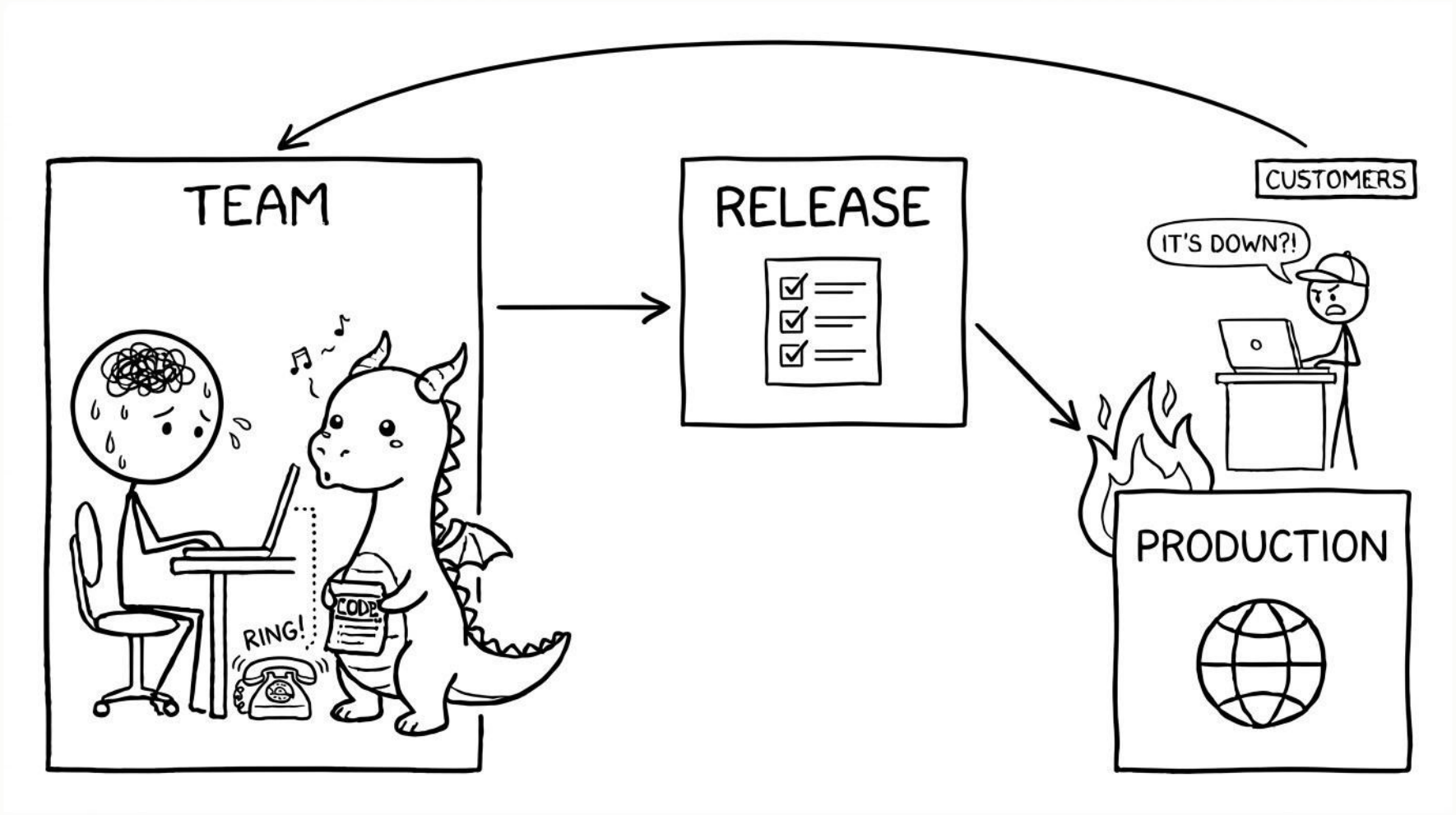

TEAM
CODE
CUSTOMERS
MEH.
ONLINE

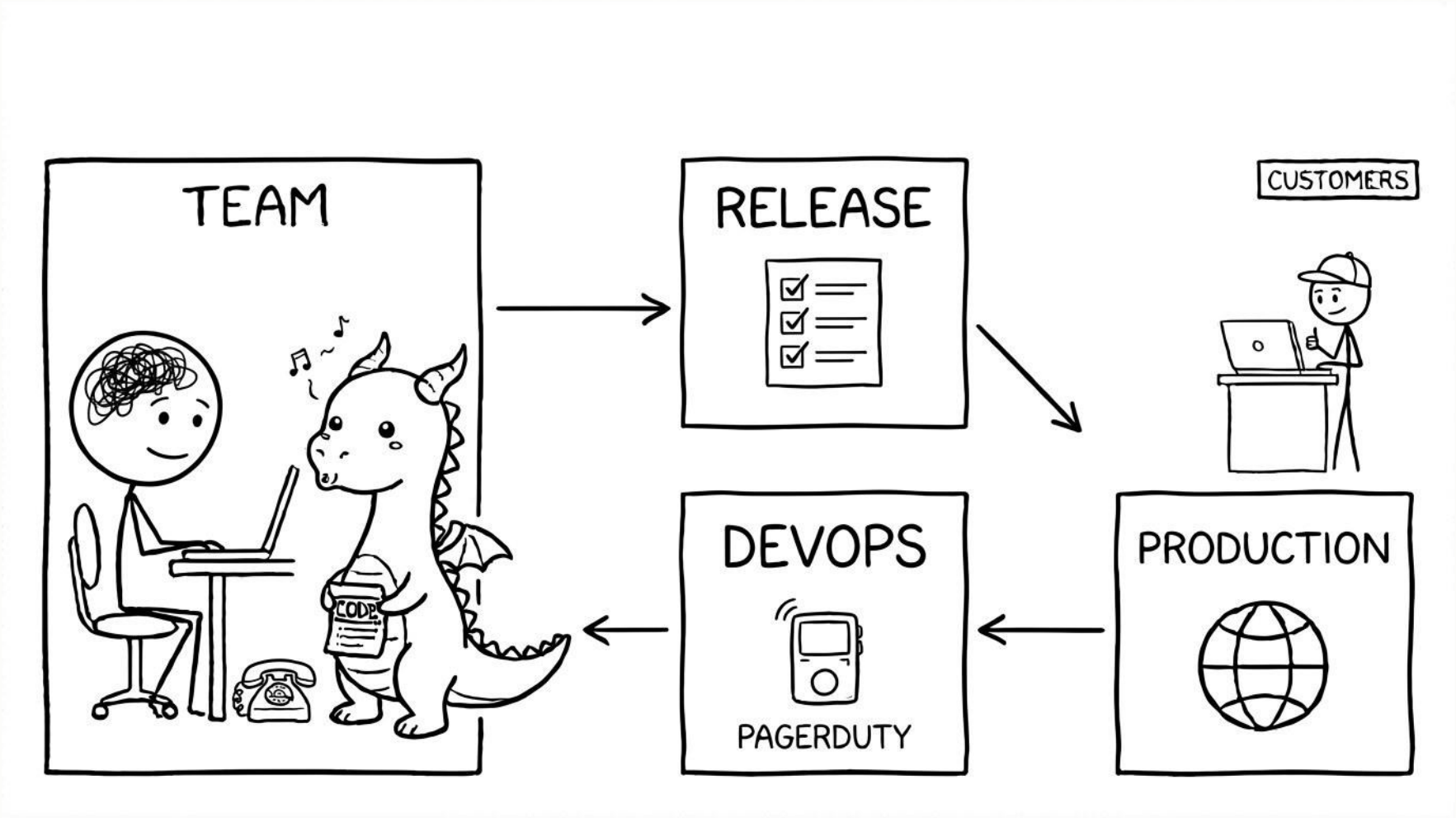

TEAM
CODE
CUSTOMERS
NAH.
ONLINE

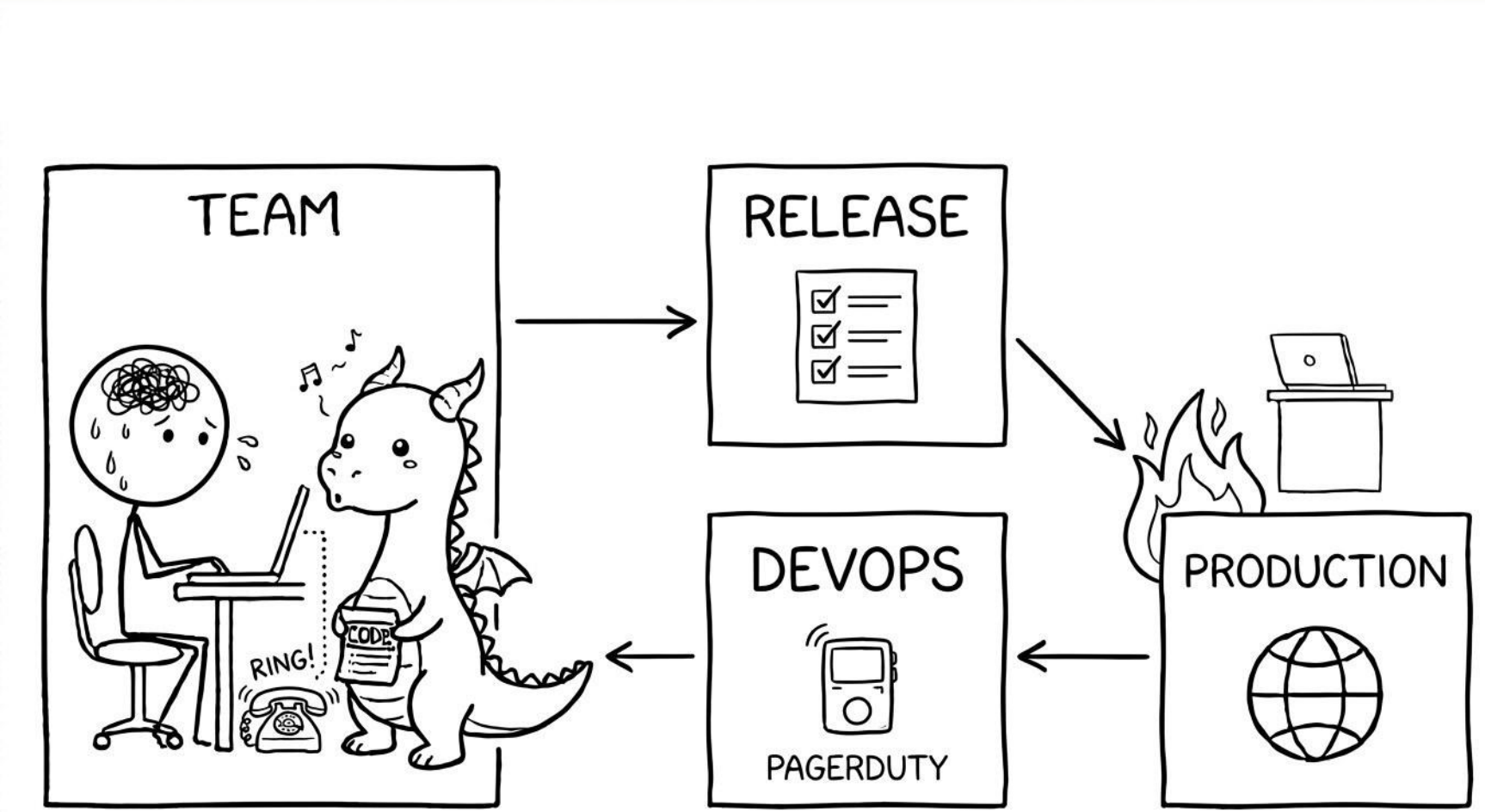

TEAM
CODE
CUSTOMERS
OH, NICE!
ONLINE

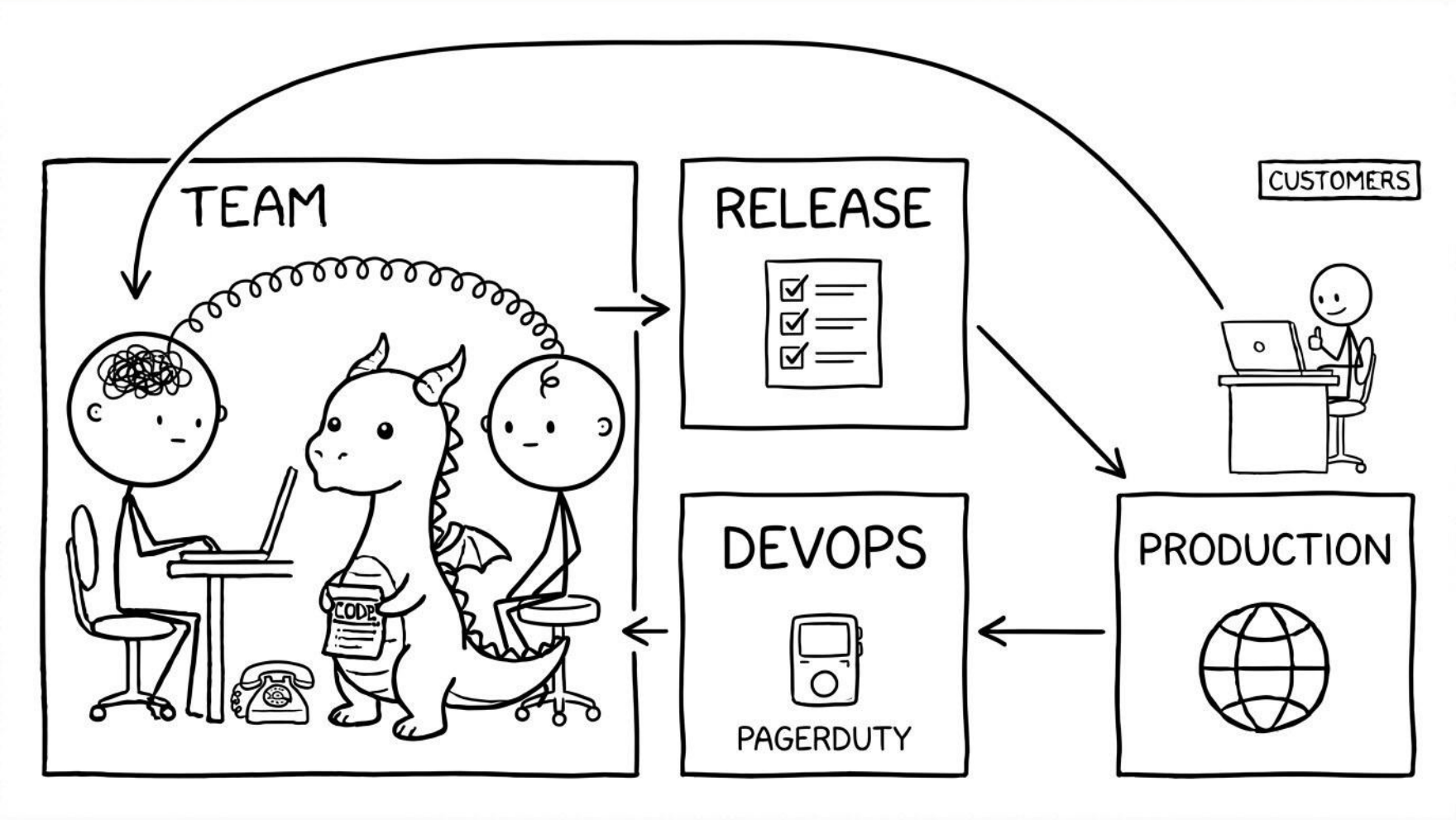

TEAM
CUSTOMERS
HEY!
IT'S BROKEN!
ONLINE

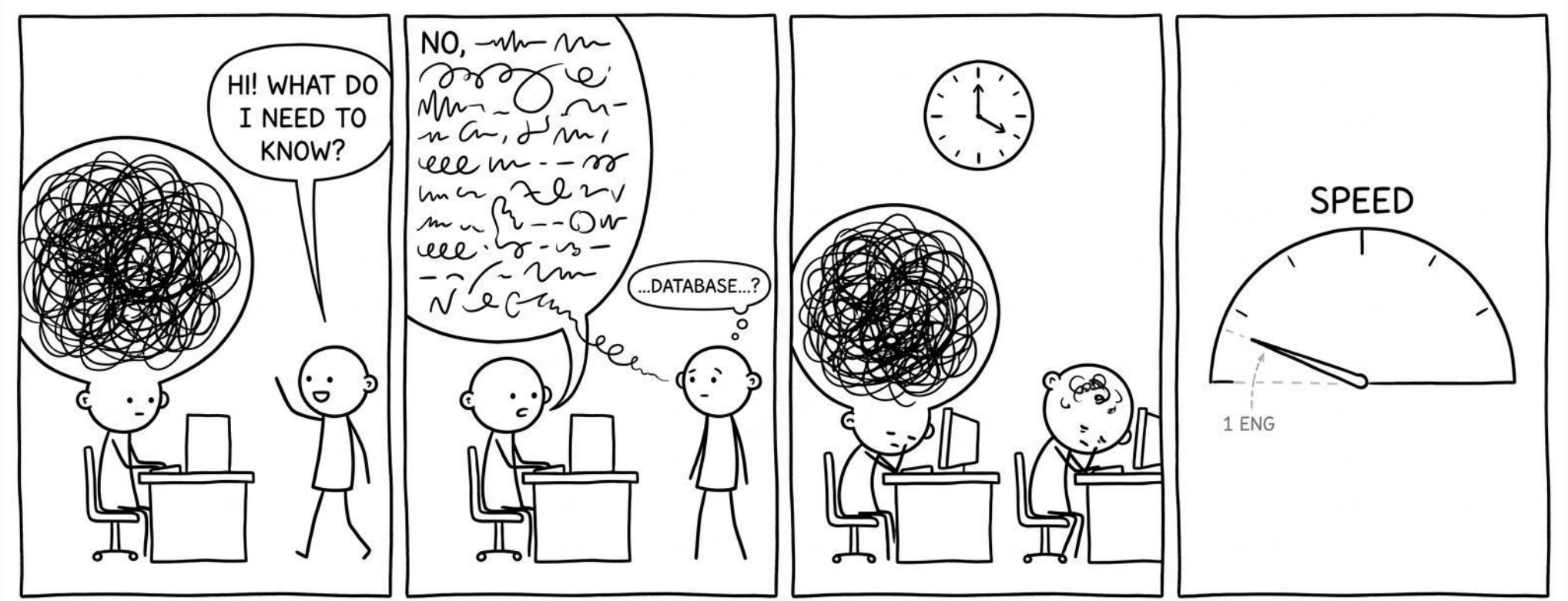

TEAM
CODE
RELEASE
ONLINE
CUSTOMERS


TEAM
RELEASE
CUSTOMERS
IT'S DOWN?!
PRODUCTION
RING!
CODE

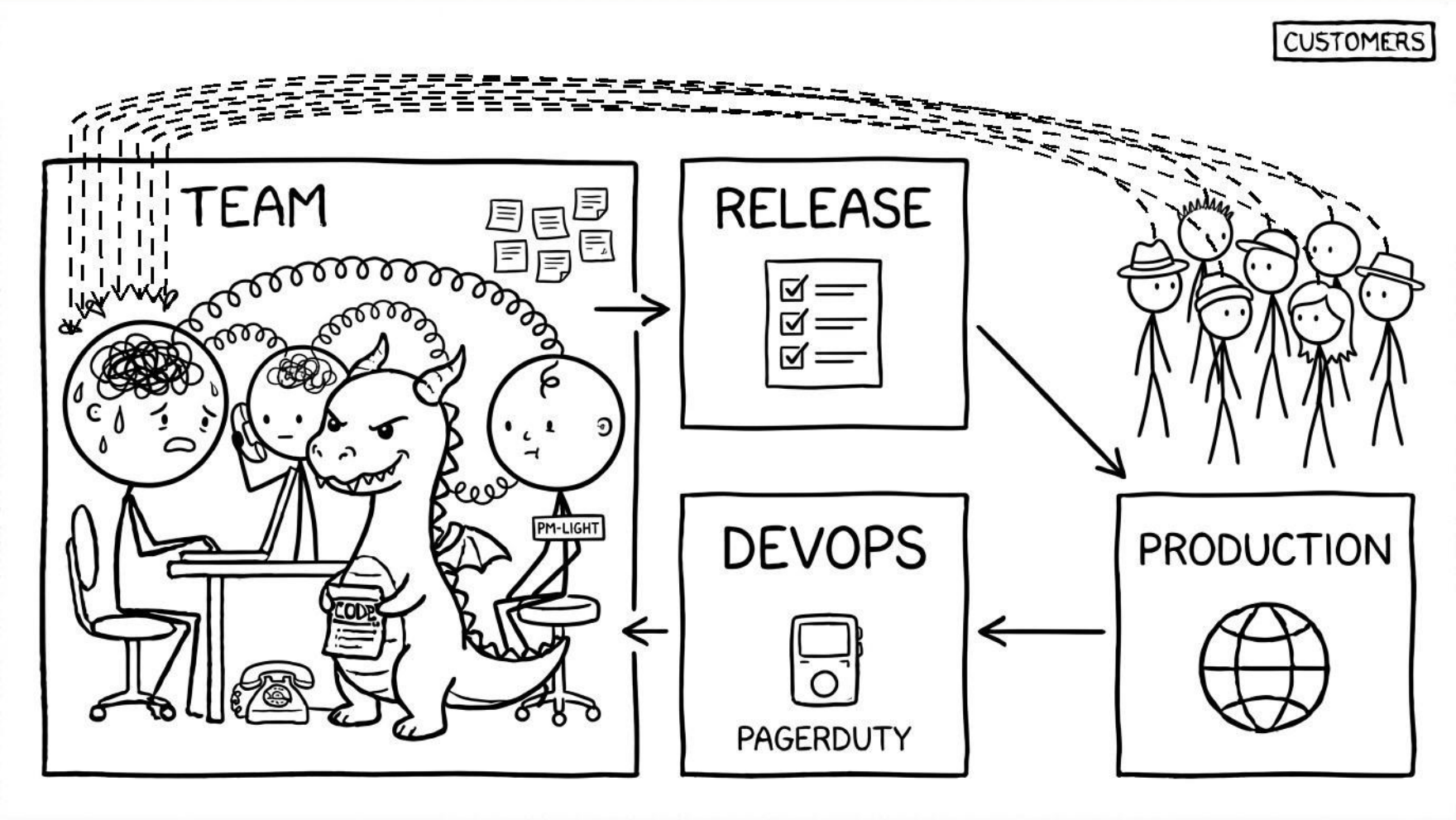

TEAM
CODE
RELEASE
CUSTOMERS
PRODUCTION
DEVOPS
PAGERDUTY

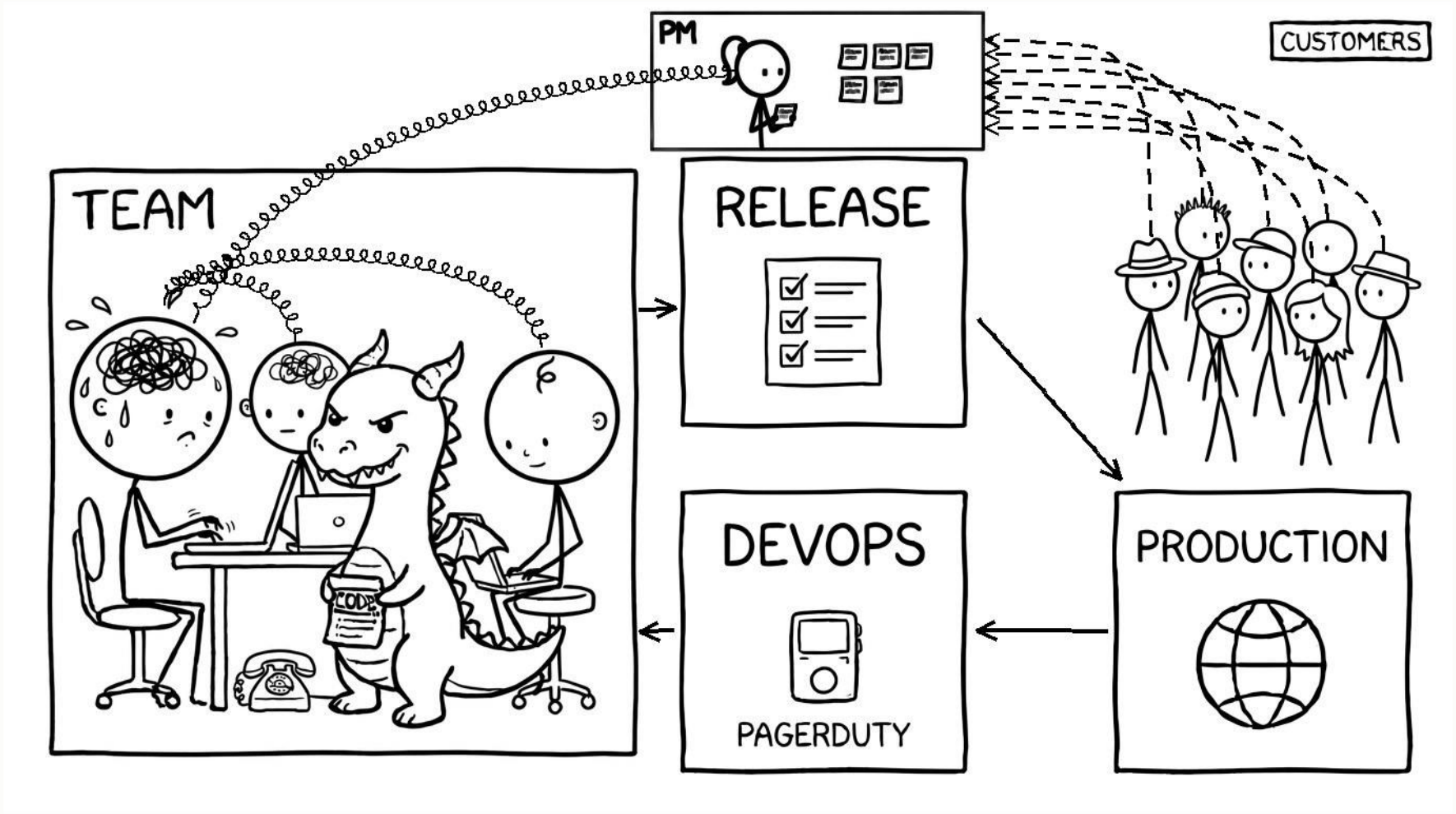

TEAM
RELEASE
PRODUCTION
DEVOPS
PAGERDUTY
RING!
CODE

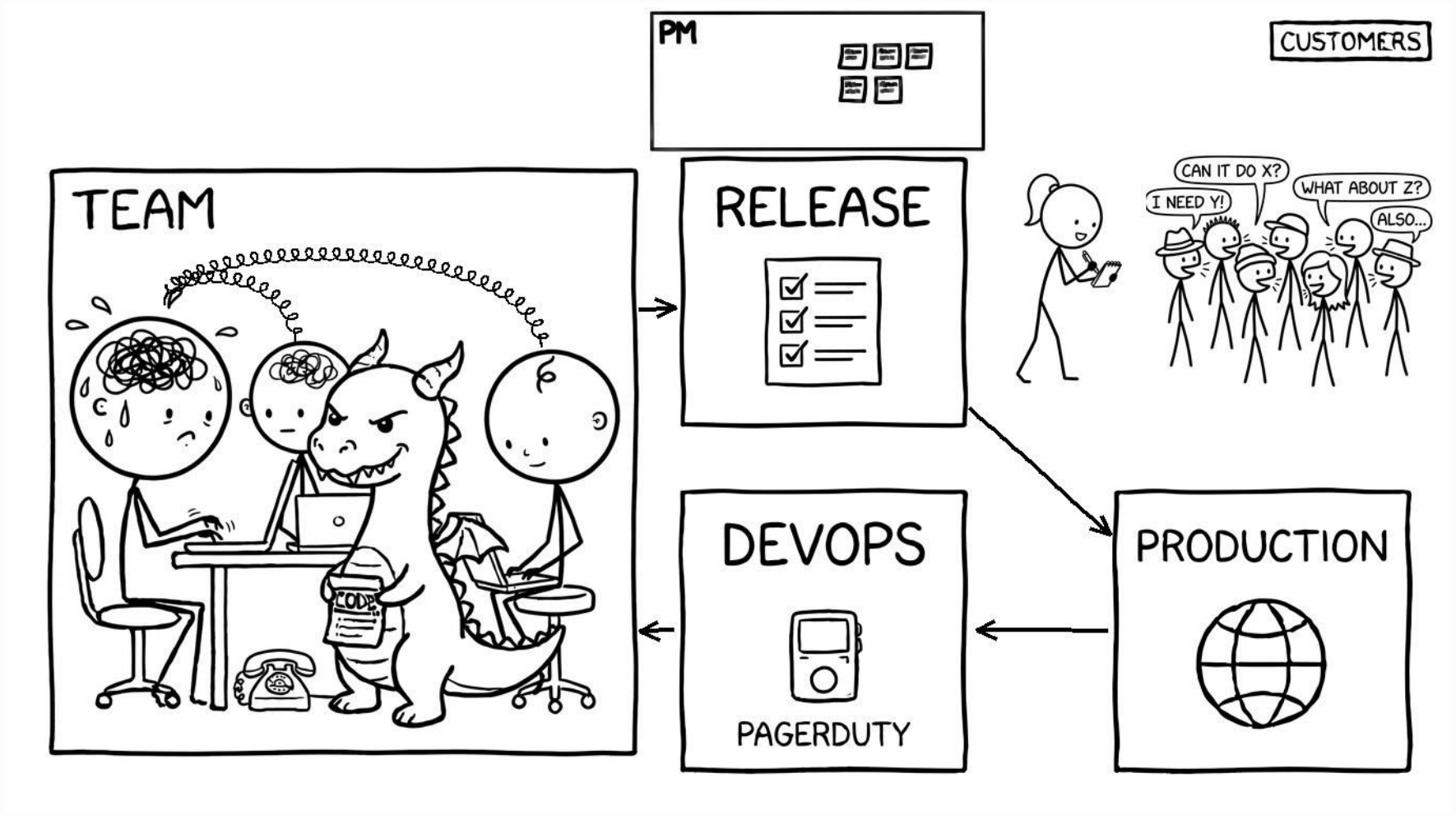

TEAM
CODE
RELEASE
CUSTOMERS
DEVOPS
PAGERDUTY
PRODUCTION

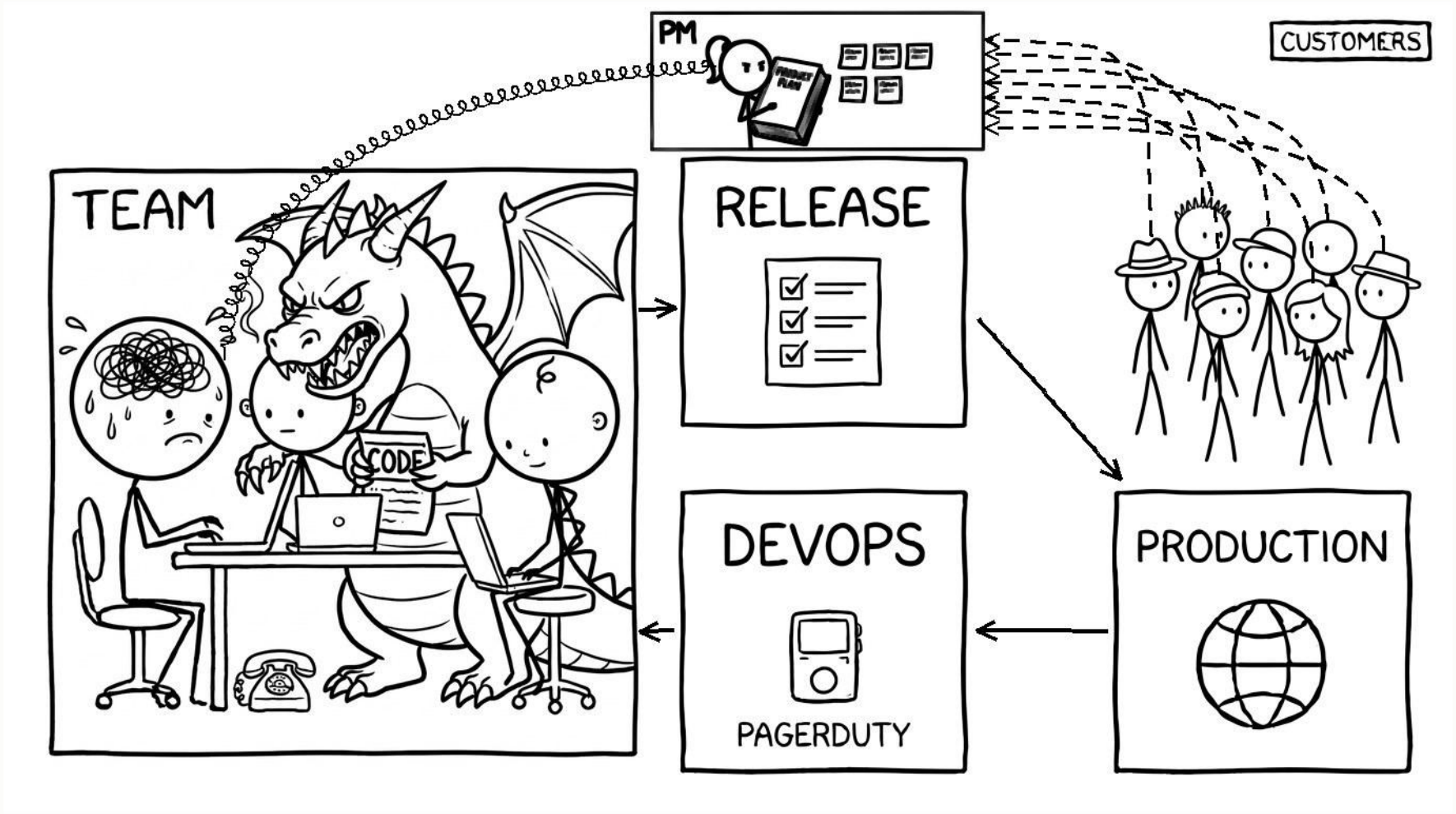


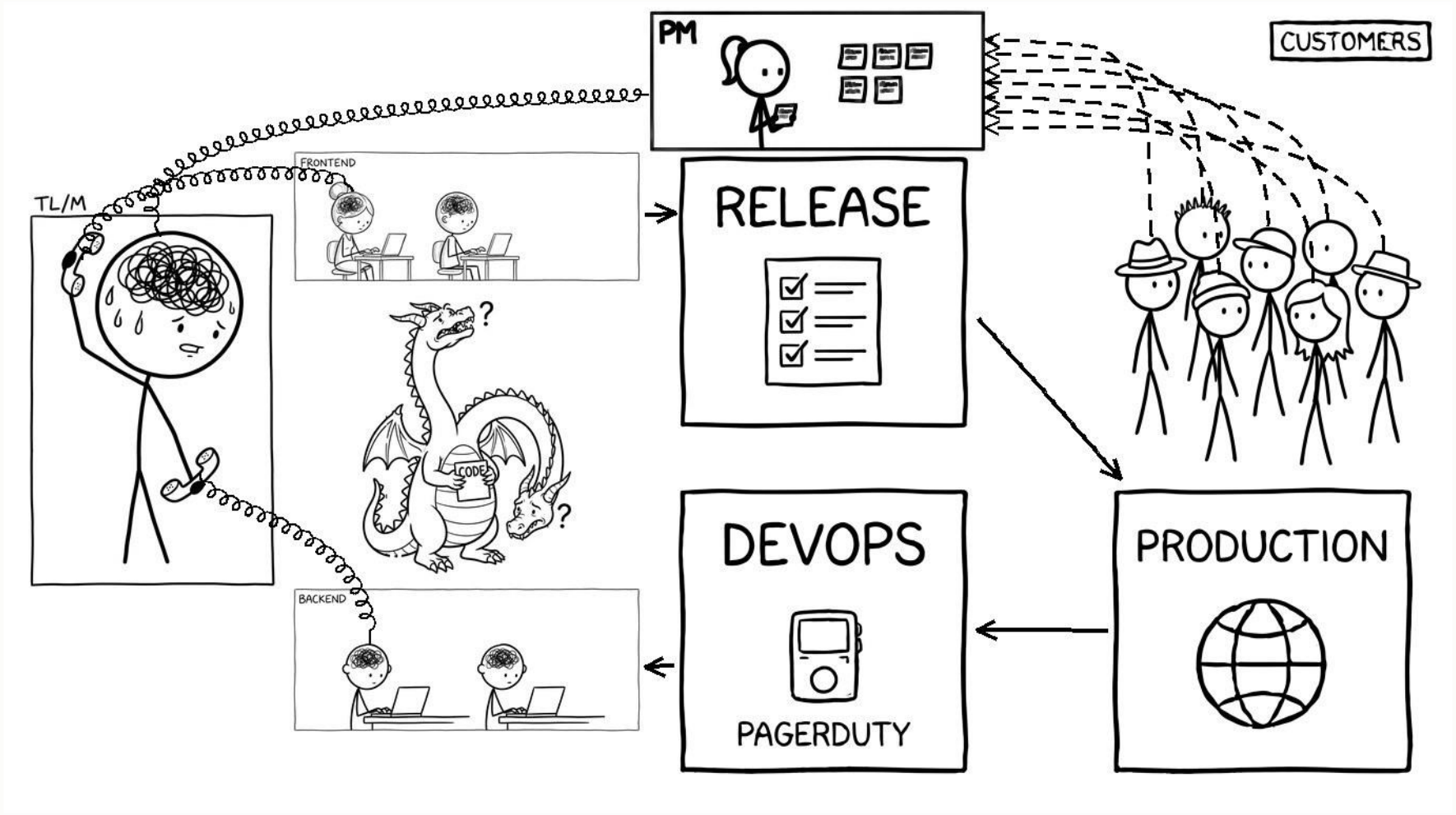
TWO ENGINEERS ≠ TWICE AS FAST.

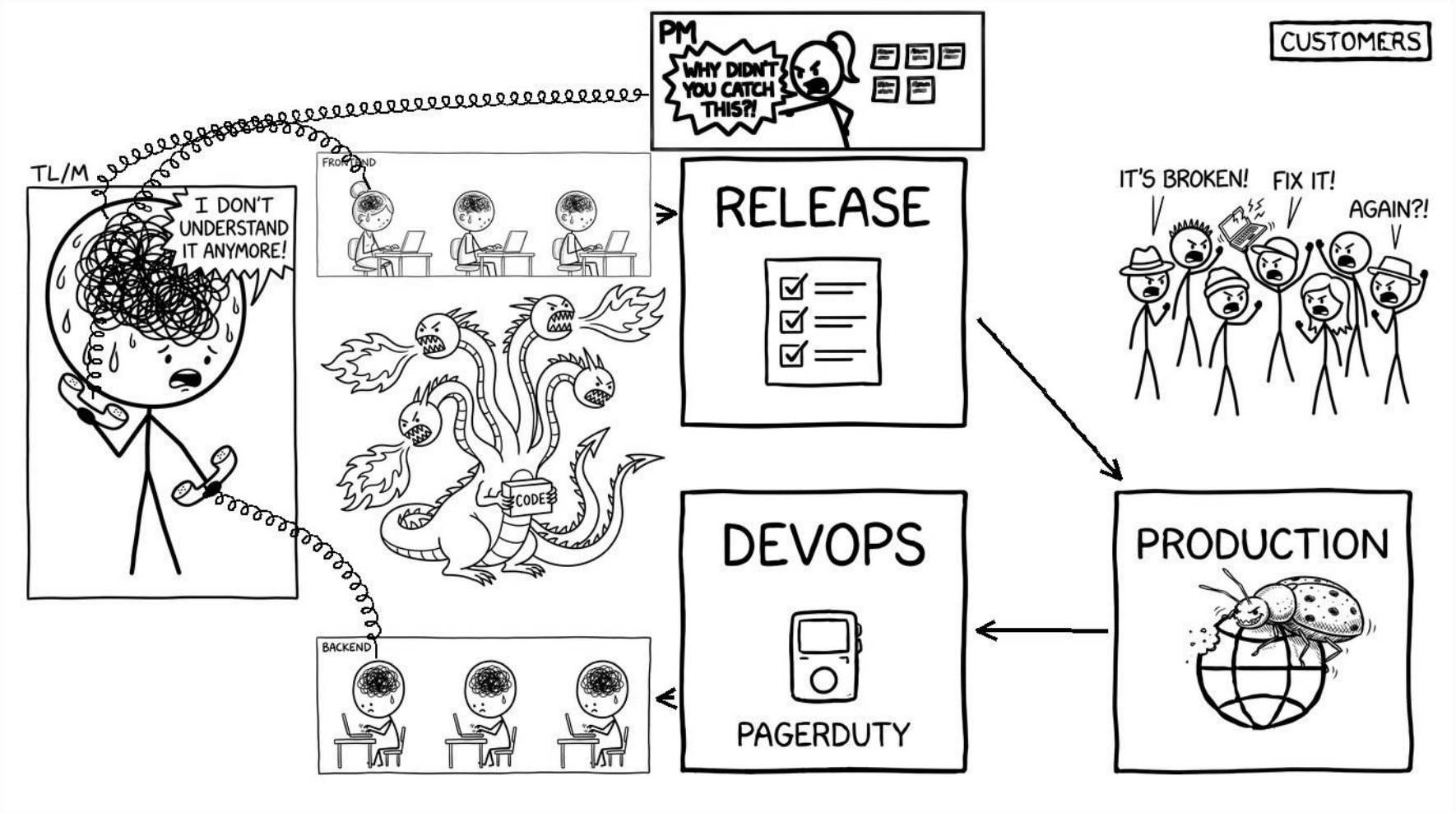

CUSTOMERS
TEAM
PM-LIGHT
RELEASE
DEVOPS
PAGERDUTY
PRODUCTION

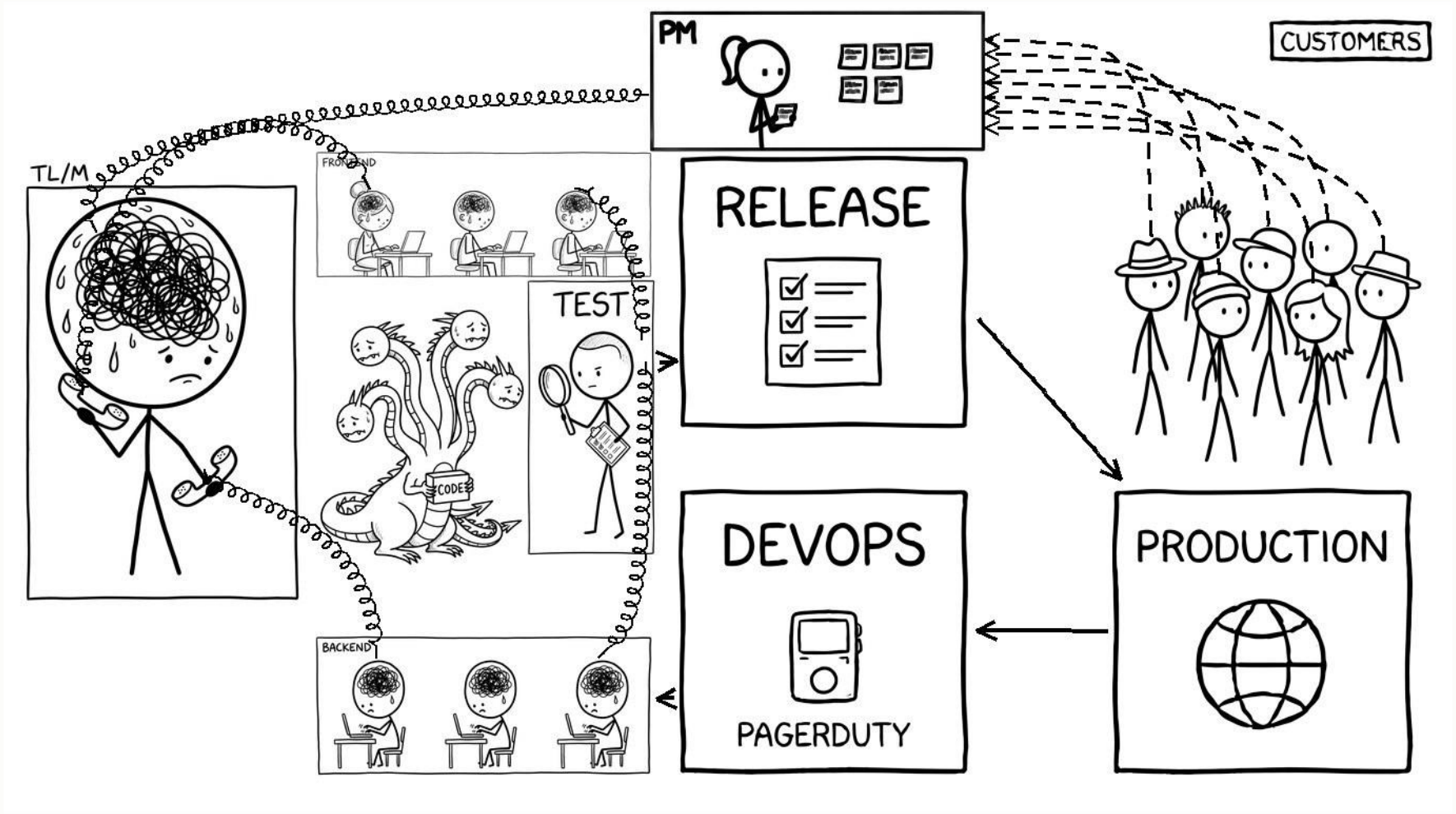

PM
CUSTOMERS
TEAM
RELEASE
DEVOPS
PAGERDUTY
PRODUCTION

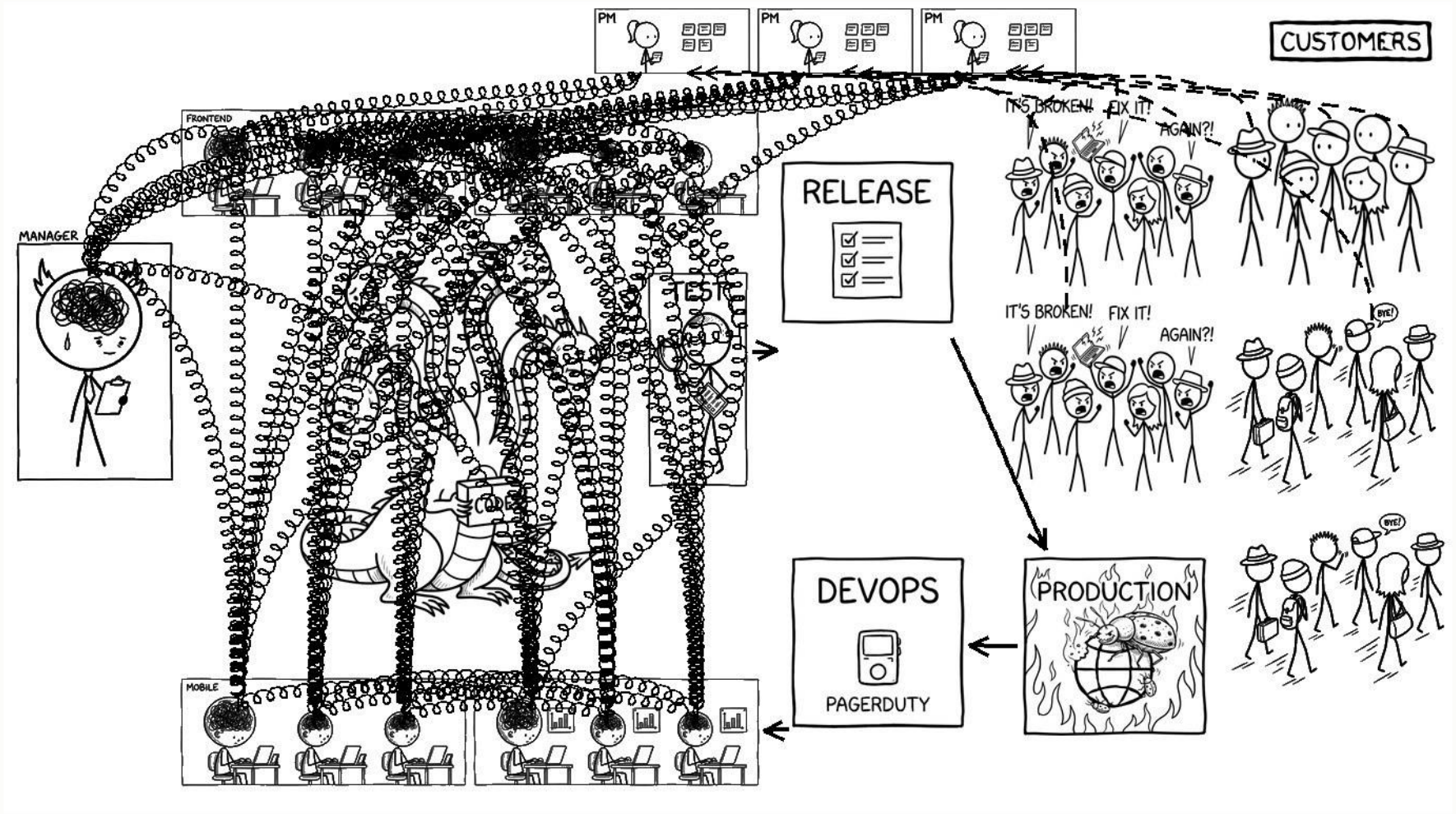

PM
CUSTOMERS
TEAM
RELEASE
CAN IT DO X?
I NEED Y!
WHAT ABOUT Z?
ALSO...
CODE
DEVOPS
PAGERDUTY
PRODUCTION

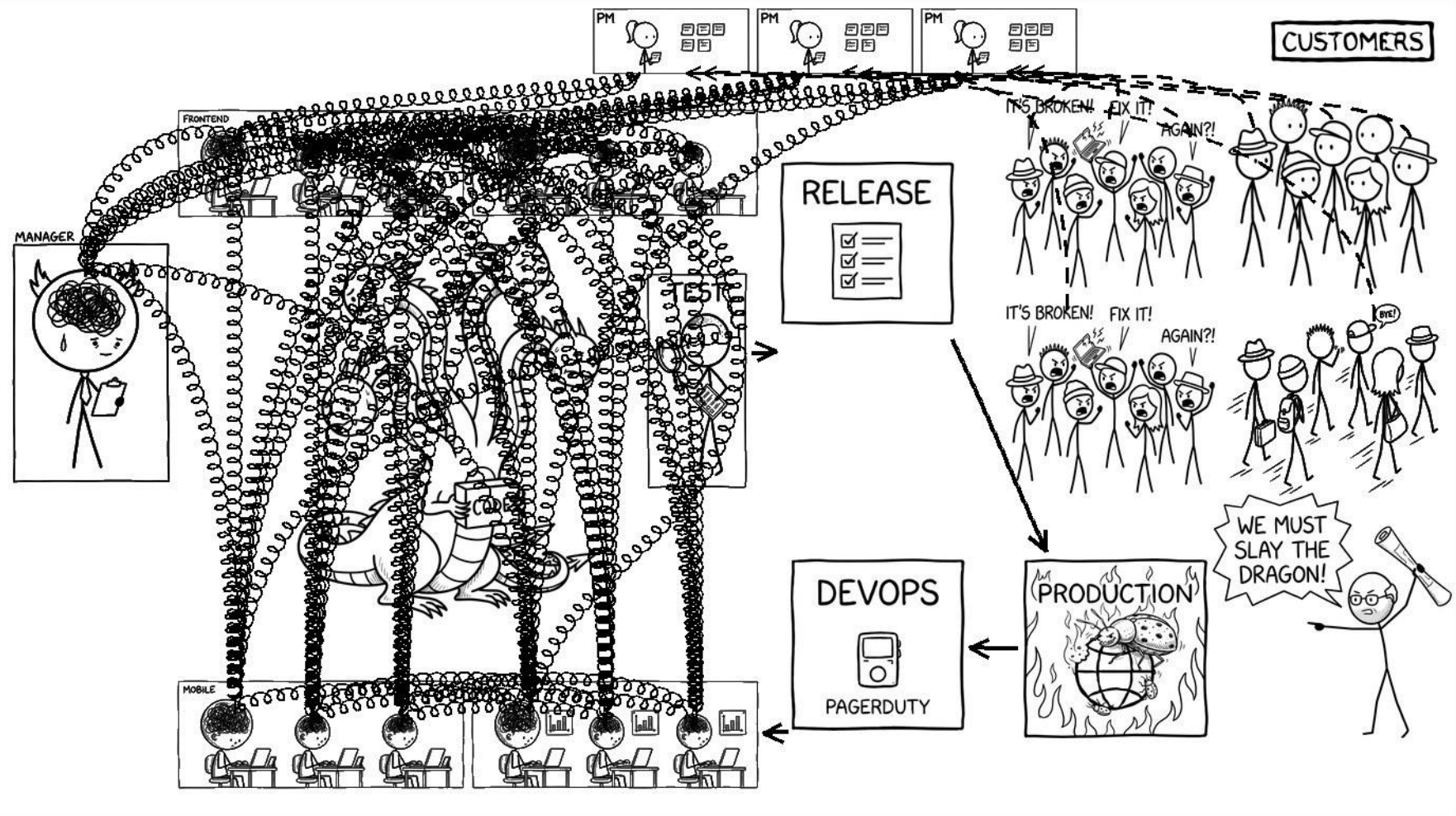

PM
PRODUCT PLAN
CUSTOMERS
TEAM
CODE
RELEASE
DEVOPS
PAGERDUTY
PRODUCTION

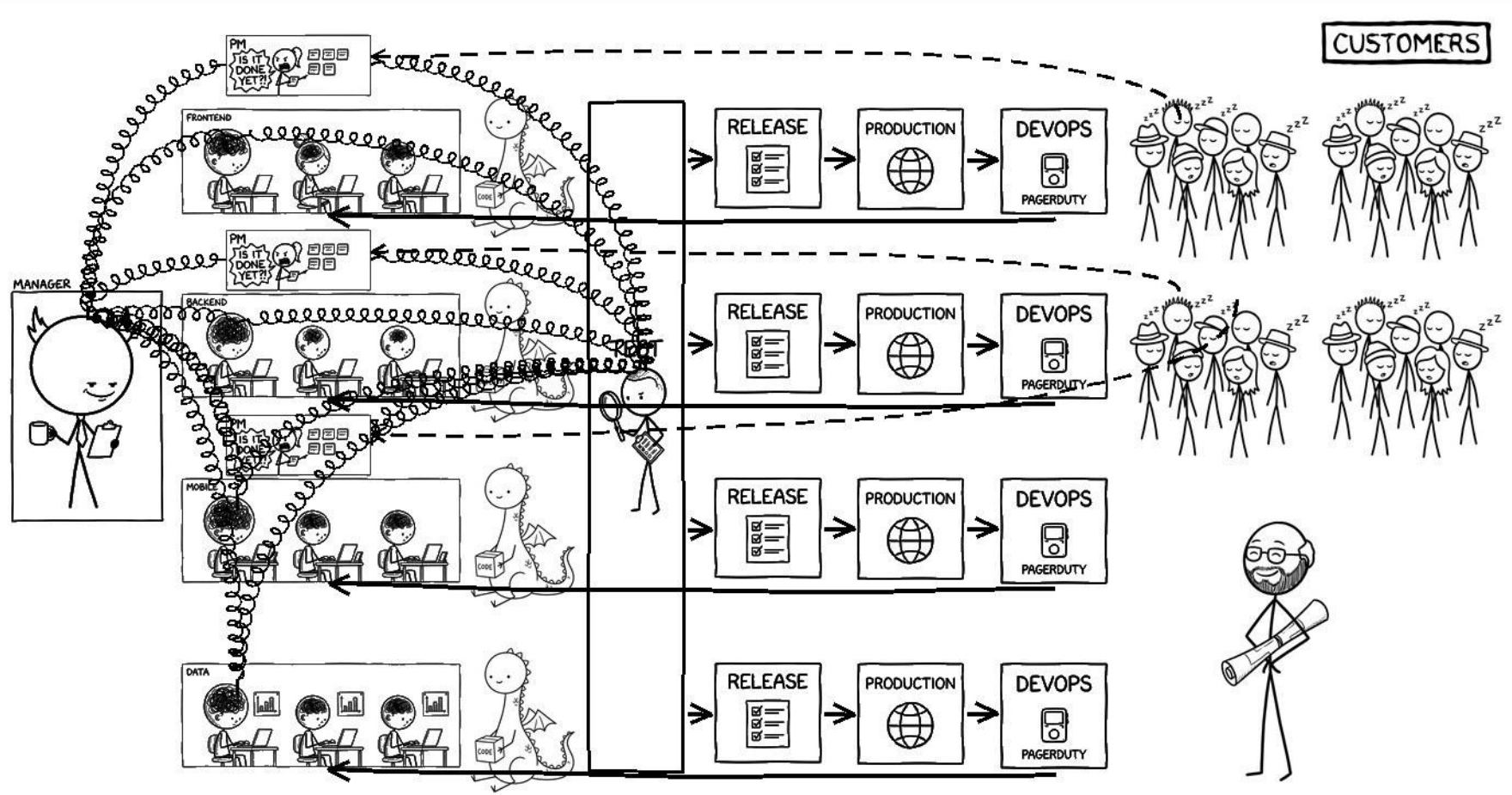

PM
CUSTOMERS
TL/M
FRONTEND
RELEASE
CODE
BACKEND
DEVOPS
PAGERDUTY
PRODUCTION

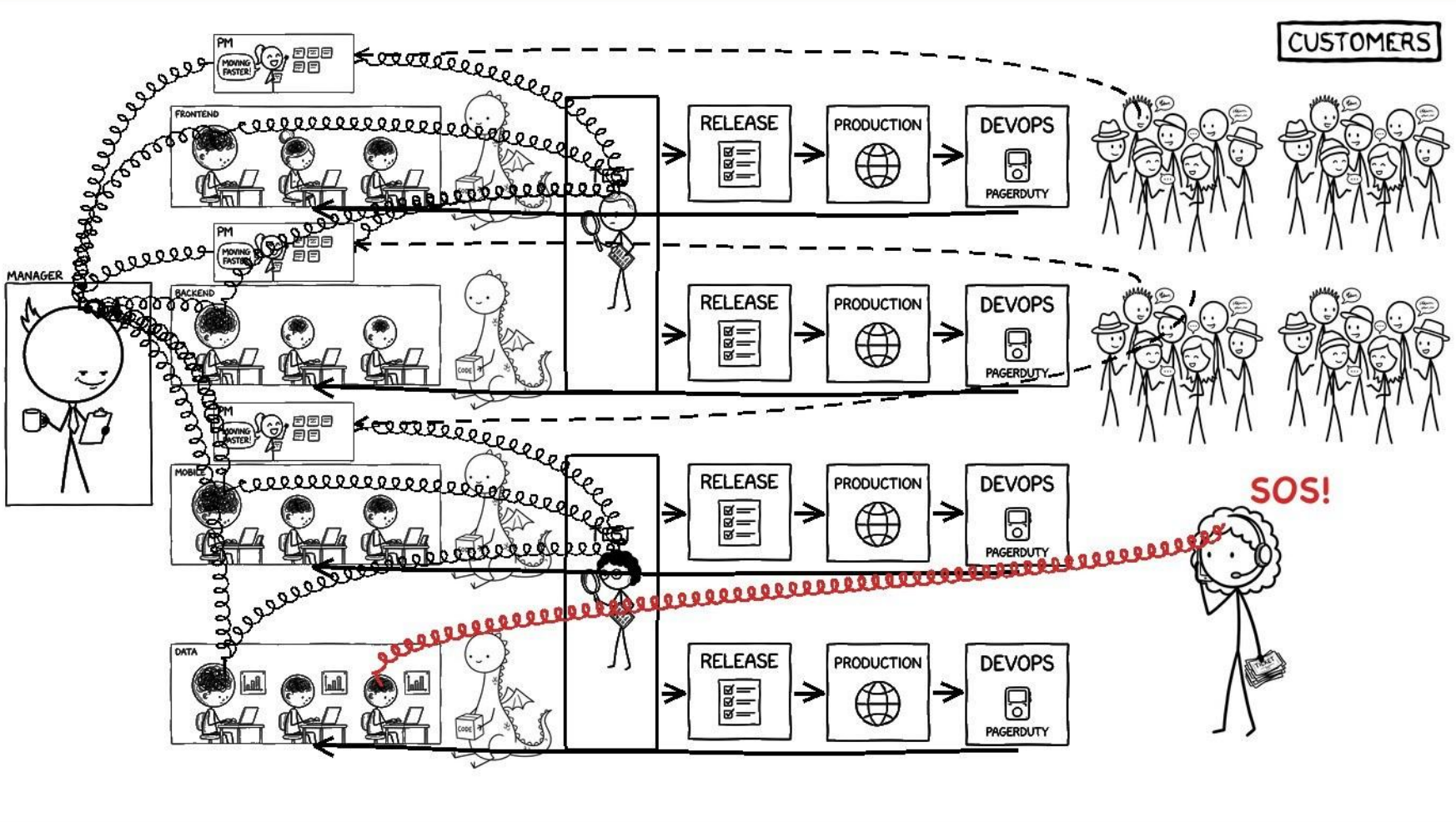

PM
WHY DIDN'T YOU CATCH THIS?!
CUSTOMERS
TL/M
I DON'T UNDERSTAND IT ANYMORE!
FRONTEND
RELEASE
IT'S BROKEN!
FIX IT!
AGAIN?!
CODE
DEVOPS
PAGERDUTY
PRODUCTION
BACKEND

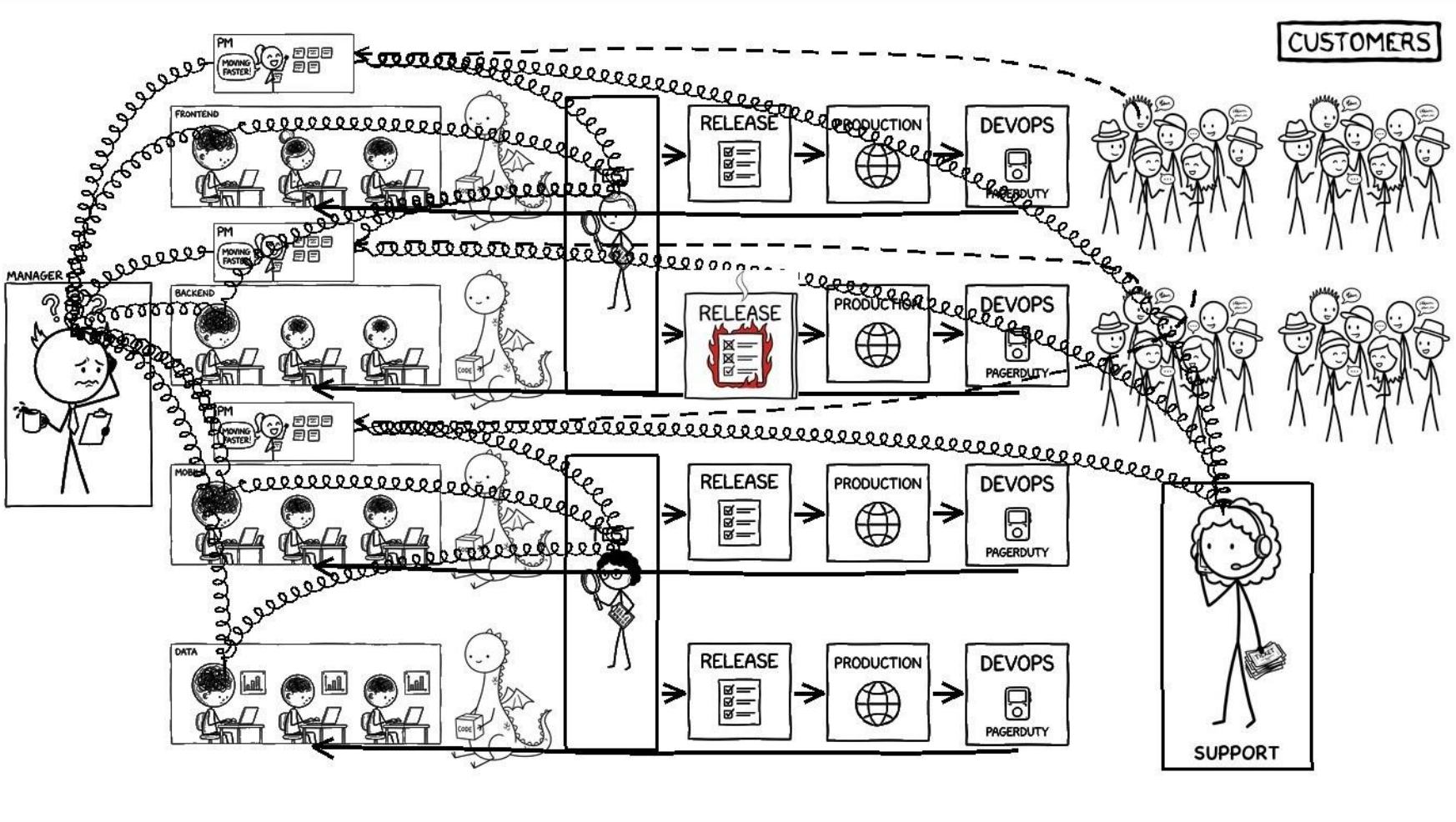

PM
CUSTOMERS
TL/M
FRONTEND
TEST
RELEASE
CODE
DEVOPS
PRODUCTION
BACKEND
PAGERDUTY

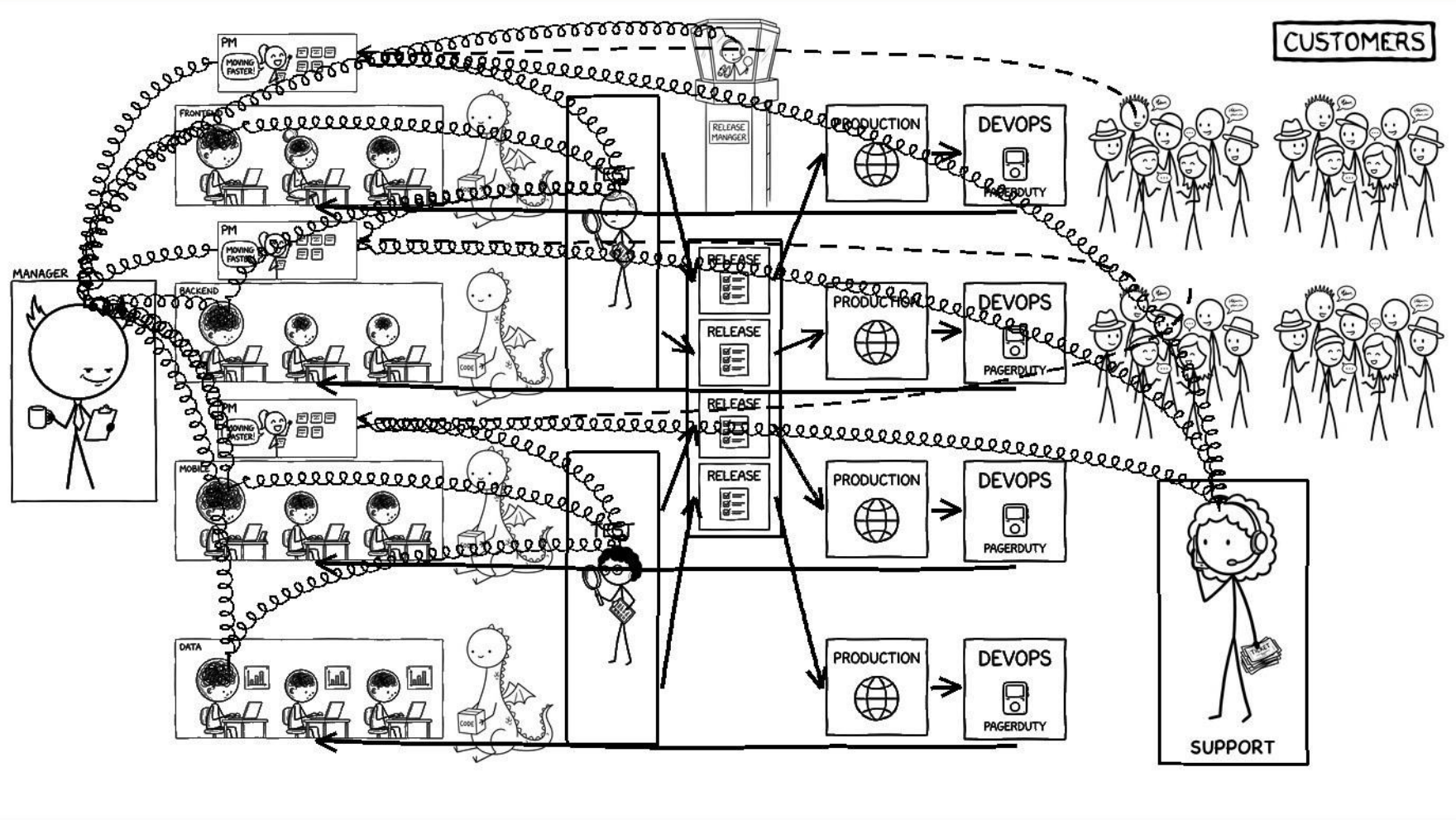

PM
PM
PM
CUSTOMERS
FRONTEND
MANAGER
TEST
RELEASE
IT'S BROKEN!
FIX IT!
AGAIN?!
IT'S BROKEN!
FIX IT!
AGAIN?!
BYE!
BYE!
DEVOPS
PAGERDUTY
PRODUCTION
MOBILE

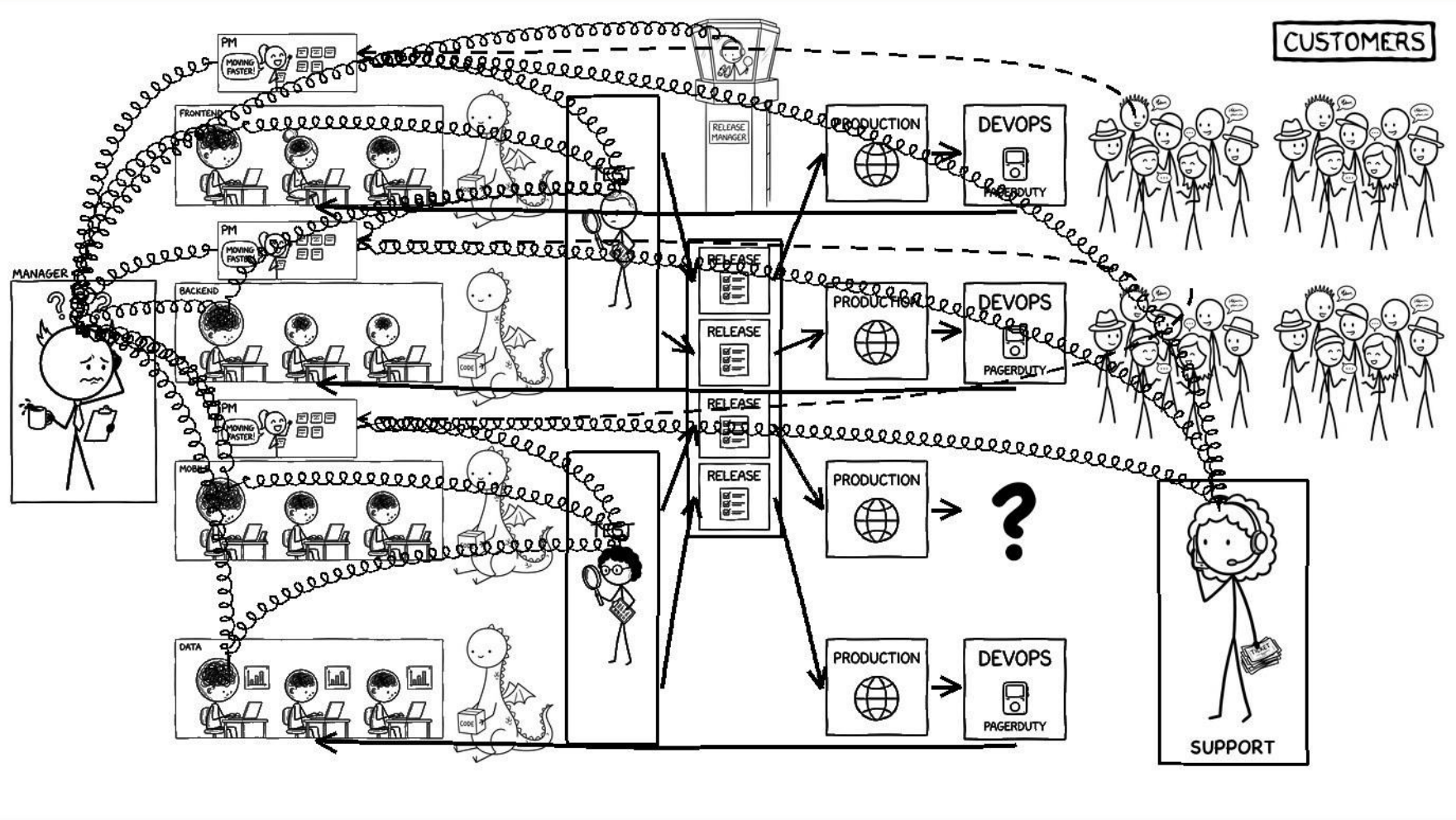

PM
PM
PM
CUSTOMERS
FRONTEND
MANAGER
TEST
RELEASE
IT'S BROKEN!
FIX IT!
AGAIN?!
IT'S BROKEN!
FIX IT!
AGAIN?!
BYE!
WE MUST SLAY THE DRAGON!
DEVOPS
PAGERDUTY
PRODUCTION
MOBILE

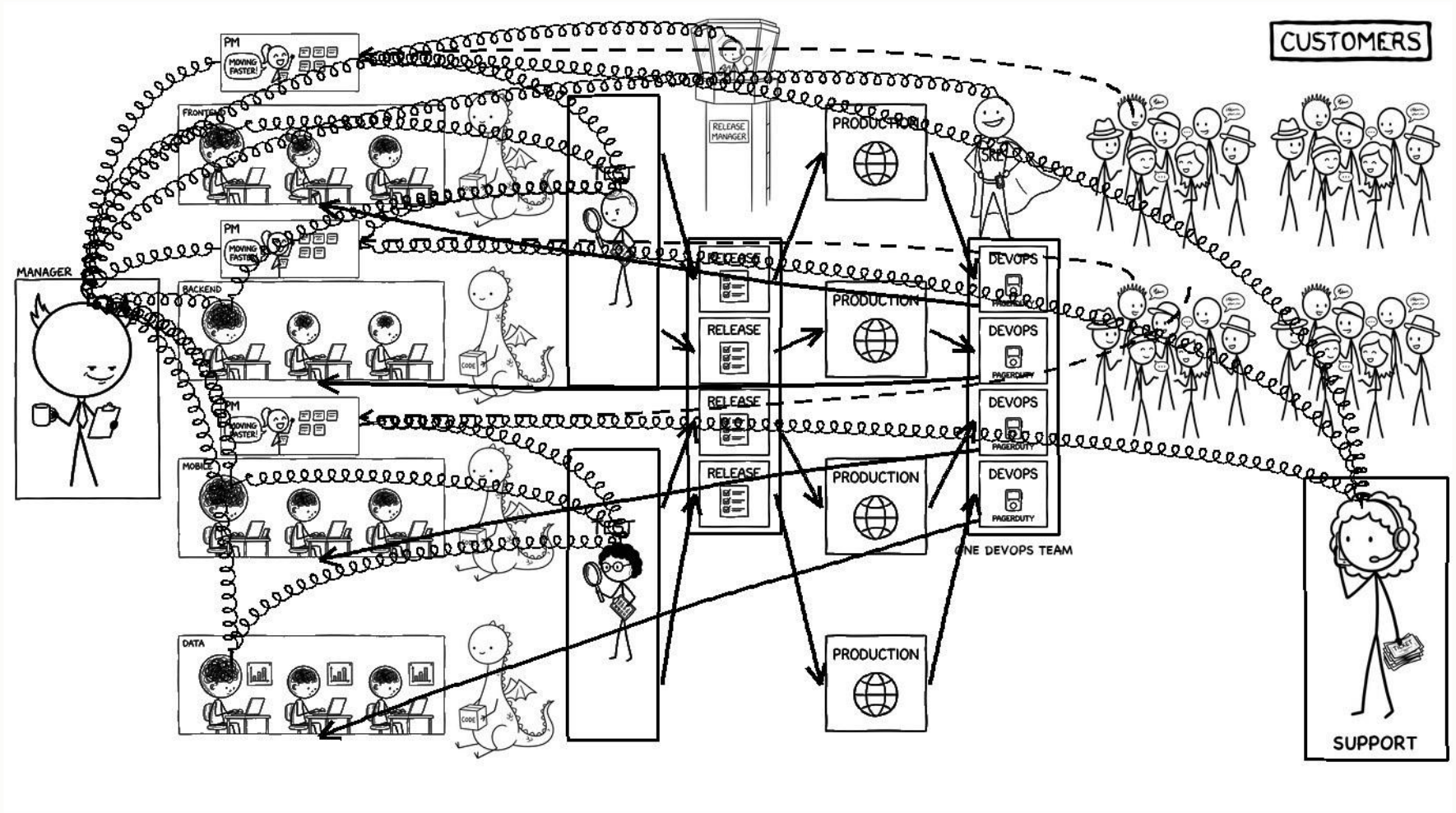

CUSTOMERS
PM
IS IT DONE YET?!
FRONTEND
BACKEND
MOBILE
DATA
MANAGER
CODE
RELEASE
PRODUCTION
DEVOPS
PAGERDUTY

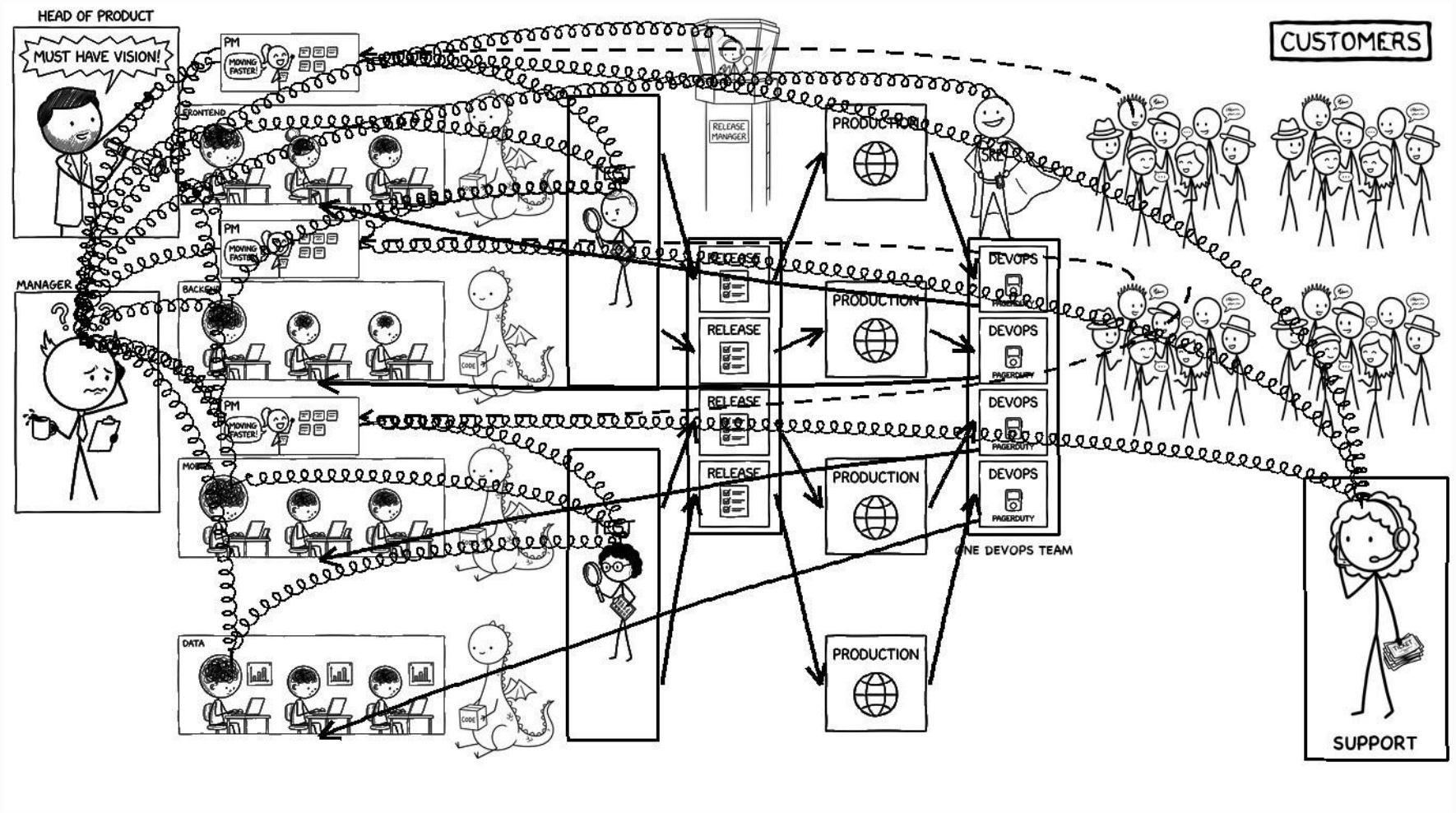

CUSTOMERS
PM
MOVING FASTER!
FRONTEND
PM
MOVING FASTER!
MANAGER
BACKEND
CODE
PM
MOVING FASTER!
MOBILE
DATA
CODE
RELEASE
PRODUCTION
DEVOPS
PAGERDUTY
RELEASE
PRODUCTION
DEVOPS
PAGERDUTY
RELEASE
PRODUCTION
DEVOPS
PAGERDUTY
RELEASE
PRODUCTION
DEVOPS
PAGERDUTY
SOS!


CUSTOMERS
PM
MOVING FASTER!
FRONTEND
RELEASE
PRODUCTION
DEVOPS
PAGERDUTY
MANAGER
BACKEND
CODE
MOBILE
DATA
SUPPORT
TICKET

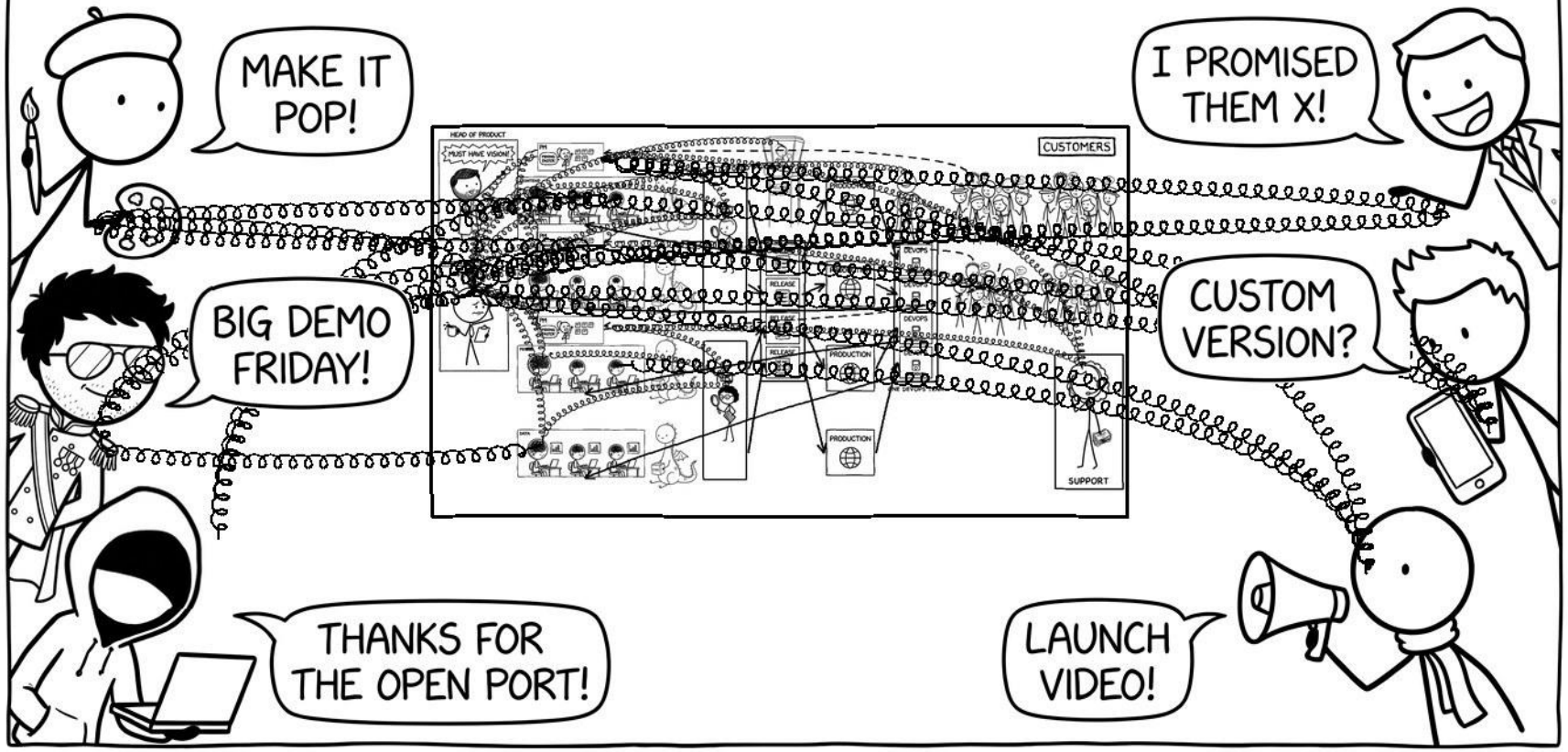

CUSTOMERS
PM
MOVING FASTER!
FRONTEND
PM
MOVING FASTER!
MANAGER
BACKEND
PM
MOVING FASTER!
MOBILE
DATA
RELEASE MANAGER
RELEASE
RELEASE
RELEASE
RELEASE
PRODUCTION
PRODUCTION
PRODUCTION
PRODUCTION
DEVOPS
PAGERDUTY
DEVOPS
PAGERDUTY
DEVOPS
PAGERDUTY
DEVOPS
PAGERDUTY
SUPPORT


CUSTOMERS
PM
MOVING FASTER!
FRONTEND
PM
MOVING FASTER!
MANAGER
BACKEND
PM
MOVING FASTER!
MOBILE
DATA
RELEASE MANAGER
RELEASE
RELEASE
RELEASE
RELEASE
PRODUCTION
DEVOPS
PAGERDUTY
PRODUCTION
DEVOPS
PAGERDUTY
PRODUCTION
?
PRODUCTION
DEVOPS
PAGERDUTY
SUPPORT


CUSTOMERS
PM
MOVING FASTER!
FRONTEND
PM
MOVING FASTER!
MANAGER
BACKEND
PM
MOVING FASTER!
MOBILE
DATA
RELEASE MANAGER
RELEASE
RELEASE
RELEASE
RELEASE
PRODUCTION
PRODUCTION
PRODUCTION
PRODUCTION
SRE
DEVOPS
PAGERDUTY
DEVOPS
PAGERDUTY
DEVOPS
PAGERDUTY
DEVOPS
PAGERDUTY
ONE DEVOPS TEAM
SUPPORT


HEAD OF PRODUCT
MUST HAVE VISION!
PM
MOVING FASTER!
FRONTEND
BACKEND
MOBILE
DATA
MANAGER
RELEASE MANAGER
RELEASE
PRODUCTION
DEVOPS
PAGERDUTY
ONE DEVOPS TEAM
CUSTOMERS
SUPPORT

OH, AND DON'T FORGET...

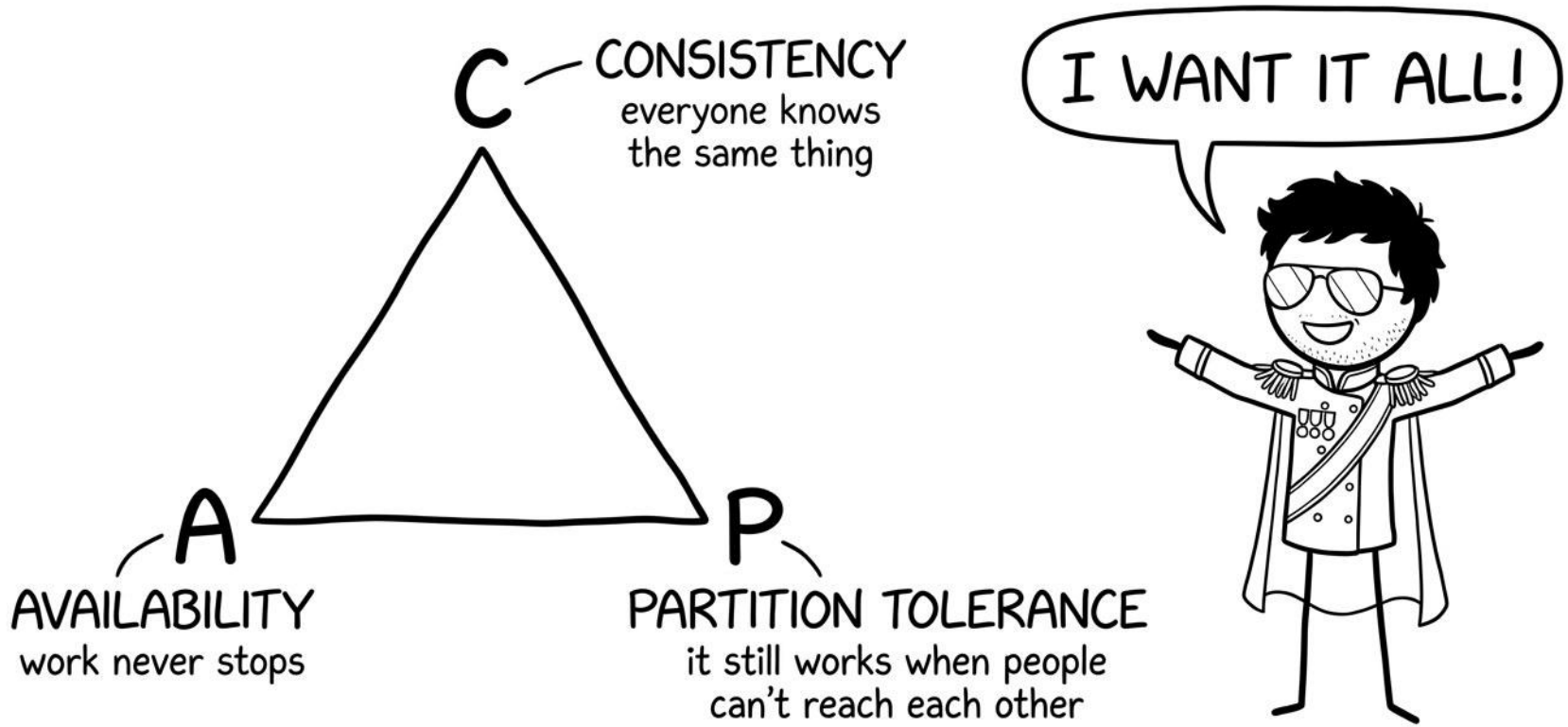


PART 2

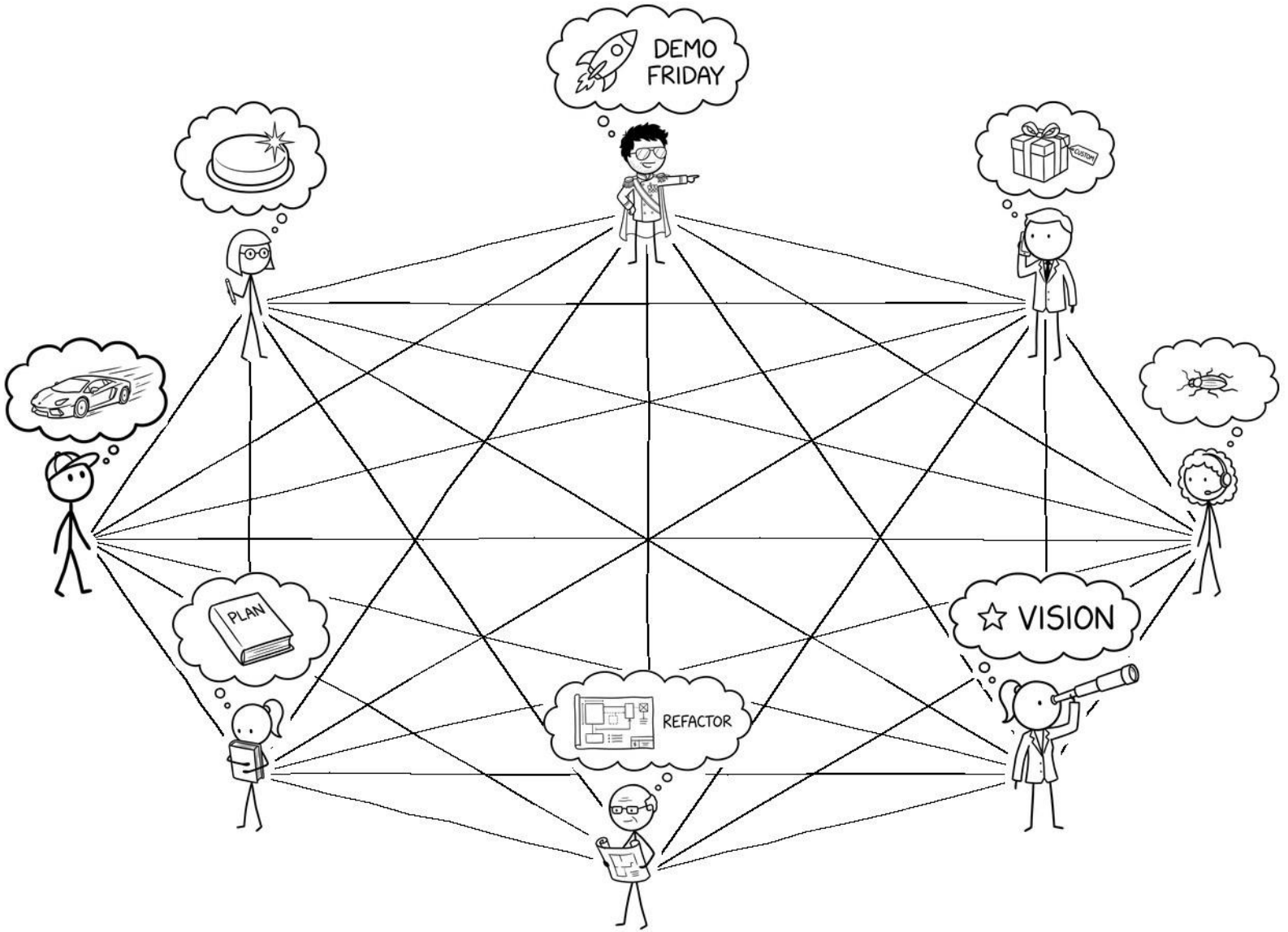
Why It's So Hard

Everyone's signal is different, and that can't be avoided

THE CAP THEOREM



CONSISTENCY: everyone knows the same thing



Every line is a conversation that has to happen.

2 people = 1 line

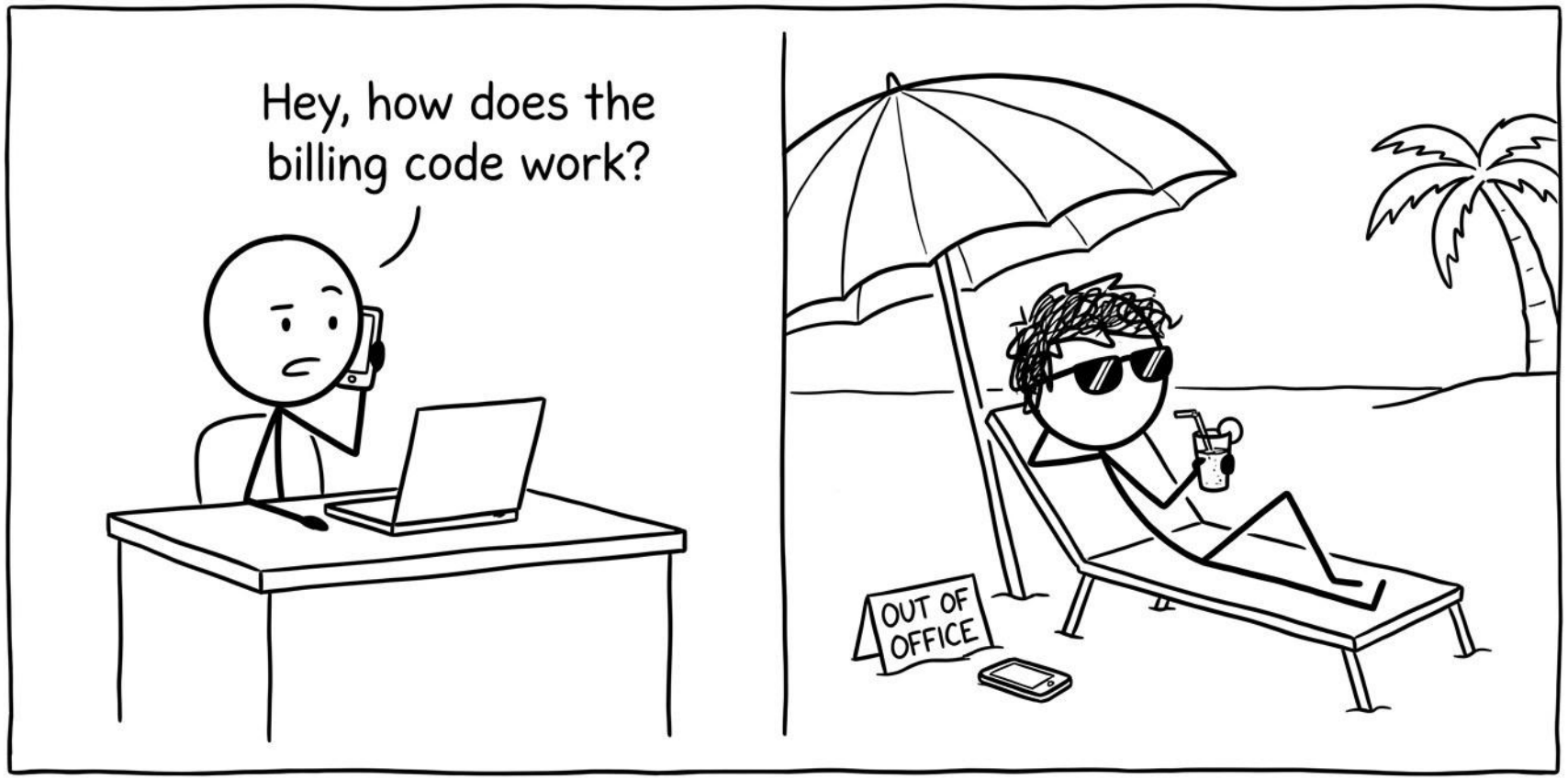
4 people = 6 lines

8 people = 28 lines

50 people = 1,225 lines

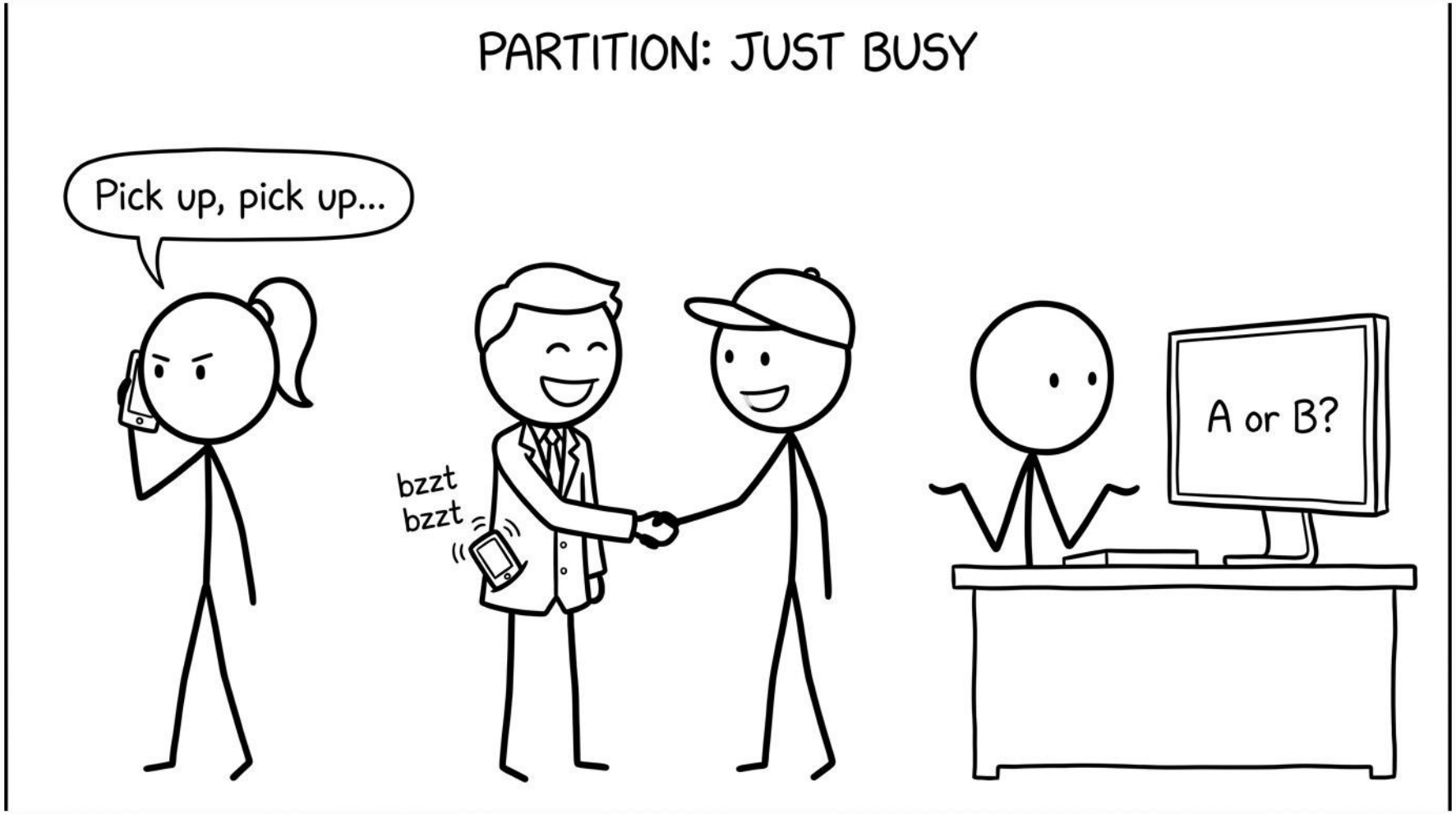
...and they all want something different.

PARTITION: ON VACATION

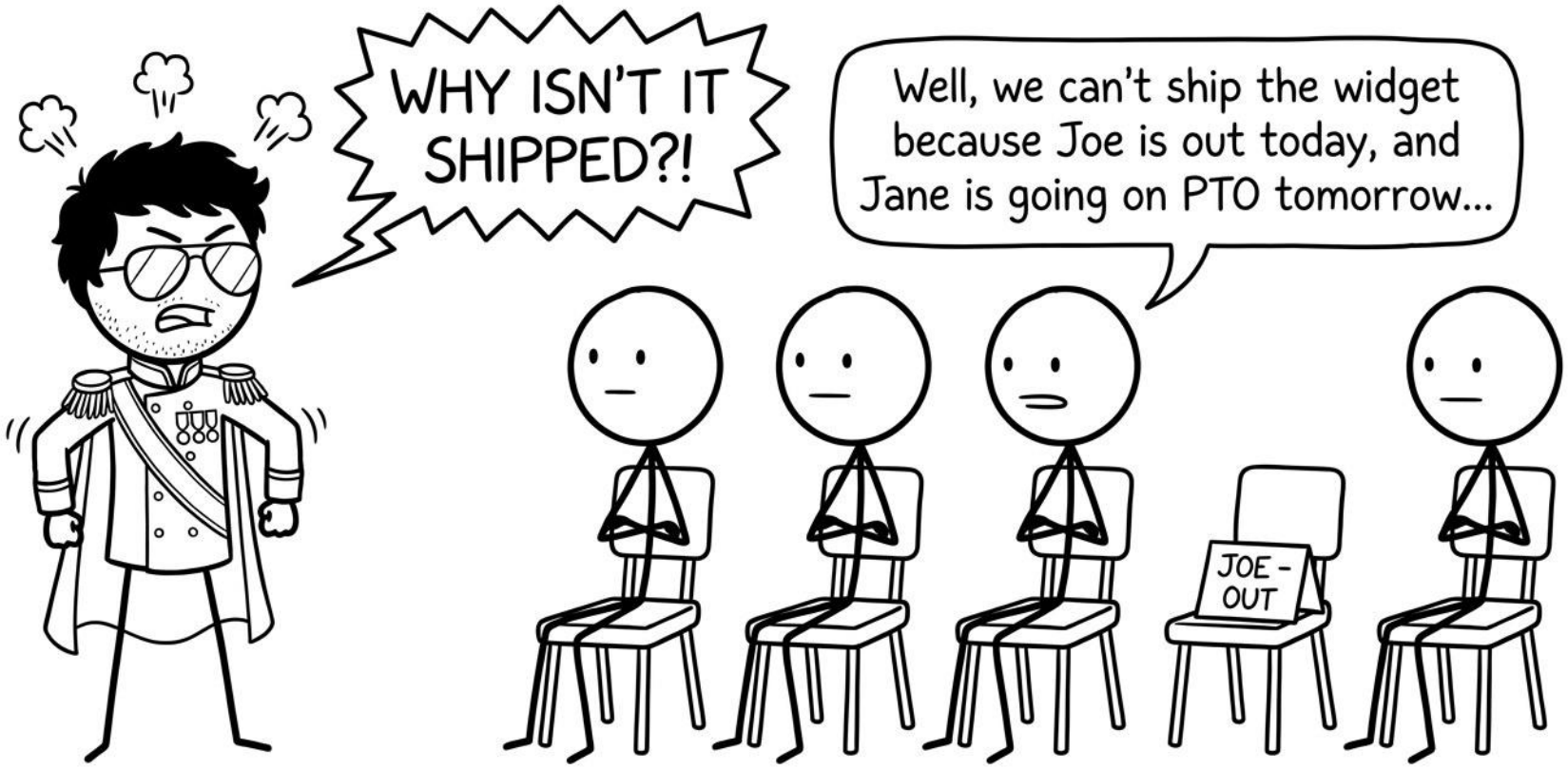


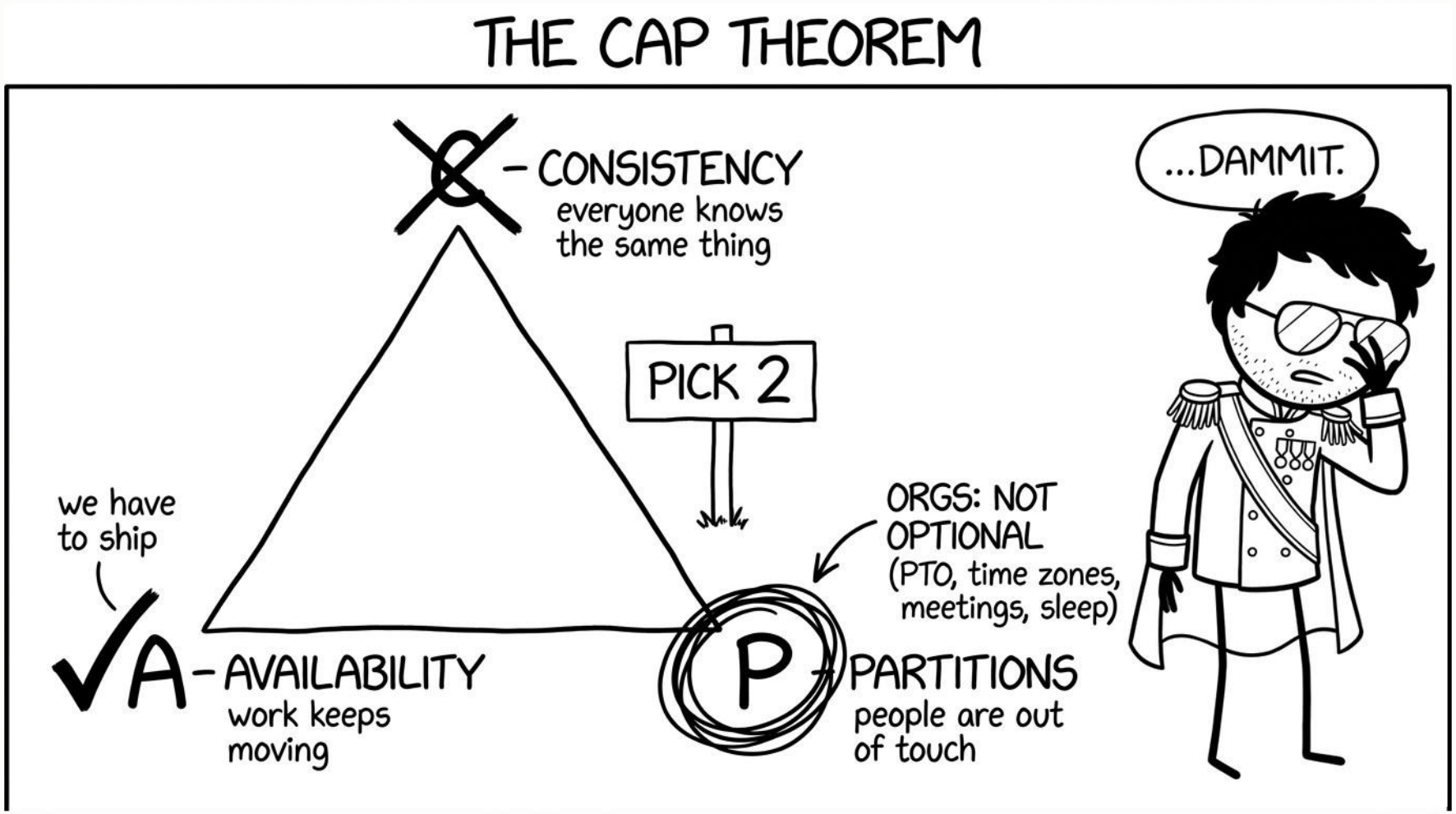
PARTITION: GONE FOR GOOD



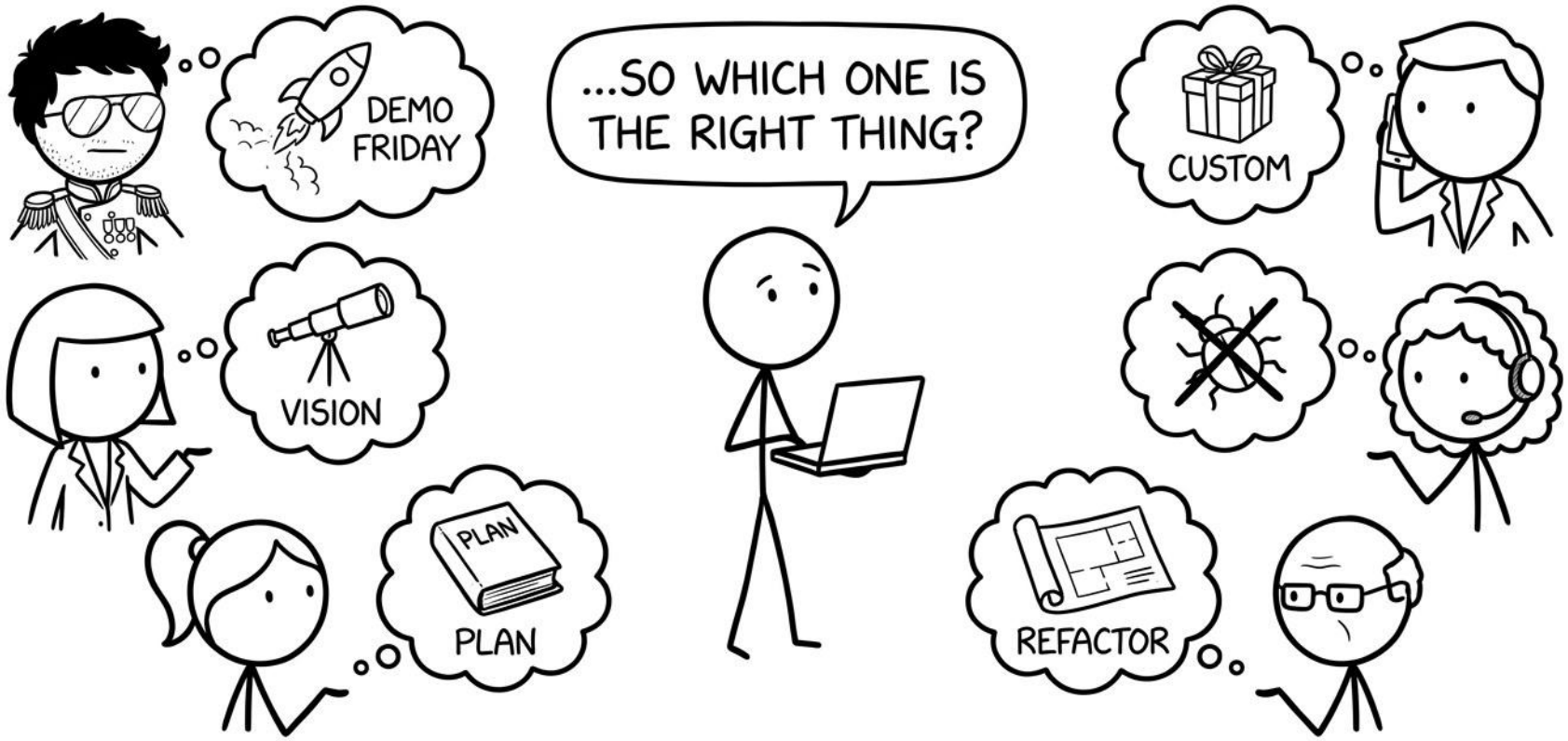

PARTITION: JUST BUSY
Pick up, pick up...
bzzt
bzzt
A or B?

AVAILABILITY: WORK NEVER STOPS... RIGHT?

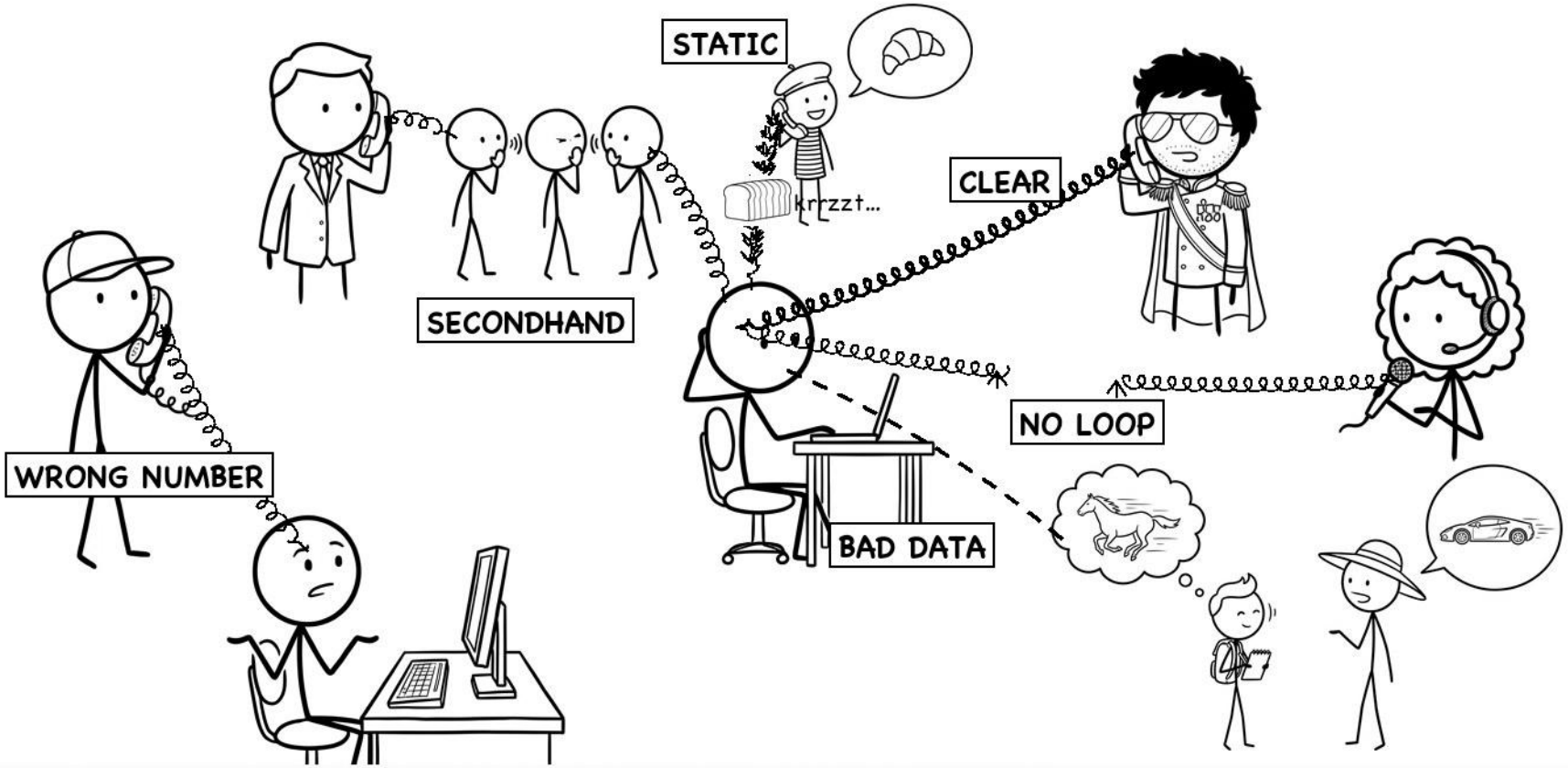



THE CAP THEOREM
C - CONSISTENCY
everyone knows the same thing
PICK 2
we have to ship
A - AVAILABILITY
work keeps moving
P - PARTITIONS
people are out of touch
ORGS: NOT OPTIONAL
(PTO, time zones, meetings, sleep)
...DAMMIT.

Everyone has a different "right thing".

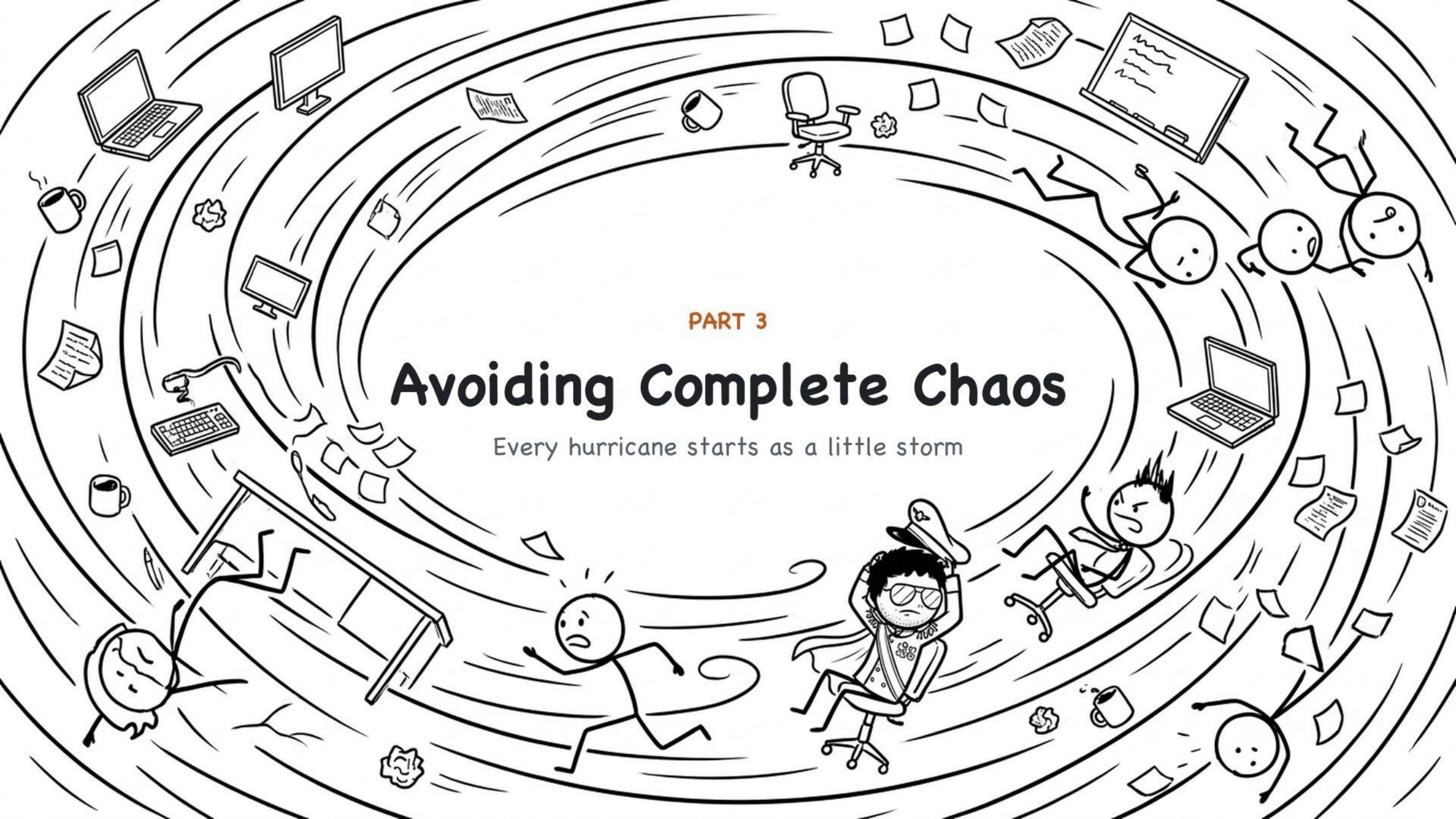


...and every signal arrives differently.





And that's why we never build "the right thing".



PART 3

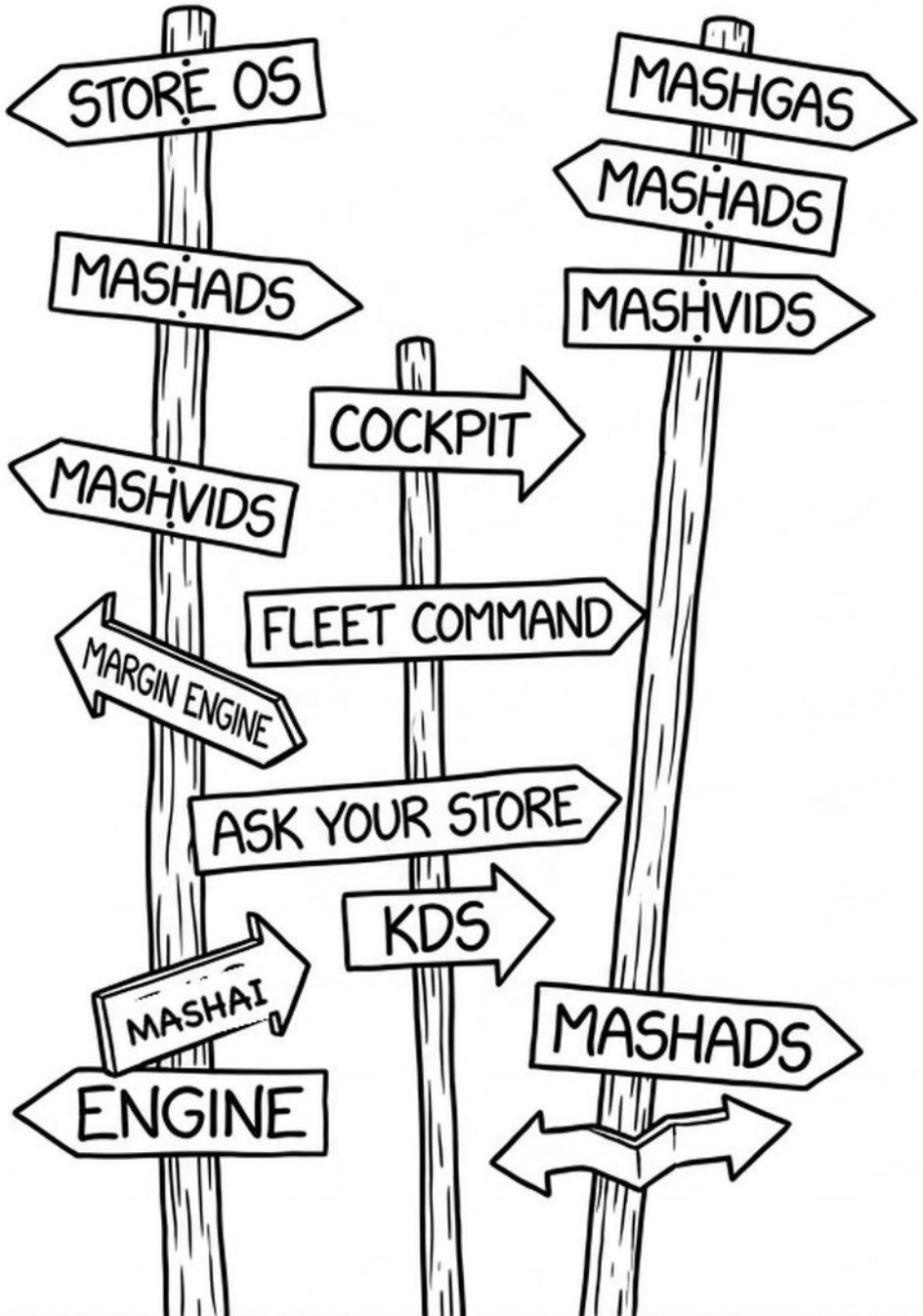
Avoiding Complete Chaos

Every hurricane starts as a little storm

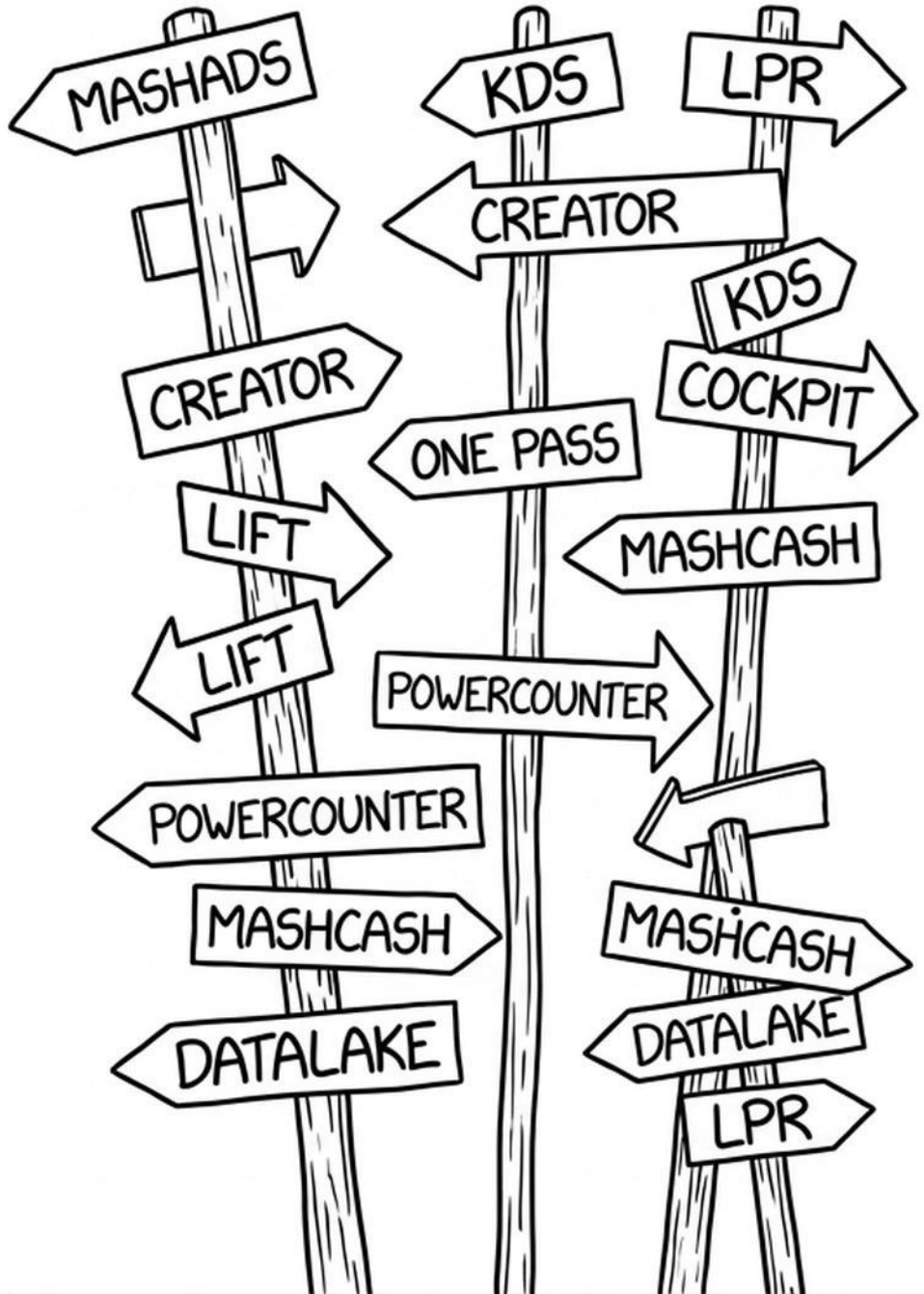
STORM 1 OF 4

Direction

"What matters?"

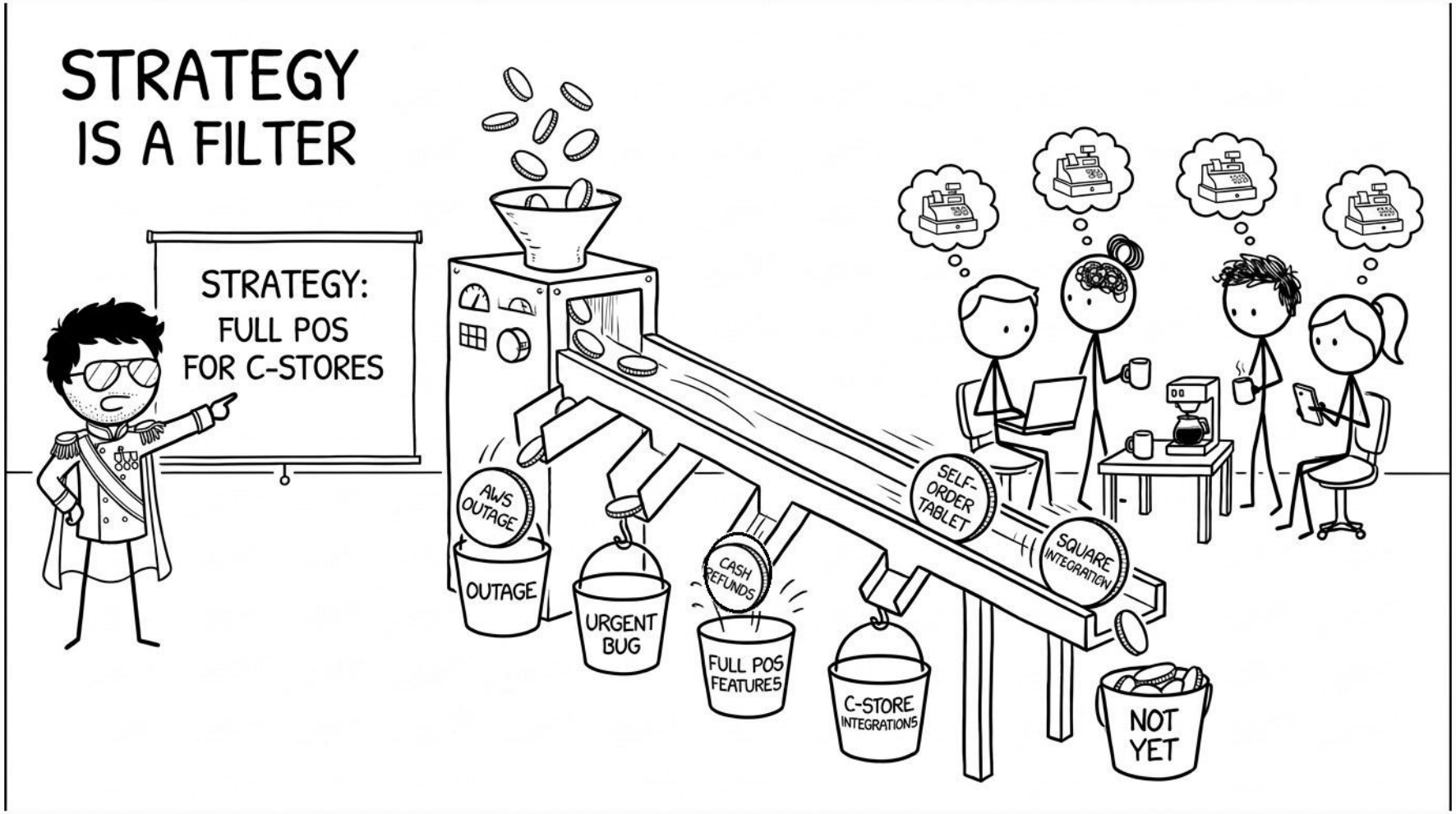






Strategy Nobody Understands




STRATEGY IS A FILTER
STRATEGY: FULL POS FOR C-STORES
AWS OUTAGE
OUTAGE
URGENT BUG
CASH REFUNDS
FULL POS FEATURES
C-STORE INTEGRATIONS
SELF-ORDER TABLET
SQUARE INTEGRATION
NOT YET

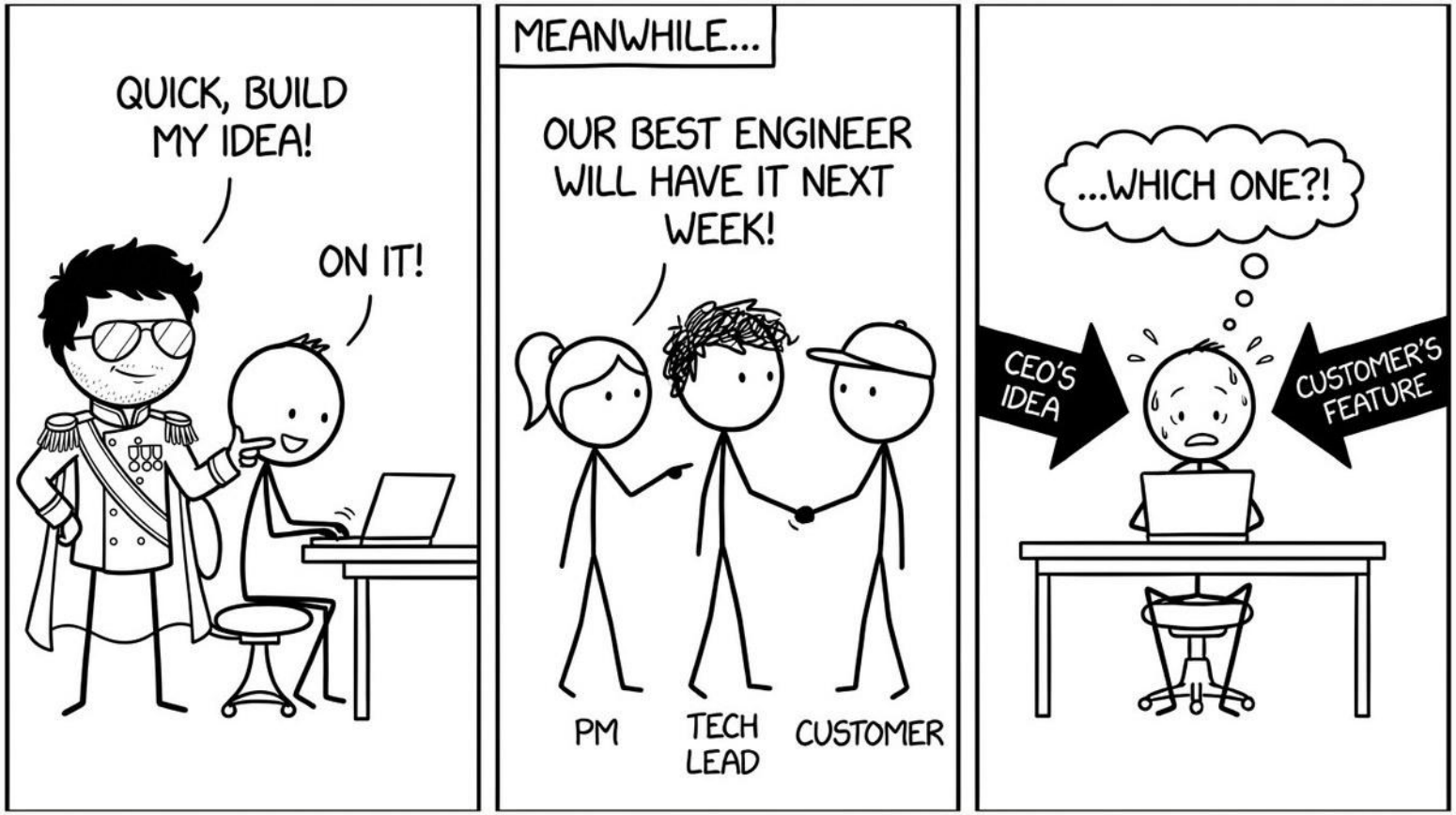
Everything Is Top Priority



ONE PRIORITIZED LIST, OUT IN THE OPEN

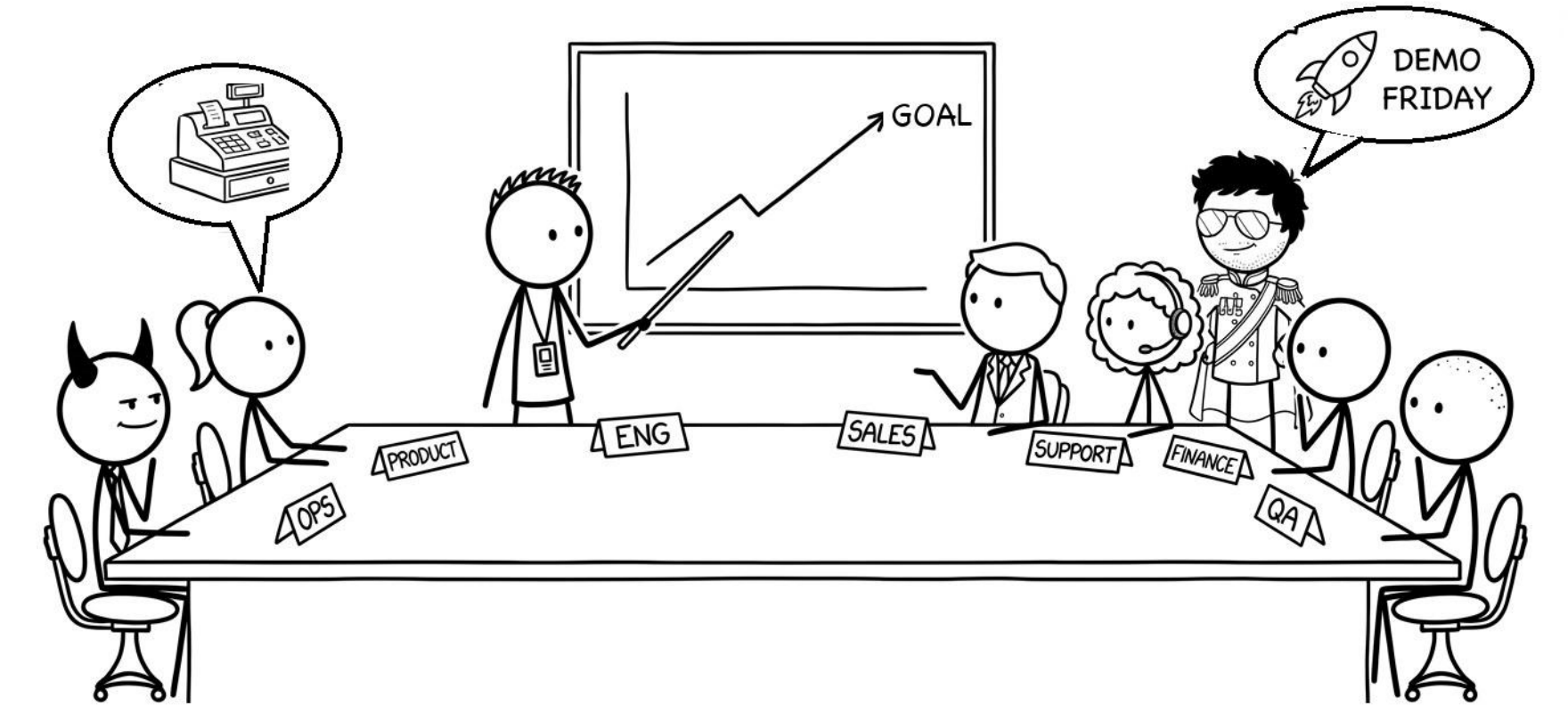


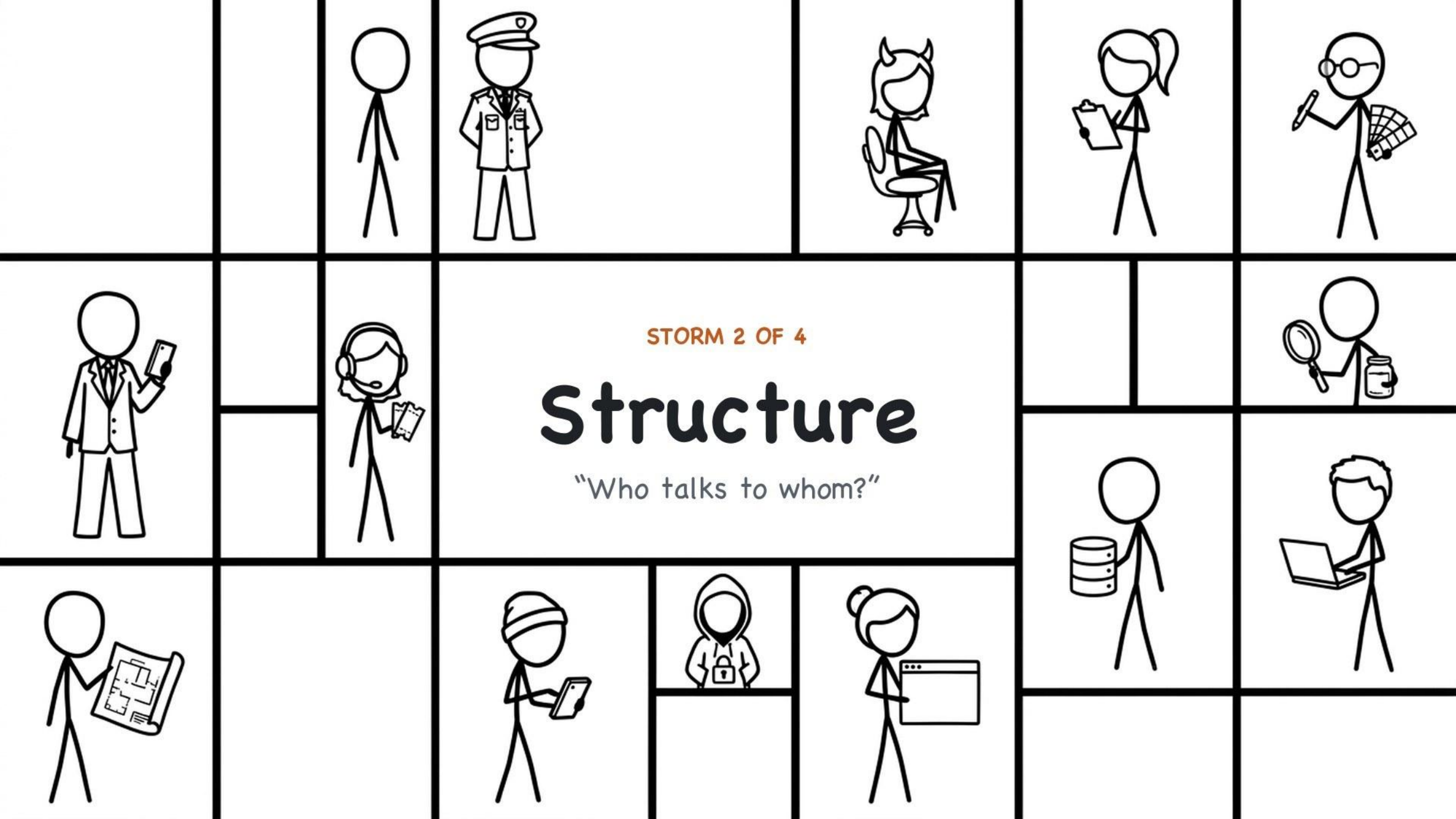
The CEO Side Door



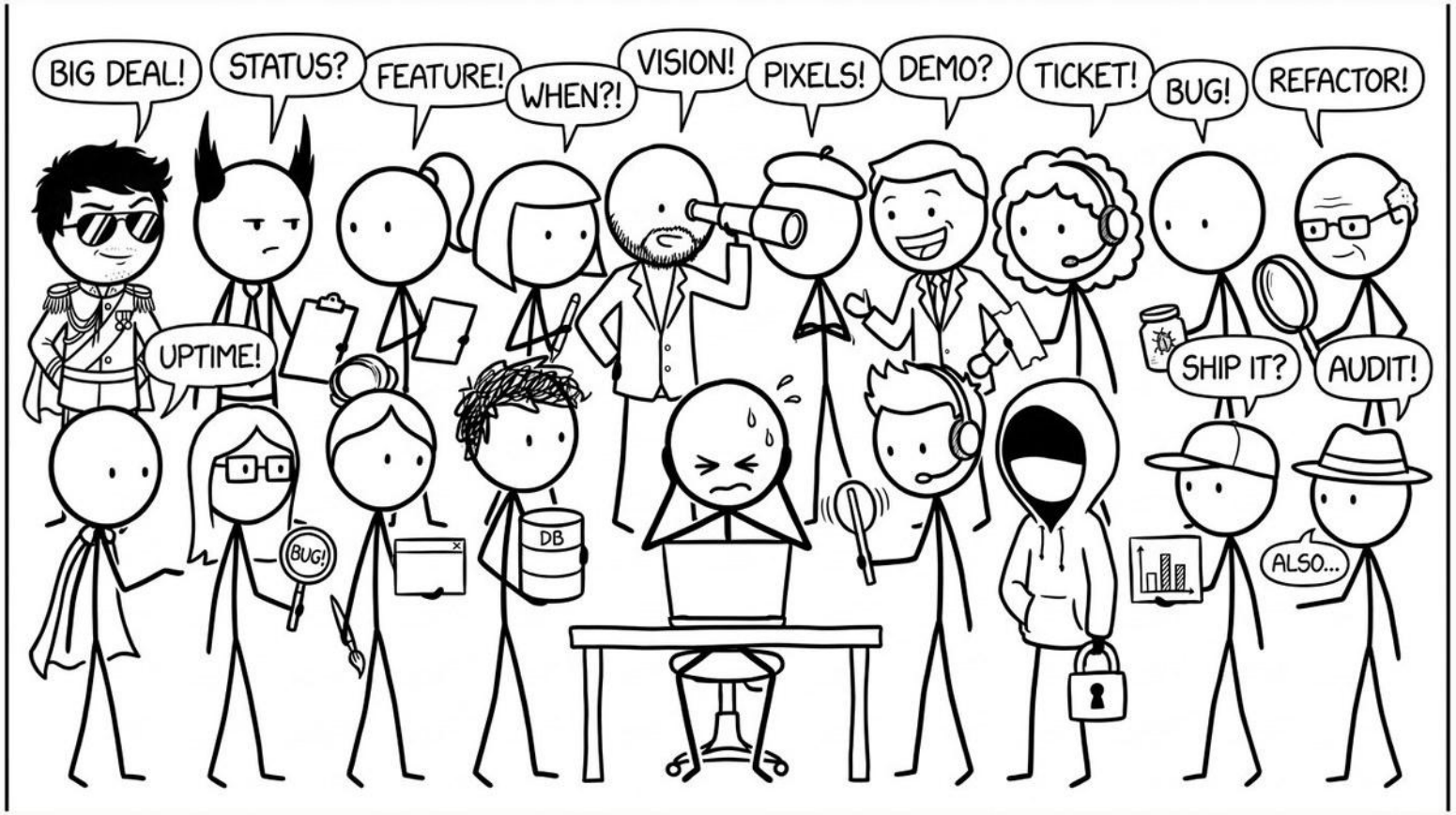
CUSTOMER WEEKLY · STORE HEALTH

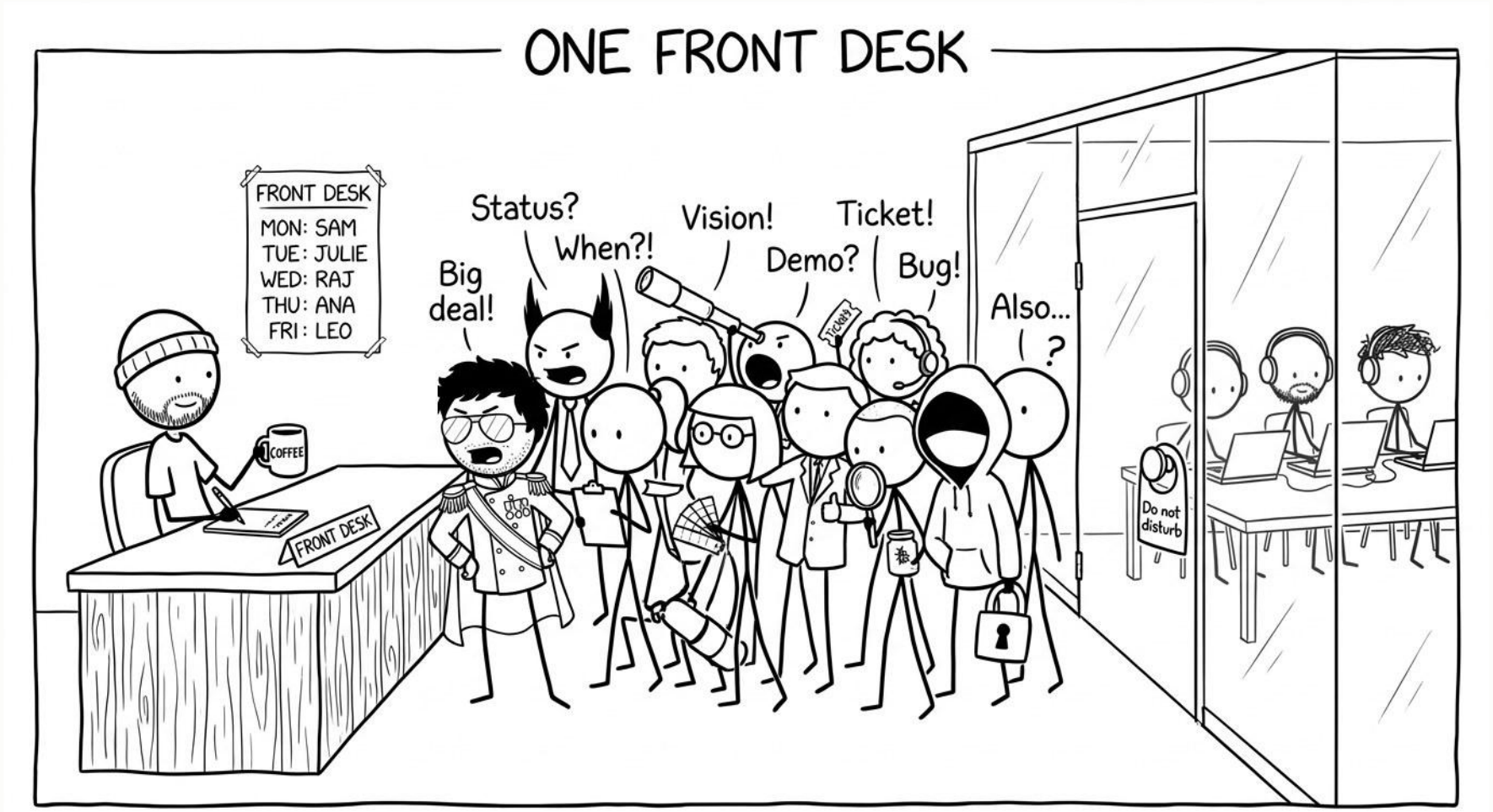
Everyone who can affect a major goal. Weekly. No excuses for the BUSINESS not to be aligned.



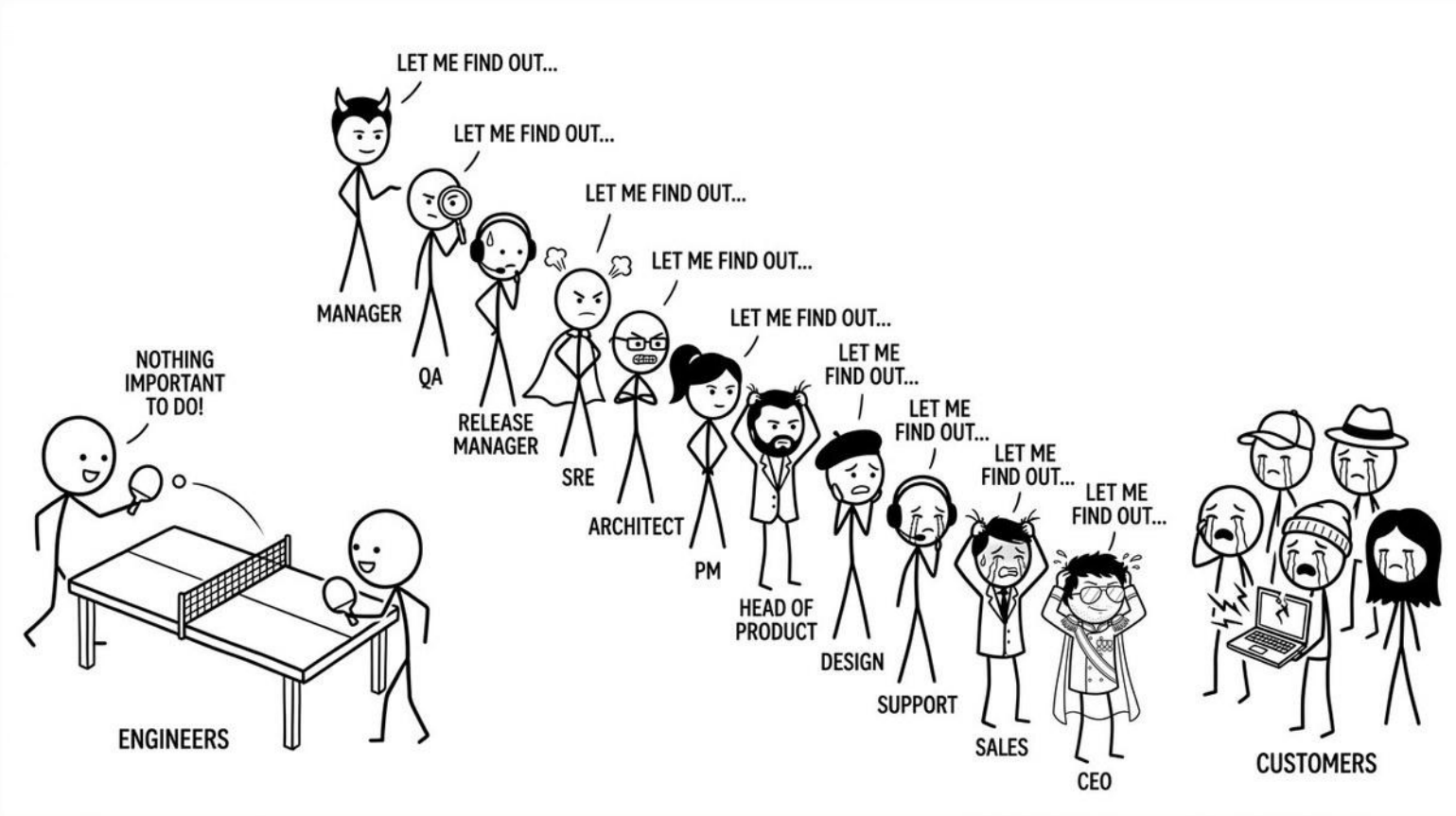

STORM 2 OF 4
Structure
"Who talks to whom?"

Everyone Piles Onto One Engineer




ONE FRONT DESK
FRONT DESK
MON: SAM
TUE: JULIE
WED: RAJ
THU: ANA
FRI: LEO
COFFEE
FRONT DESK
Big deal!
Status?
When?!
Vision!
Demo?
Ticket!
Bug!
Also...
?
TICKETS
Do not disturb

Too Many Layers



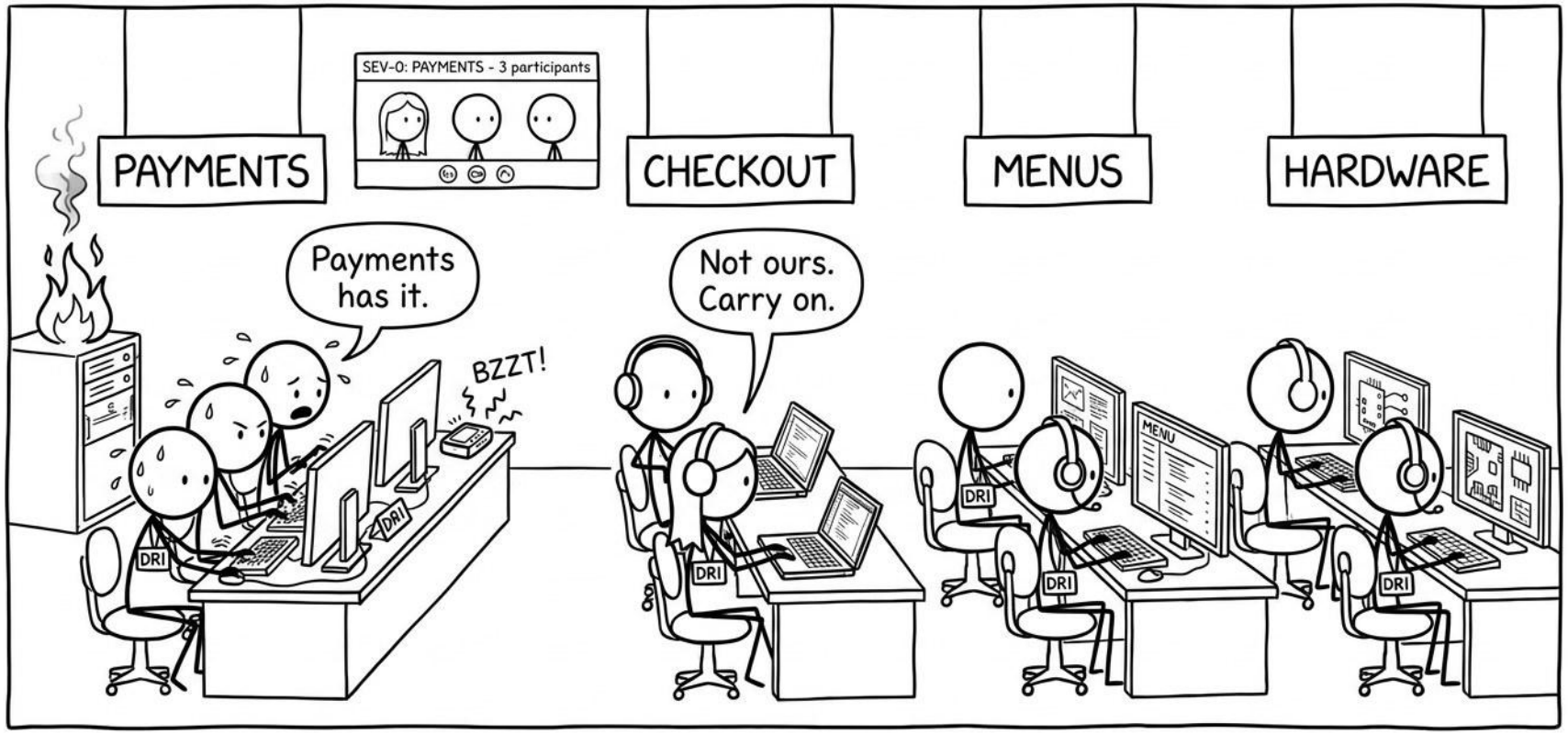
SOMETIMES, SHORTCUT THROUGH THE STRUCTURE



Everyone Is Responsible for Everything

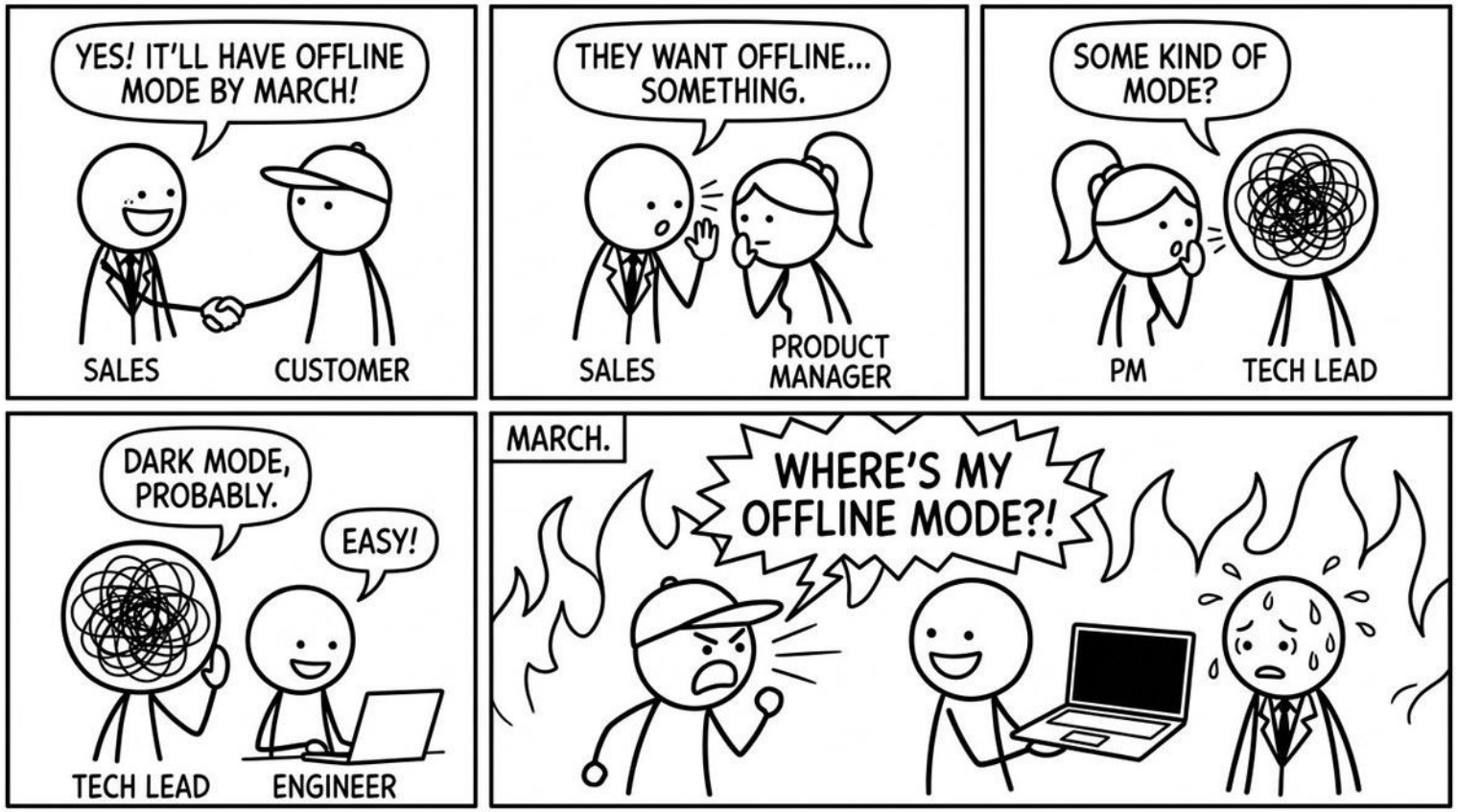


EVERY TEAM OWNS ITS OWN PIECE




STORM 3 OF 4
Communication
"How does it travel?"

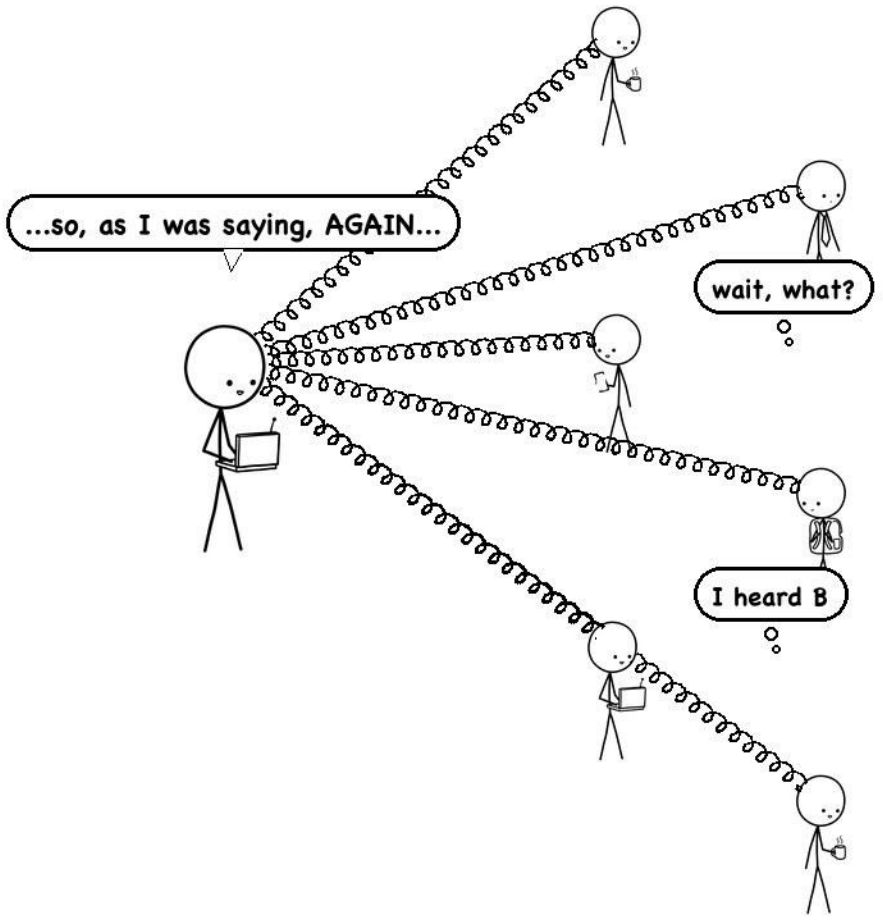
The Telephone Game



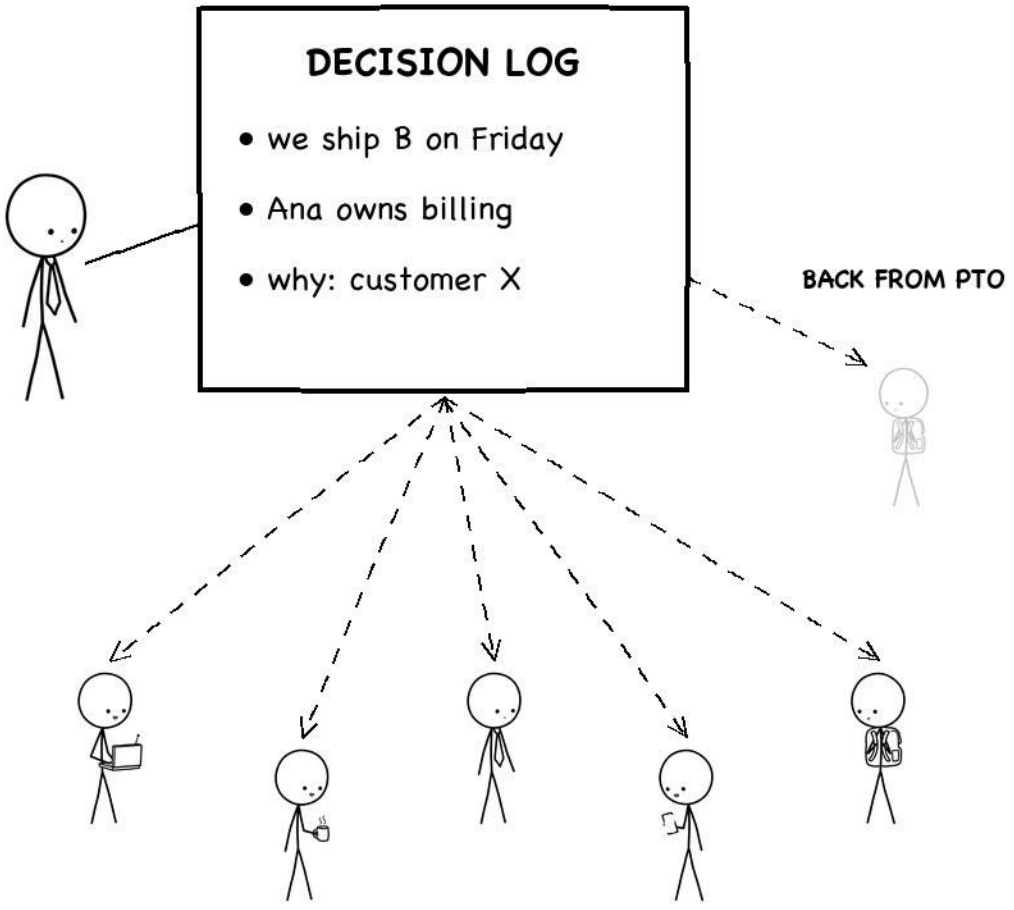
WRITE IT DOWN: BROADCAST, DON'T RPC

One write, many reads. Async. Still there when someone is on PTO.

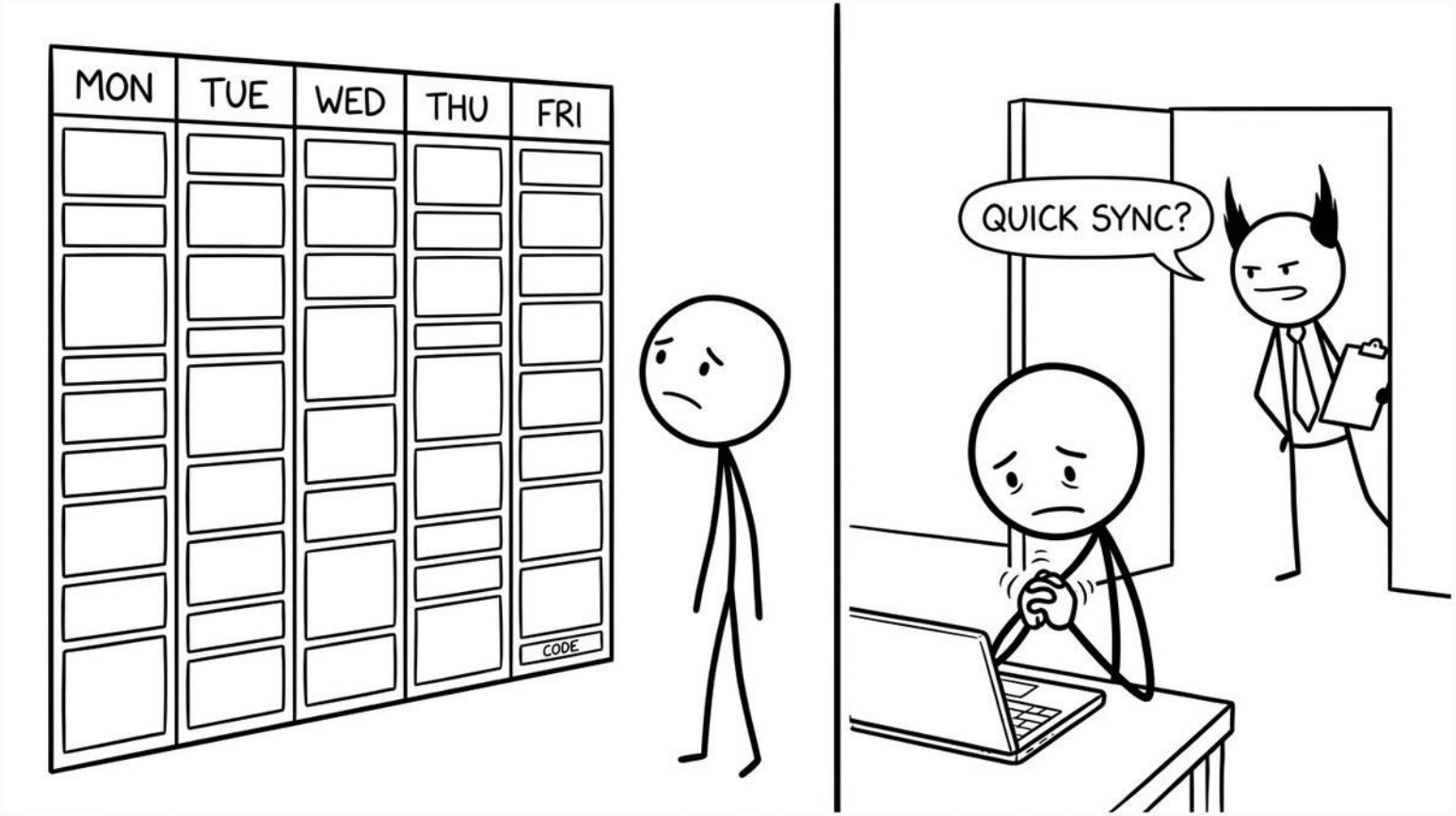
TELLING EVERYONE, ONE BY ONE



WRITE IT DOWN ONCE



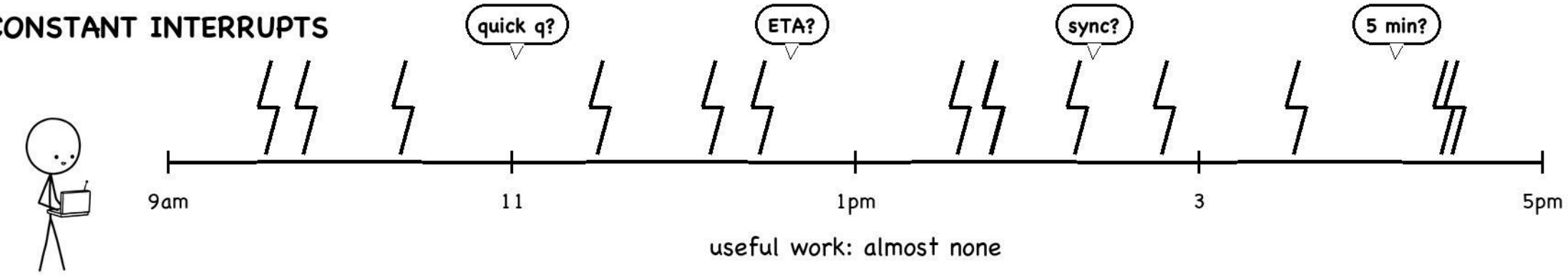
Meeting Creep



HEARTBEATS & BATCHING

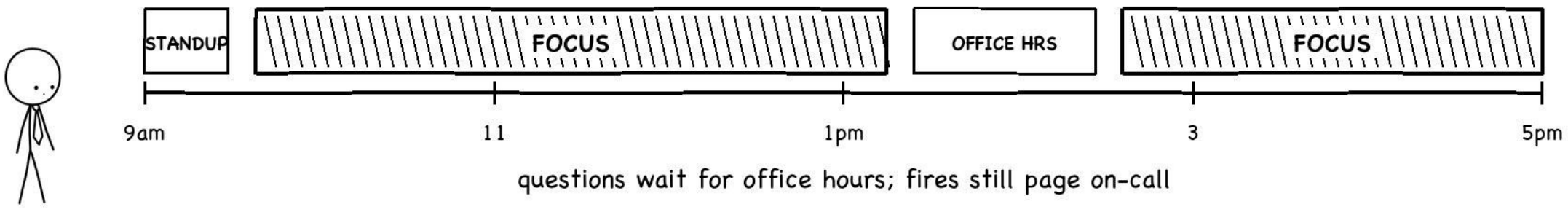
Predictable rituals beat constant interrupts.

CONSTANT INTERRUPTS



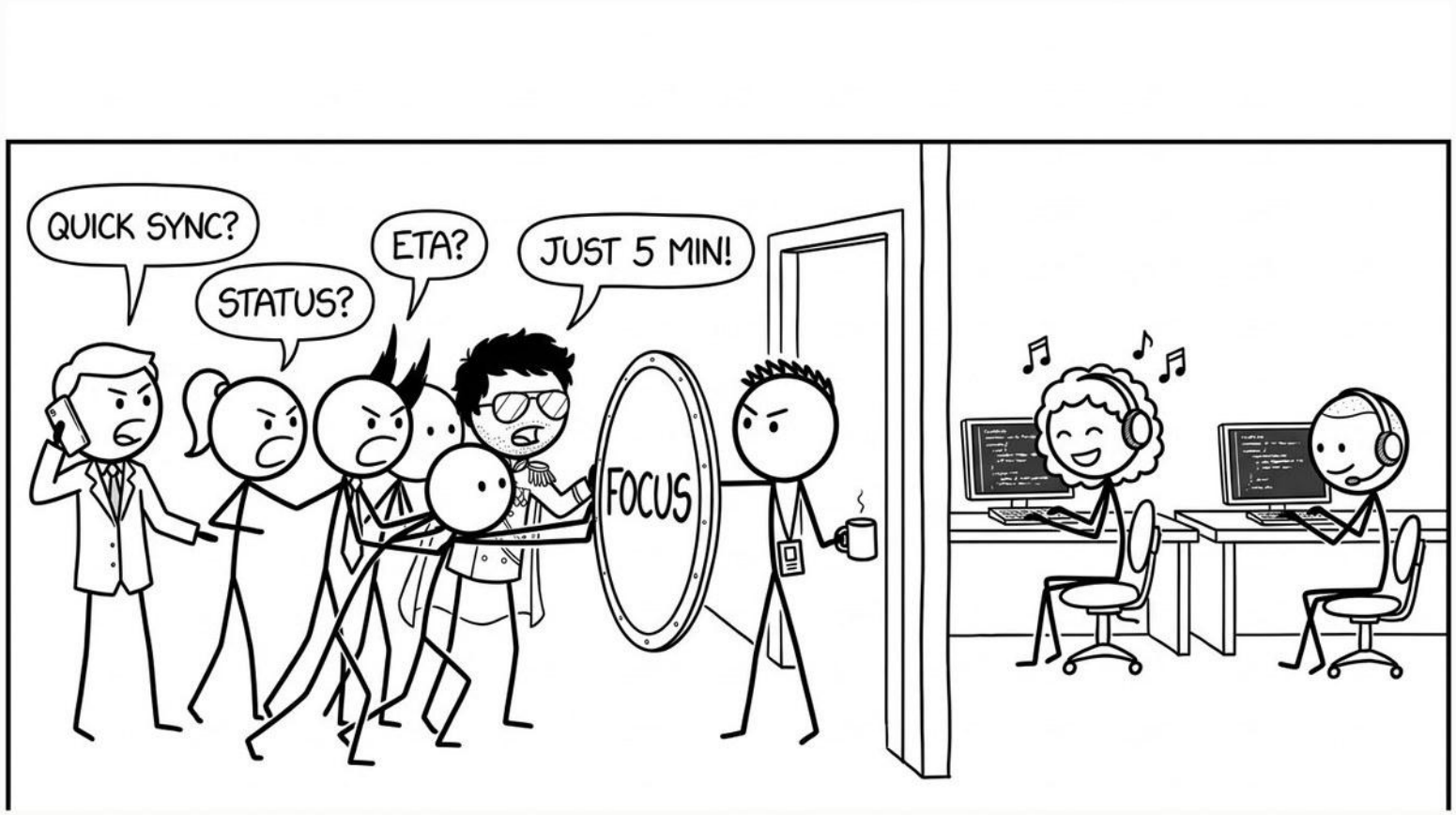
useful work: almost none

HEARTBEATS + BATCHING



questions wait for office hours; fires still page on-call

Protect the CPU



PICK THE CONSISTENCY LEVEL

Not everything needs everyone to know right now.

STRONG

everyone, right now

- incidents
- priority changes
- who owns what

HOW:

say it live AND write it down

cost: $$$

EVENTUAL

the right people, this week

- roadmap changes
- decisions + why
- new hires, departures

HOW:

weekly notes, decision log

cost: $$

NEED-TO-KNOW

whoever touches it

- implementation details
- how the cache works
- edge cases

HOW:

in the code, PR and docs

cost: $

...and that includes YOU. You don't need to know everything.

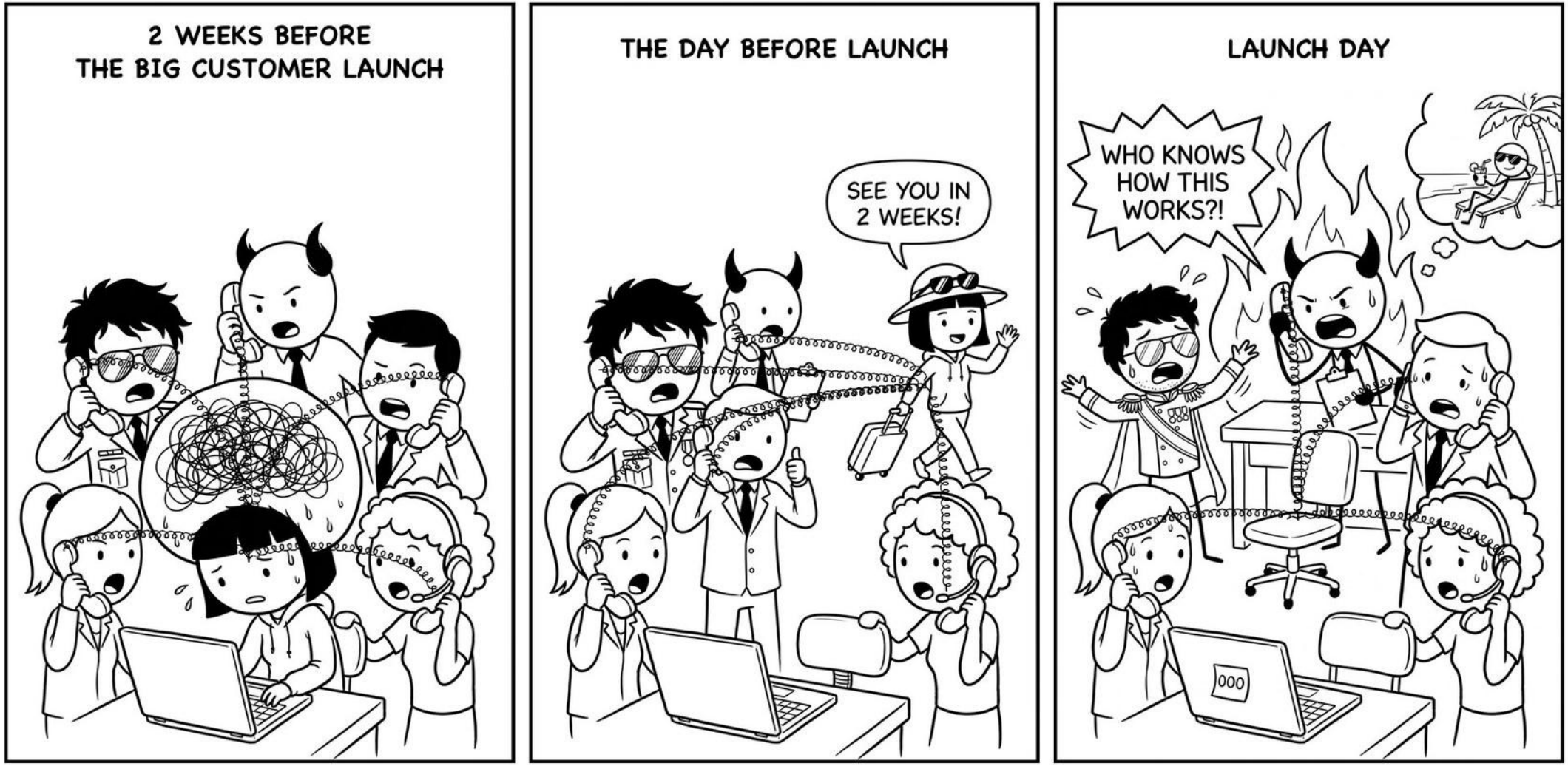
Don't burn all your cycles on info acquisition.

STORM 4 OF 4

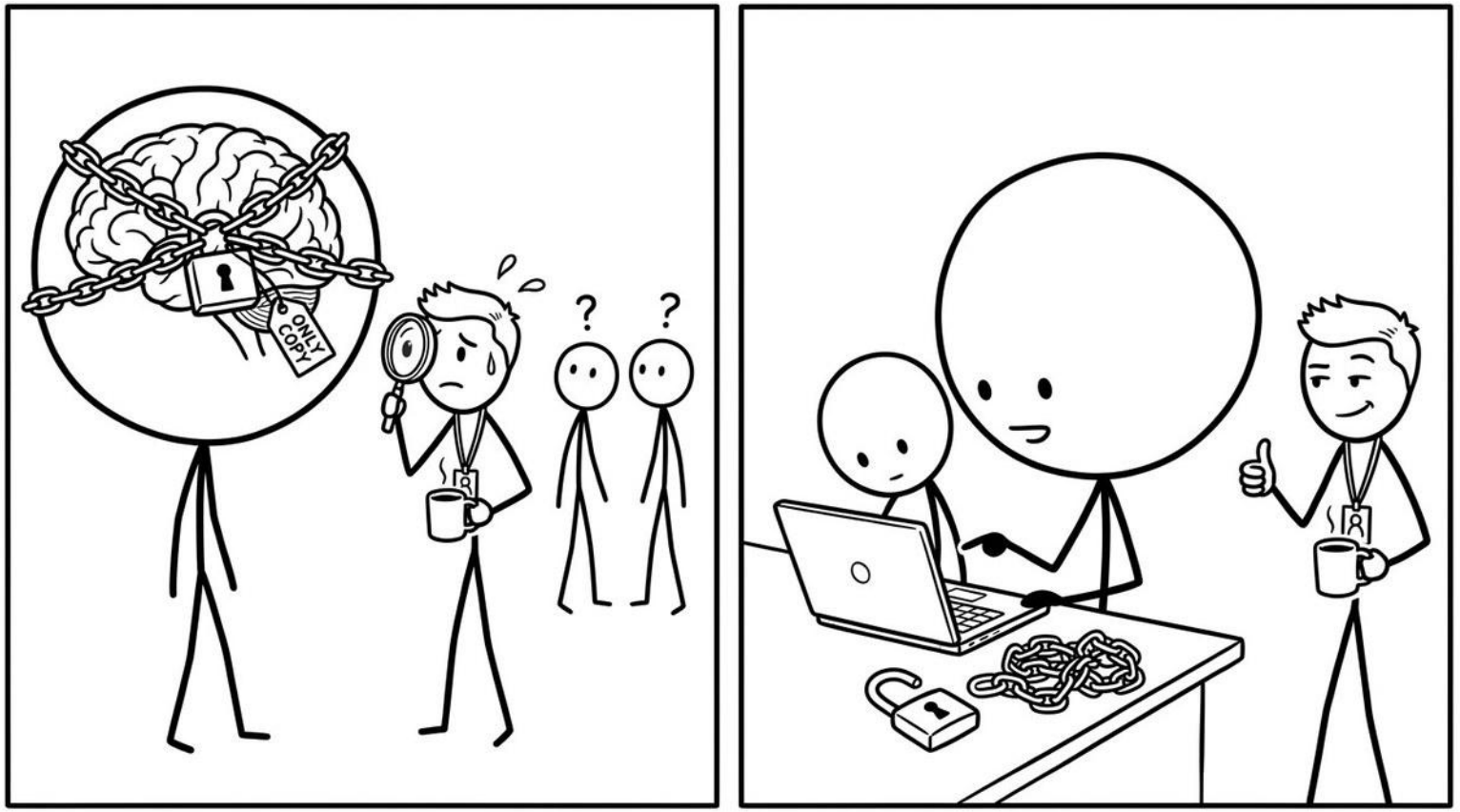
Knowledge

"Where does it live?"

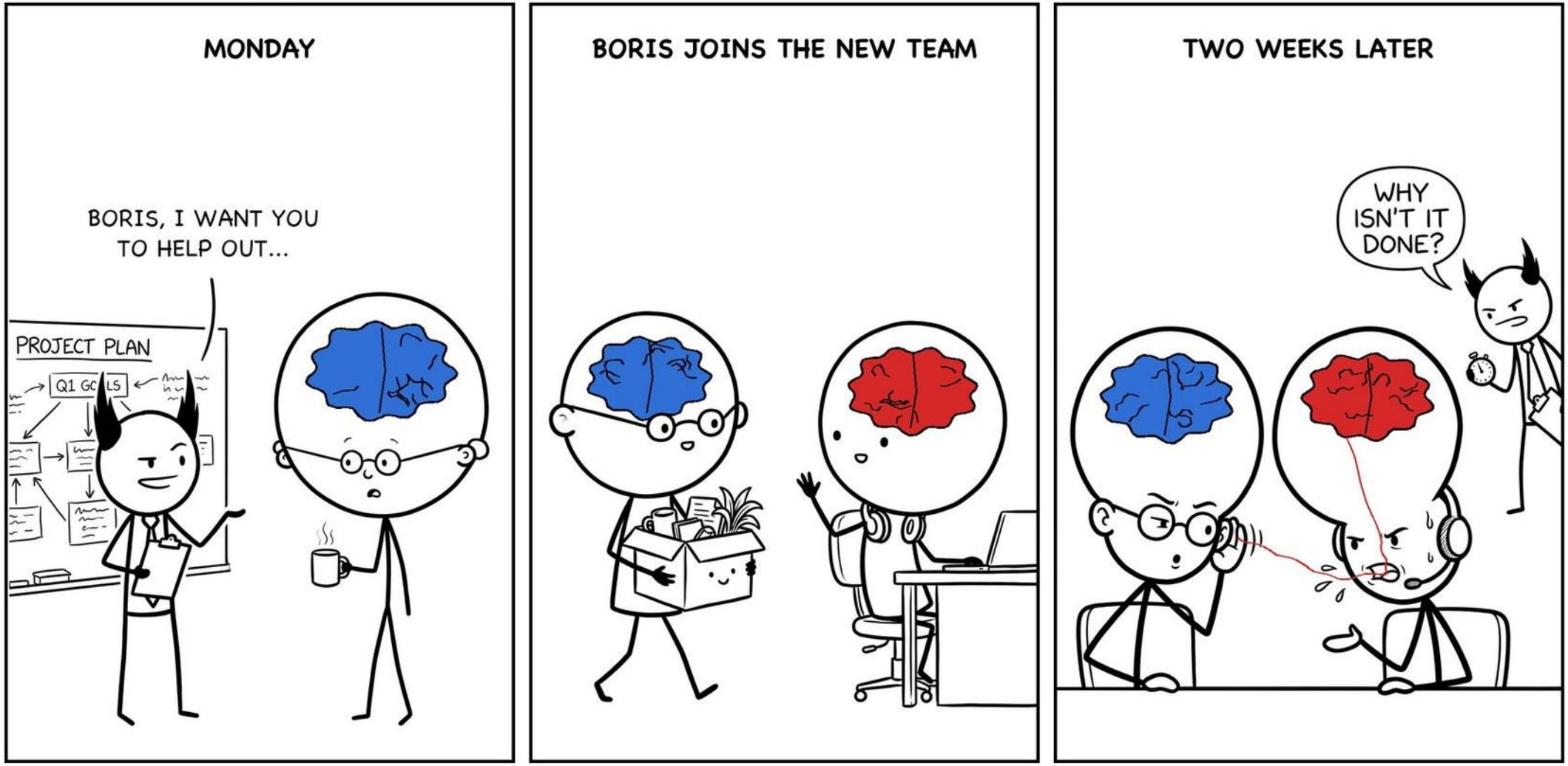
The Hero Dependency



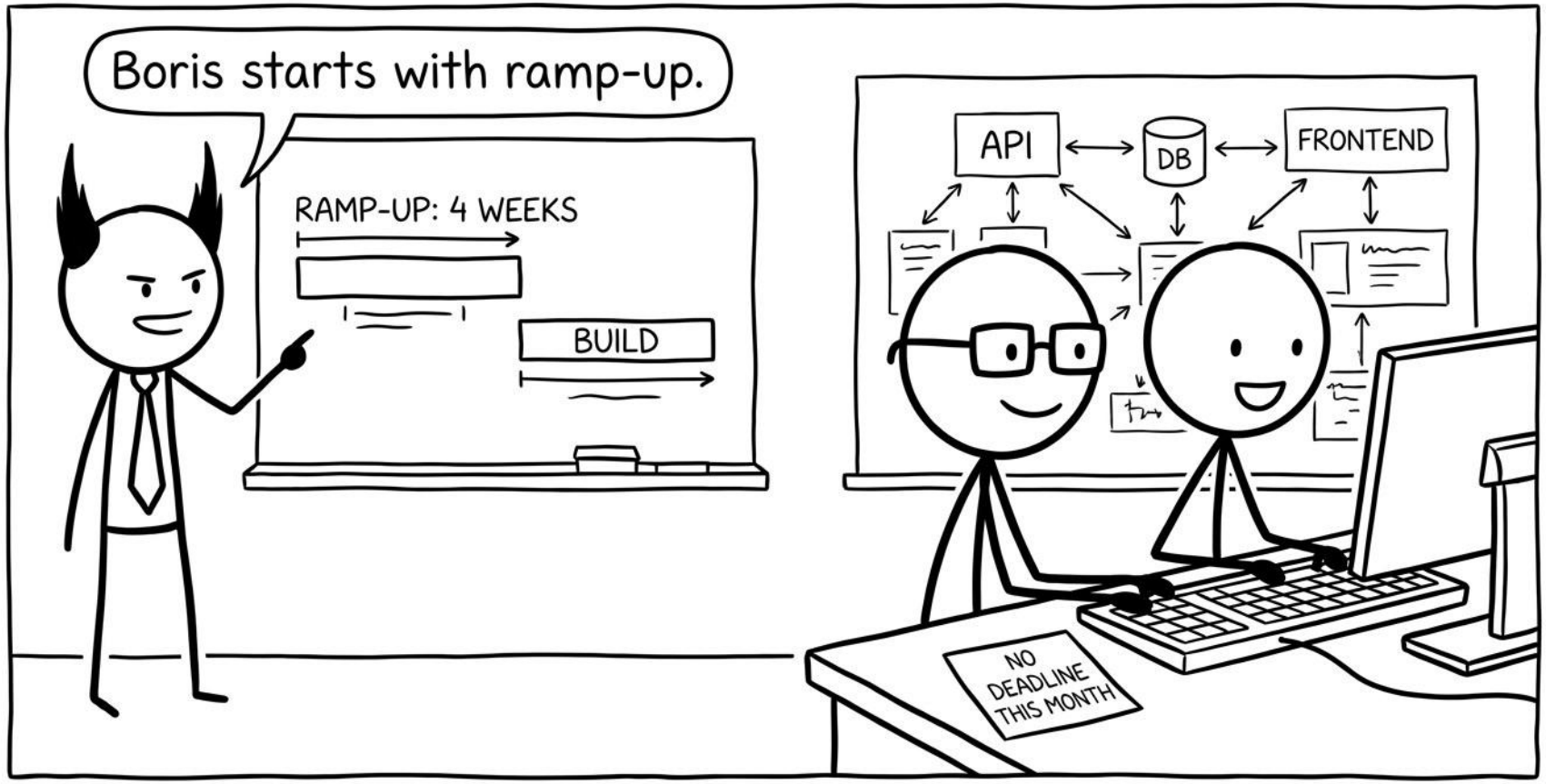
Find Where Knowledge Is Stuck



Engineers Aren't Fungible



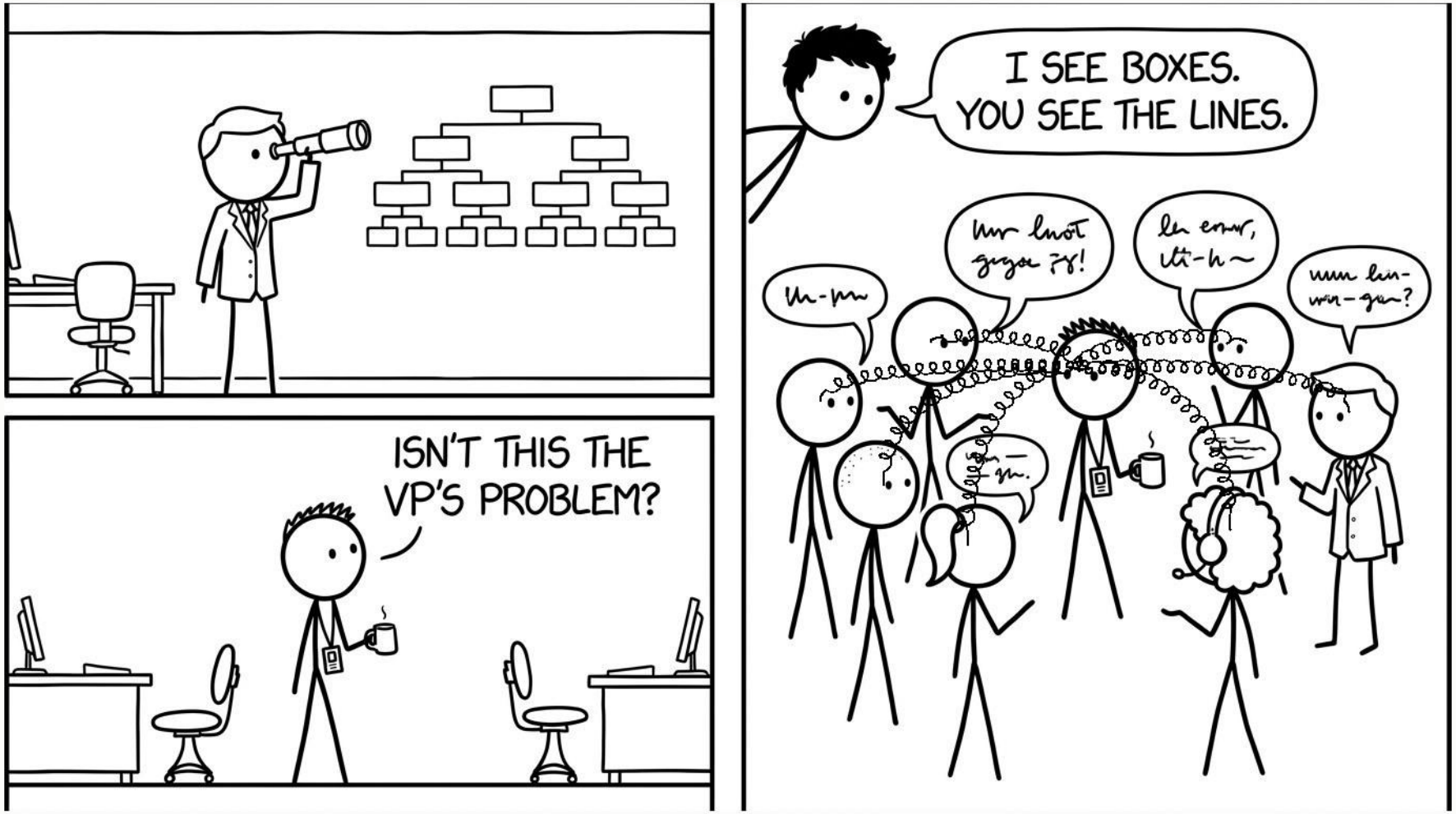
BUDGET THE RAMP-UP



PART 4

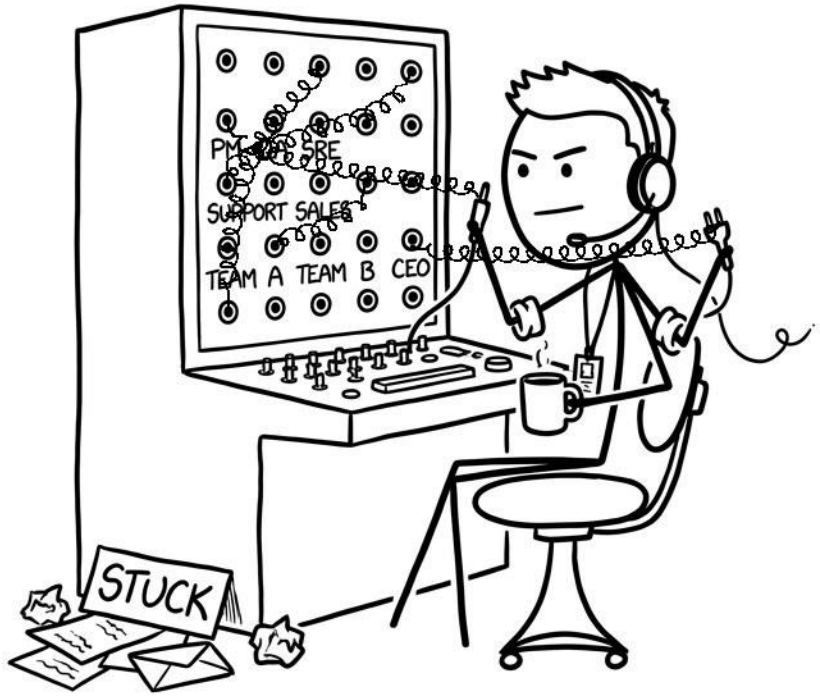
The EM's Responsibility

(Yes, you.)


ISN'T THIS THE VP'S PROBLEM?
I SEE BOXES. YOU SEE THE LINES.

WHO needs to know WHAT, WHEN — and WHY?

WHERE are they going to learn it?

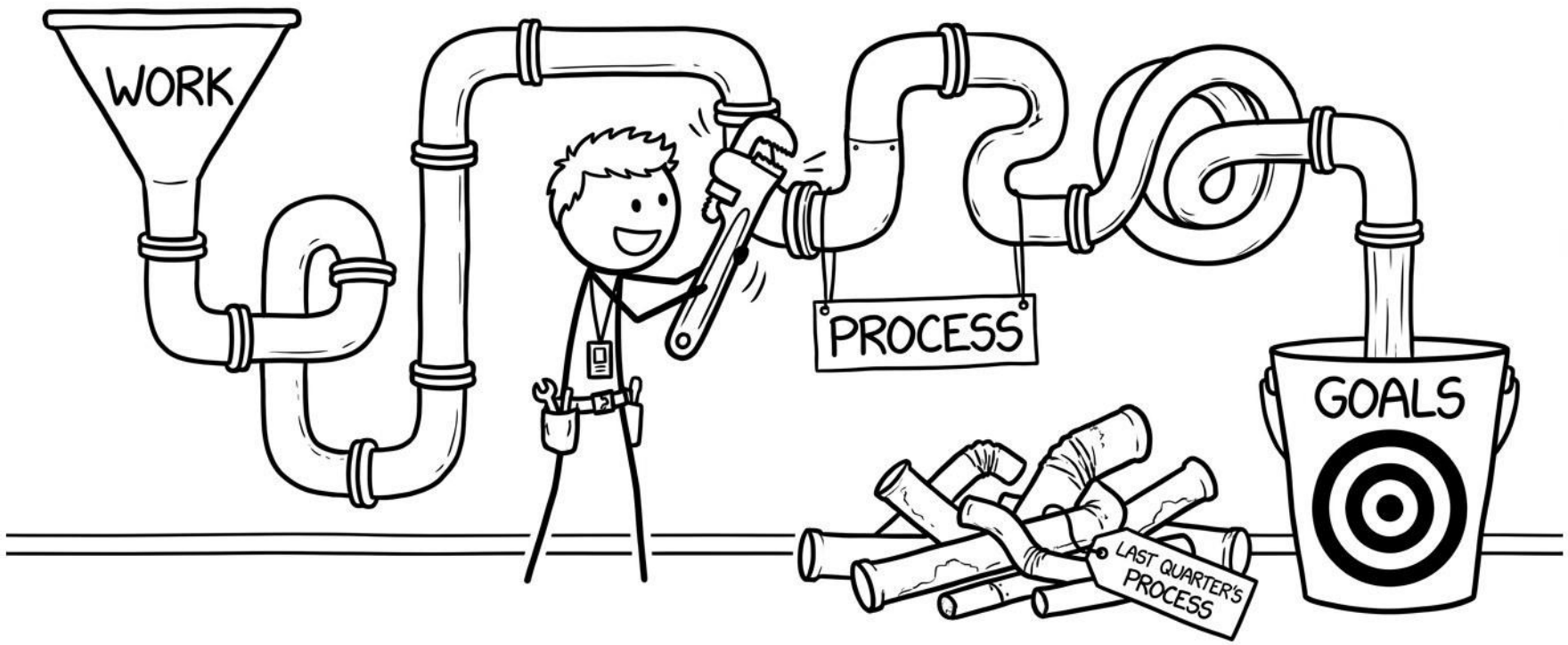


Yes, you're just
a fancy router.

(Sorry.)

SHAPE PROCESS TO FIT GOALS

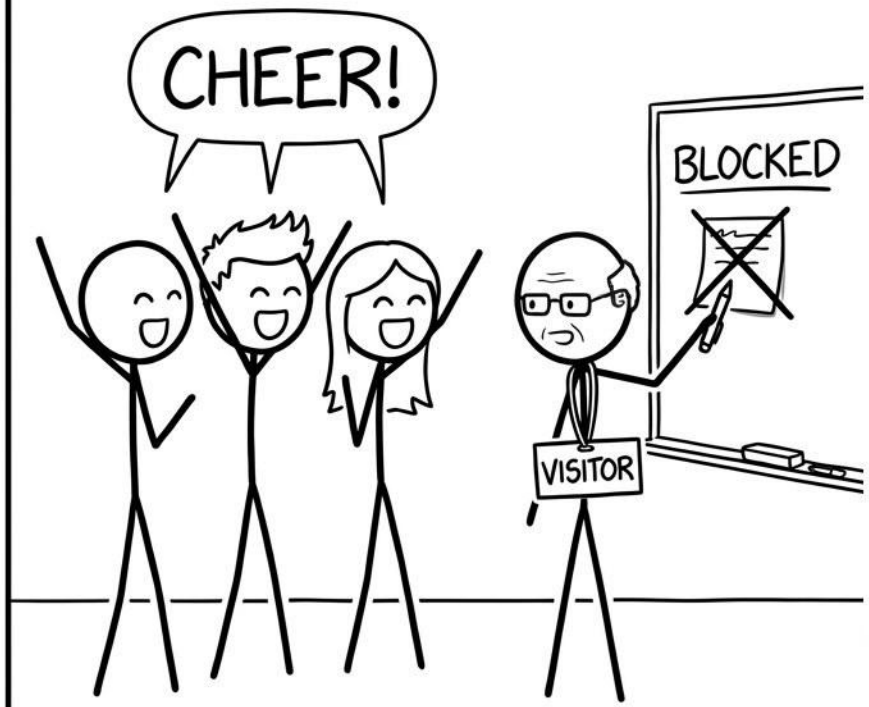
...and be comfortable changing it.



STANDUPS: SUPER HIGH BANDWIDTH, SUPER EXPENSIVE. USE JUDICIOUSLY.



VIRTUAL TEAMS: BORROW (FOR A WHILE) WHOEVER UNBLOCKS YOU.



THE EM'S CHECKLIST

Four storms to watch for, every week.

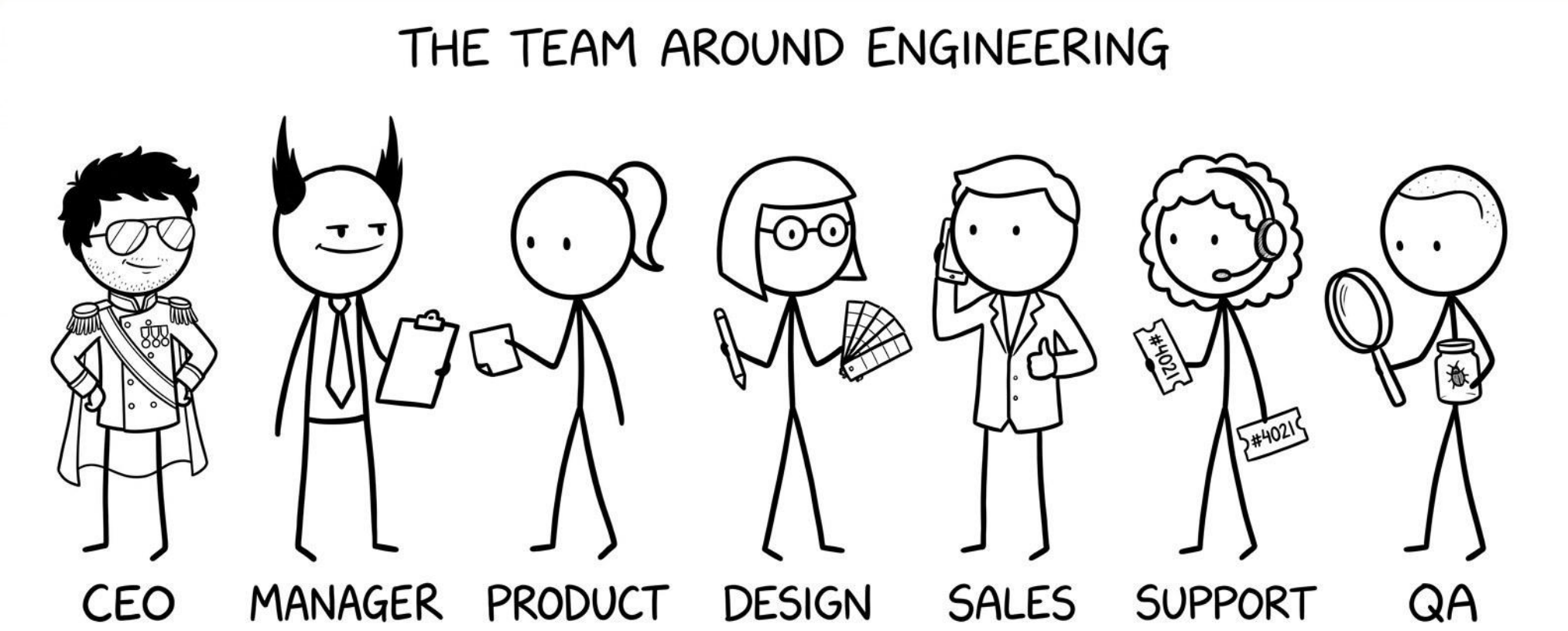
- [x] **DIRECTION**
  Does everyone know what matters, and why?

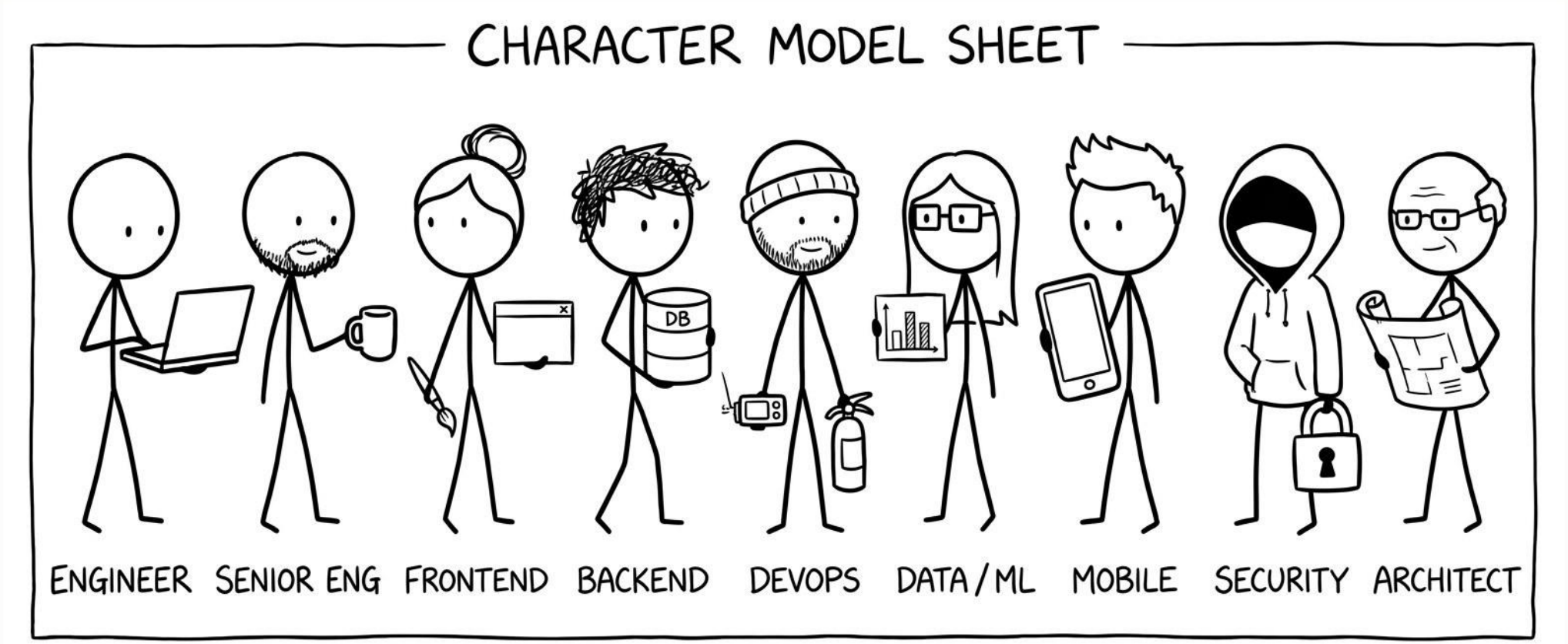
- [x] **STRUCTURE**
  Who talks to whom? Small teams, clear owners?

- [x] **COMMUNICATION**
  Is it written down? Syncing too much, or too little?

- [x] **KNOWLEDGE**
  Where is it stuck? What if they go on PTO tomorrow?




THE TEAM AROUND ENGINEERING
#4021
#4021
CEO
MANAGER
PRODUCT
DESIGN
SALES
SUPPORT
QA

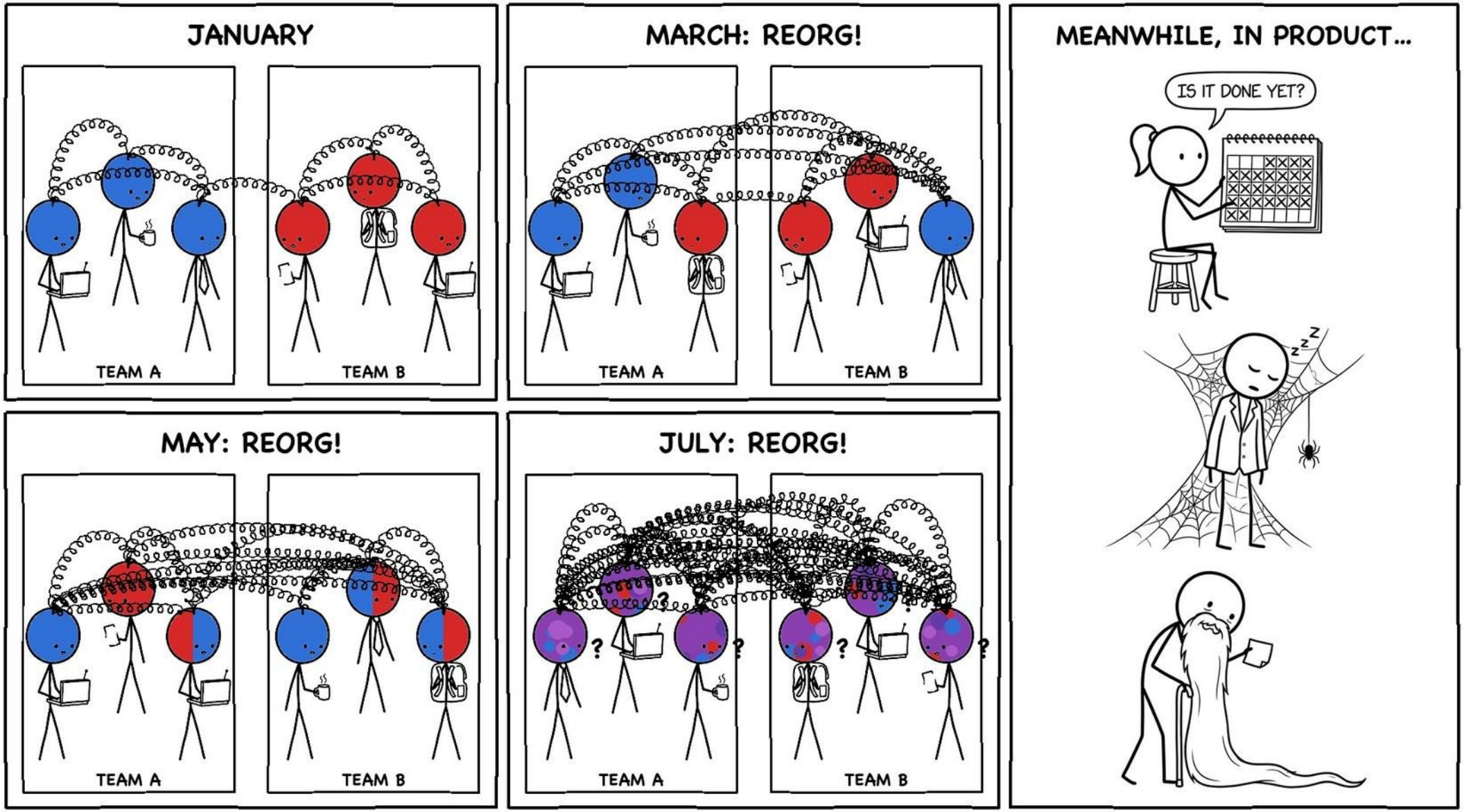

CHARACTER MODEL SHEET
DB
ENGINEER
SENIOR ENG
FRONTEND
BACKEND
DEVOPS
DATA / ML
MOBILE
SECURITY
ARCHITECT

APPENDIX

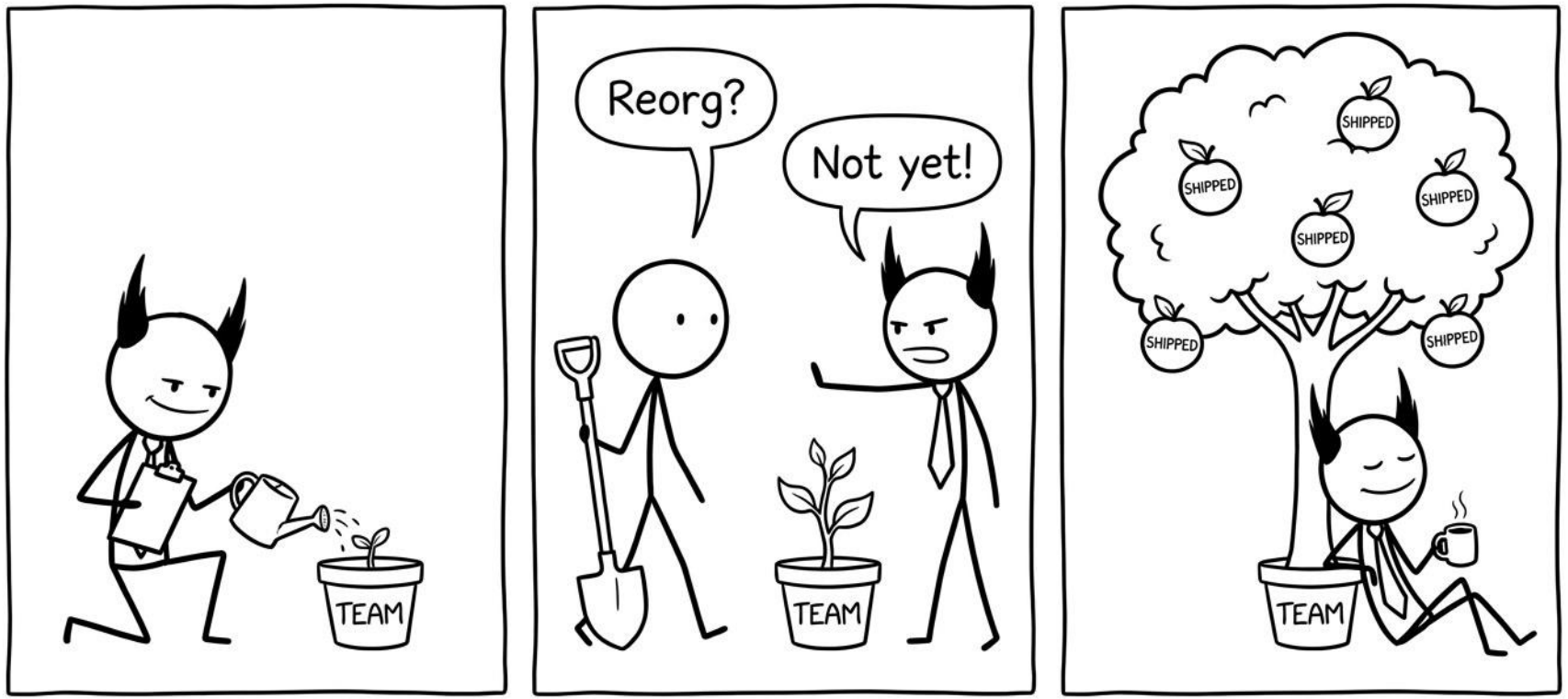
Extra Slides

Not usually shown

Reorg Whiplash



CHANGE THE WIRING SLOWLY



MONTH 1 MONTH 3 MONTH 6